SSC JE

GENERAL AWARENESS

Topper-Style Concept Notes

Paper-I (CBT-1) General Awareness – 50 Marks • Concept + Key Facts + PYQ-Trend Revision

Ancient • Medieval • Modern History • Art & Culture • Polity • Indian & World Geography • Economy • Physics • Chemistry • Biology • Sports • Books • Awards • Days • Organisations • Science & Tech • Computers

Concept-based notes – no MCQ practice. "PYQ-Trend" boxes list concepts that SSC JE has repeatedly tested in past years. Always cross-check with official papers.

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INTRODUCTION

How to Use These Notes & GA Blueprint

What Part-B (General Awareness) Tests

In SSC JE Paper-I (CBT-1), General Awareness carries 50 questions of 1 mark each (−0.25 for a wrong answer in recent notifications; confirm in yours). Questions are short factual recalls: one line, one concept. SSC repeats the same concepts across years with the wording changed, so knowing a concept well covers several possible questions.

Area	Typical questions (out of 50)	Nature
History (Ancient + Medieval + Modern)	6–9	Rulers, battles, movements, sessions of INC, Governors-General
Art & Culture	3–5	Dances, festivals, musicians, temples, paintings
Polity	5–8	Articles, schedules, amendments, bodies, Parliament
Geography (India + World)	5–8	Rivers, passes, national parks, soils, latitudes, straits
Economy	3–5	Banking, RBI, budget, taxes, five-year plans, schemes
Physics	3–5	Units, laws, optics, sound, electricity
Chemistry	3–5	Elements, compounds, acids, metals, common names
Biology	4–6	Cells, vitamins, diseases, human body, plants
Static GK (sports, books, awards, days, organisations)	5–8	Pure memory facts
Current affairs & Science-Tech	3–6	Recent events (read monthly separately)

How Each Chapter Is Built

- Concept points in compact bullets, with tables for lists that SSC loves (rulers, dances, rivers, vitamins).
- Yellow strips = key facts to memorise exactly (dates, numbers, names). They are collected again in Appendix A.
- Blue PYQ-TREND boxes = concepts SSC has asked repeatedly in past papers. They are collected again in Appendix B. They describe what gets asked; they are not reproduced questions.
- Green TOPPER TRICK boxes = memory tricks. Red COMMON MISTAKE boxes = traps where students lose marks.

- No MCQs are included. Test yourself with the recall bank near the end instead.

45-Day GA Plan (1–1.5 hours a day)

Days	Target
1–8	History (Ancient, Medieval, Modern)
9–11	Art & Culture
12–17	Polity
18–23	Geography (India + World)
24–26	Economy
27–34	General Science (Physics, Chemistry, Biology)
35–39	Static GK + Science-Tech + Computers
40–45	Recall bank, appendices, last-minute facts, plus current affairs of the last 6–8 months

✔ **TOPPER TRICK:** GA is the fastest section in the exam: 50 questions can be finished in 10–12 minutes. The time you save goes to reasoning and the technical part. Attempt GA first, answer what you know, and don't stare at unknown questions.

✖ **COMMON MISTAKE:** Static facts that change (office holders, latest award winners, record holders, rankings) must be checked against a current-affairs source. These notes deliberately focus on stable facts.

CHAPTER 1

Ancient Indian History

1.1 Prehistoric Period

- Palaeolithic (Old Stone Age): hunter-gatherers; quartzite tools; Bhimbetka rock shelters (Madhya Pradesh) have paintings (UNESCO site). Soan valley (Pakistan), Belan valley (UP), Attirampakkam (Tamil Nadu).
- Mesolithic (Middle Stone Age): microliths; first domestication of animals (Adamgarh in MP, Bagor in Rajasthan).
- Neolithic (New Stone Age): agriculture and settled life; polished tools. Mehrgarh (Baluchistan) – earliest evidence of agriculture (wheat, barley) in the subcontinent. Burzahom (Kashmir) – pit dwellings and burial of dogs with masters. Chirand (Bihar) – bone tools. Koldihwa (UP) – early rice.
- Chalcolithic (Copper-Stone Age): first use of copper; Ahar (Banas culture), Jorwe, Malwa, Kayatha cultures; Inamgaon, Navdatoli.

Bhimbetka – rock paintings (MP, UNESCO) | Mehrgarh – earliest agriculture | Burzahom – pit houses, dog burials | Koldihwa – early rice

★ **PYQ-TREND:** Bhimbetka location and its rock paintings. Mehrgarh as earliest farming site. Burzahom pit dwellings. Microliths belong to the Mesolithic age.

1.2 Indus Valley (Harappan) Civilisation

- Period ≈ 2500–1750 BCE (mature phase). Bronze Age civilisation. Discovered in 1921 at Harappa (Daya Ram Sahni) and 1922 at Mohenjo-daro (R.D. Banerji). John Marshall was Director-General of ASI then. Named Harappan after the first site found.
- Features: town planning with grid streets, covered drains, burnt bricks (standard ratio 1:2:4), citadel and lower town, granaries, no temples found, script undeciphered (boustrophedon in part; mostly right to left), seals mainly of steatite, standard weights (binary/16 multiples).
- Crops: wheat, barley, peas, sesame; cotton was first grown here (Greeks called it Sindon). Rice husk at Lothal and Rangpur. No iron. Horse remains doubtful (Surkotada).

Site	Location (today)	Remember for
Harappa	Ravi, Punjab (Pakistan)	First excavated; granaries, cemetery R-37
Mohenjo-daro	Indus, Sindh (Pakistan)	Great Bath, Great Granary, bronze dancing girl, priest-king bust; 'mound of the dead'
Lothal	Bhogavo, Gujarat	Dockyard (earliest), rice husk, bead-making
Kalibangan	Ghaggar, Rajasthan	Ploughed field, fire altars, 'black bangles'
Dholavira	Rann of Kutch, Gujarat	Water reservoirs, 3-part city, signboard; UNESCO site (2021)

Site	Location (today)	Remember for
Rakhigarhi	Haryana	Largest Harappan site in India
Chanhudaro	Sindh	Only city without a citadel; bead factory
Banawali	Haryana	Toy plough, barley
Surkotada	Gujarat	Horse bones (disputed)
Ropar	Punjab (India)	Dog buried with human; first site excavated after independence
Alamgirpur	UP	Easternmost site
Sutkagendor	Baluchistan	Westernmost site; trade post
Manda	J&K	Northernmost site
Daimabad	Maharashtra	Southernmost site; bronze chariot

- Religion: Mother Goddess, Pashupati seal (proto-Shiva with animals: elephant, tiger, rhinoceros, buffalo; deer below), worship of trees (peepal) and animals (humped bull / unicorn on seals).

★ **PYQ-TREND:** Great Bath at Mohenjo-daro, dockyard at Lothal, ploughed field at Kalibangan, Dholavira's water management and UNESCO status, Rakhigarhi largest Indian site, discoverers of Harappa and Mohenjo-daro, Pashupati seal animals, cotton first grown here.

✔ **TOPPER TRICK:** "Lothal = Loading port (dockyard)". "Kalibangan = Kali (black) bangles + ploughed field".

✖ **COMMON MISTAKE:** There is no evidence of temples, iron, or a deciphered script in the Harappan civilisation. Options with these are wrong.

1.3 Vedic Period

- Aryans; Early (Rig) Vedic ≈ 1500–1000 BCE in Sapta Sindhu region; Later Vedic ≈ 1000–600 BCE in the Ganga plains.
- Four Vedas: Rigveda (oldest; 10 mandalas, 1028 hymns; Gayatri Mantra in 3rd mandala addressed to Savitri; Purusha Sukta in 10th mandala mentions four varnas), Samaveda (melodies – origin of Indian music), Yajurveda (sacrificial formulae; prose and verse), Atharvaveda (magic, charms, medicine).
- Other texts: Brahmanas (rituals), Aranyakas (forest texts), Upanishads (philosophy; 108; 'Satyameva Jayate' from Mundaka Upanishad). Vedangas: 6 (Shiksha, Kalpa, Vyakarana, Nirukta, Chhanda, Jyotisha).
- Rig Vedic gods: Indra (most hymns, 'Purandara'), Agni, Varuna (cosmic order 'rita'), Soma. Later Vedic: Prajapati, Vishnu, Rudra rose in importance.
- Polity: tribal (jana), assemblies Sabha (elders) and Samiti (people), Vidatha. Battle of Ten Kings (Dasharajna) on the Parushni (Ravi) – Sudas of the Bharata clan won.
- Economy: Rig Vedic – pastoral (cattle = wealth, 'gavishti' = war for cows). Later Vedic – agriculture, iron (shyama ayas / krishna ayas), Painted Grey Ware.
- Rivers in Rigveda: Sindhu most mentioned; Saraswati 'best of rivers' (naditama). Vitasta = Jhelum, Asikni = Chenab, Parushni = Ravi, Vipasha = Beas, Shutudri = Sutlej.
- Six schools of philosophy (Shad Darshana): Samkhya (Kapila), Yoga (Patanjali), Nyaya (Gautama), Vaisheshika (Kanada – atom theory), Purva Mimamsa (Jaimini), Uttara

Mimamsa/Vedanta (Badarayana).

Rigveda – oldest, 10 mandalas, 1028 hymns | Gayatri Mantra – Rigveda 3rd mandala | Satyameva Jayate – Mundaka Upanishad | Samaveda – music

★ **PYQ-TREND:** Satyameva Jayate source (Mundaka Upanishad). Samaveda linked to music. Gayatri Mantra in Rigveda. Ancient names of Punjab rivers. Founders of the six schools of philosophy (Kanada – Vaisheshika, Kapila – Samkhya).

1.4 Mahajanapadas and Rise of Magadha

- Anguttara Nikaya (Buddhist) lists 16 Mahajanapadas in the 6th century BCE. Only Assaka (Ashmaka) lay in South India (on the Godavari). Vajji and Malla were republics (ganas).

Mahajanapada	Capital
Magadha	Rajagriha (Girivraja), later Pataliputra
Kosala	Shravasti
Vatsa	Kaushambi
Avanti	Ujjain (north), Mahishmati (south)
Anga	Champa
Kashi	Varanasi
Vajji	Vaishali (Lichchhavis)
Malla	Kushinagar, Pava
Gandhara	Taxila
Kamboja	Rajapura
Kuru	Indraprastha
Panchala	Ahichchhatra, Kampilya
Chedi	Shuktimati
Matsya	Viratnagar
Surasena	Mathura
Assaka	Potali / Paithan

- Magadha dynasties: Haryanka (Bimbisara – contemporary of Buddha, matrimonial alliances; Ajatashatru – killed his father, First Buddhist Council; Udayin – founded Pataliputra), Shishunaga (Shishunaga; Kalashoka – Second Buddhist Council at Vaishali), Nanda (Mahapadma Nanda – 'first empire builder', 'Ekarat'; Dhanananda – contemporary of Alexander).
- Alexander invaded in 326 BCE; Battle of Hydaspes (Jhelum) with Porus.

★ **PYQ-TREND:** Capitals of Mahajanapadas (especially Avanti–Ujjain, Kosala–Shravasti, Vatsa–Kaushambi, Anga–Champa). Only southern Mahajanapada = Assaka. Udayin founded Pataliputra. Battle of Hydaspes.

1.5 Buddhism and Jainism

Buddhism

- Gautama Buddha (Siddhartha), born 563 BCE (traditional) at Lumbini (Nepal), Shakya clan; father Suddhodana, mother Mahamaya; wife Yashodhara, son Rahul. Charioteer Channa, horse Kanthaka.

Event	Place	Symbol
Birth	Lumbini	Lotus and bull
Great Renunciation (Mahabhinishkramana)	–	Horse
Enlightenment (Nirvana)	Bodh Gaya, under peepal tree, river Niranjana	Bodhi tree
First sermon (Dharmachakra-pravartana)	Sarnath (Deer Park)	Wheel
Death (Mahaparinirvana)	Kushinagar, 483 BCE	Stupa

- Teachings: Four Noble Truths, Eightfold Path (Ashtangika Marga), Middle Path. Tripitaka (Pali): Vinaya (monastic rules), Sutta (discourses), Abhidhamma (philosophy).

Buddhist council	Year	Place	King	Chair
First	483 BCE	Rajagriha (Saptaparni cave)	Ajatashatru	Mahakassapa
Second	383 BCE	Vaishali	Kalashoka	Sabakami
Third	250 BCE	Pataliputra	Ashoka	Moggaliputta Tissa
Fourth	72 CE (approx.)	Kundalvana, Kashmir	Kanishka	Vasumitra (Ashvaghosha vice) – split into Hinayana & Mahayana

Jainism

- 24 Tirthankaras. First – Rishabhanatha (Adinath); 23rd – Parshvanatha (four vows); 24th – Vardhamana Mahavira.
- Mahavira: born 540 BCE at Kundagrama (near Vaishali), Jnatrika clan; father Siddhartha, mother Trishala; attained Kaivalya at Jrimbhikagrama on the Rijupalika river; died at Pavapuri (527 BCE traditional, aged 72).
- Mahavira added the fifth vow, Brahmacharya, to Parshvanatha's four (non-violence, truth, non-stealing, non-possession). Three jewels (Triratna): right faith, right knowledge, right conduct.
- Councils: first at Pataliputra (Sthulabhadra, c. 300 BCE), second at Vallabhi (Devardhi Kshamasramana, 5th century CE) – texts compiled in Prakrit (Ardhamagadhi). Sects: Digambara (sky-clad; Bhadrabahu) and Shvetambara (white-clad; Sthulabhadra).
- Chandragupta Maurya embraced Jainism and died at Shravanabelagola (Karnataka). Gomateshwara (Bahubali) statue at Shravanabelagola built by Chamundaraya (Ganga minister).

Buddha: Lumbini birth – Bodh Gaya enlightenment – Sarnath first sermon – Kushinagar death | Mahavira: Kundagrama – Jrimbhikagrama – Pavapuri

★ **PYQ-TREND:** Place of first sermon (Sarnath). Fourth council under Kanishka and the Hinayana–Mahayana split. Third council under Ashoka. Mahavira added Brahmacharya. 23rd and 24th Tirthankaras. Tripitaka components. Shravanabelagola and Gomateshwara.

✔ **TOPPER TRICK:** Buddhist councils by place: "Raja-Vai-Pata-Kash" (Rajagriha, Vaishali, Pataliputra, Kashmir) in order.

1.6 Mauryan Empire (321–185 BCE)

- Chandragupta Maurya founded it with Chanakya (Kautilya/Vishnugupta – author of Arthashastra, on statecraft). Defeated Seleucus Nikator (305 BCE); Megasthenes (Seleucus's ambassador) wrote Indica.
- Bindusara ('Amitraghata'); sent for sweet wine, dried figs and a philosopher from Antiochus.
- Ashoka (273–232 BCE): Kalinga war 261 BCE (in 8th regnal year) changed him; adopted Buddhism under Upagupta/Nigrodha; Dhamma policy; sent son Mahendra and daughter Sanghamitra to Sri Lanka.
- Ashokan edicts: deciphered by James Prinsep (1837) in Brahmi script. Scripts used: Brahmi (most), Kharosthi (north-west; right to left), Greek and Aramaic (Afghanistan). Major Rock Edict XIII – Kalinga war. Names 'Devanampiya Piyadasi'; the name 'Ashoka' appears in Maski and Gujarra edicts. Lion capital of Sarnath = national emblem (four lions; 'Satyameva Jayate' added below).
- Administration: highly centralised; officials – Samaharta (revenue collector), Sannidhata (treasurer), Dhamma Mahamattas (Ashoka). Megasthenes described seven castes and Pataliputra's administration by six committees of five members each.
- Last ruler Brihadratha killed by Pushyamitra Shunga (185 BCE).

Arthashastra – Kautilya | Indica – Megasthenes | Kalinga war 261 BCE | Edicts deciphered by James Prinsep 1837 | Rock Edict XIII – Kalinga

★ **PYQ-TREND:** Author of Arthashastra and Indica. Kalinga war and Rock Edict XIII. Decipherment by James Prinsep. Script of most edicts (Brahmi). Sarnath lion capital as national emblem. Ashoka's children sent to Sri Lanka.

1.7 Post-Mauryan Period

- Shungas (Pushyamitra – performed Ashvamedha; Patanjali wrote Mahabhashya), Kanvas.
- Satavahanas (Deccan): founder Simuka; Gautamiputra Satakarni greatest; capital Pratishthana (Paithan); matronymics; first to grant land to Brahmins/monks; Prakrit; Karle and Nasik caves.
- Indo-Greeks: first to issue gold coins in India and coins with the king's image; Menander (Milinda) – 'Milindapanho' dialogue with Nagasena.
- Shakas: Rudradaman – Junagadh (Girnar) inscription (first long Sanskrit inscription), repaired Sudarshana lake. Vikram Samvat (57 BCE) started by Vikramaditya after defeating Shakas; Shaka Samvat (78 CE) – India's national calendar (Chaitra 1 = March 22).
- Kushanas: Kanishka (began Shaka era 78 CE), capital Purushapura (Peshawar) and Mathura; patron of Mahayana; court – Ashvaghosha (Buddhacharita), Nagarjuna ('Indian Einstein', Madhyamika), Vasumitra, Charaka (physician). Gandhara art (Greco-Roman influence, grey stone) and Mathura art (red sandstone, indigenous).
- Sangam Age (Tamil): three Sangams at Madurai under the Pandyas; Cheras (Vanji/Karur; Senguttuvan – Pattini/Kannagi cult), Cholas (Uraiyur, Puhar; Karikala – Kallanai dam on Kaveri), Pandyas (Madurai). Texts: Tolkappiyam (grammar), Tirukkural (Thiruvalluvar – 'Tamil Veda'), epics Silappatikaram (Ilango Adigal) and Manimekalai (Sattanar).

Shaka Samvat 78 CE = national calendar | Vikram Samvat 57 BCE | Junagadh inscription – Rudradaman (first Sanskrit) | Tirukkural – Thiruvalluvar

★ **PYQ-TREND:** Kanishka and Shaka era; Fourth Buddhist Council. Gandhara vs Mathura art. Rudradaman's Junagadh inscription. Indo-Greeks first gold coins. Silappatikaram and Manimekalai authors. Karikala Chola built Kallanai.

1.8 Gupta Age (c. 319–550 CE) – 'Golden Age'

- Sri Gupta (founder), Chandragupta I (Gupta era 319–320 CE; married Lichchhavi princess Kumaradevi), Samudragupta ('Napoleon of India' – V.A. Smith; Allahabad Pillar inscription/Prayag Prashasti by Harisena; played veena on coins), Chandragupta II 'Vikramaditya' (defeated Shakas; Fa-Hien visited; Navaratnas including Kalidasa; Mehrauli iron pillar), Kumaragupta I (founded Nalanda University), Skandagupta (repelled Hunas).
- Navaratnas: Kalidasa (Abhijnanashakuntalam, Meghaduta, Raghuvamsha, Kumarasambhava), Amarasimha (Amarakosha), Varahamihira (Brihat Samhita, Panchasiddhantika), Dhanvantari (medicine), Vetala Bhatta, Shanku, Ghatakarpara, Kshapanaka, Vararuchi.
- Science: Aryabhata (Aryabhatiya; earth rotates on its axis; value of π; zero/place value), Brahmagupta (Brahmasphutasiddhanta – gravity hint, zero rules), Sushruta (surgery – Sushruta Samhita; 'father of surgery'), Charaka (Kushana period – medicine).
- Art: Ajanta caves (mostly Gupta-Vakataka paintings), Dashavatara temple at Deogarh (early stone temple, Nagara), Sarnath Buddha. Gold coins (dinara) most numerous.
- Harsha (606–647 CE): Pushyabhuti dynasty, capital Thanesar then Kannauj; Hiuen Tsang (Xuanzang) visited ('prince of pilgrims', Si-Yu-Ki); Banabhatta wrote Harshacharita and Kadambari; Harsha wrote Ratnavali, Nagananda, Priyadarshika; held assemblies at Kannauj and Prayag; defeated by Pulakeshin II (Chalukya) on the Narmada.

Prayag Prashasti – Harisena (Samudragupta) | Nalanda – Kumaragupta I | Fa-Hien – Chandragupta II | Hiuen Tsang – Harsha | Harshacharita – Banabhatta

★ **PYQ-TREND:** Samudragupta 'Napoleon of India'. Kalidasa's works. Aryabhata's Aryabhatiya. Nalanda founder. Chinese travellers and their kings (Fa-Hien–Chandragupta II, Hiuen Tsang–Harsha). Harsha defeated by Pulakeshin II.

1.9 Early Medieval South & Regional Kingdoms

Dynasty	Capital	Key facts
Pallavas	Kanchipuram	Mahabalipuram rathas & Shore temple (Narasimhavarman II/Rajasimha); Narasimhavarman I 'Mamalla' defeated Pulakeshin II; Kailasanathar temple, Kanchi
Chalukyas of Badami	Vatapi (Badami)	Pulakeshin II – Aihole inscription by Ravikirti; temples at Aihole ('cradle of Indian temple architecture'), Badami, Pattadakal (UNESCO)
Rashtrakutas	Manyakheta	Dantidurga founder; Krishna I built Kailasa temple (Ellora, cave 16, monolithic); Amoghavarsha wrote Kavirajamarga (Kannada)
Cholas (imperial)	Thanjavur, Gangaikonda Cholapuram	Vijayalaya founder; Rajaraja I built Brihadeshwara temple; Rajendra I conquered up to Ganga & Srivijaya ('Gangaikondachola'); Uttaramerur inscriptions – village self-government (Ur, Sabha)

Dynasty	Capital	Key facts
Palas (Bengal)	–	Gopala (elected) founder; Dharmapala founded Vikramashila university
Pratiharas	Kannauj	Tripartite struggle for Kannauj (Palas, Pratiharas, Rashtrakutas)
Senas	Bengal	Lakshmana Sena; Jayadeva wrote Gita Govinda

Kailasa temple Ellora – Krishna I | Brihadeshwara – Rajaraja I | Aihole inscription – Ravikirti (Pulakeshin II) | Vikramashila – Dharmapala

★ **PYQ-TREND:** Builder of Kailasa temple at Ellora and Brihadeshwara temple. Tripartite struggle participants. Uttaramerur inscription on local self-government. Mahabalipuram under Pallavas. Vikramashila founder.

CHAPTER 2

Medieval Indian History

2.1 Arab and Turkish Invasions

- Muhammad bin Qasim conquered Sindh in 712 CE, defeating Raja Dahir – the first Arab conquest.
- Mahmud of Ghazni raided India 17 times (1000–1027 CE); plundered Somnath temple (1025). Court scholar Al-Biruni wrote Kitab-ul-Hind (Tahqiq-i-Hind); poet Firdausi wrote Shahnama.
- Muhammad Ghori: First Battle of Tarain (1191) – Prithviraj Chauhan won; Second Battle of Tarain (1192) – Ghori won, laying the foundation of Turkish rule. Battle of Chandawar (1194) – defeated Jaichand of Kannauj. Prithviraj's court poet Chand Bardai wrote Prithviraj Raso.

712 – Sindh (Qasim) | 1025 – Somnath raid (Mahmud) | 1191/1192 – Tarain I/II | 1194 – Chandawar

★ **PYQ-TREND:** Battles of Tarain and their winners. Al-Biruni's Kitab-ul-Hind. Muhammad bin Qasim and Sindh. Prithviraj Raso by Chand Bardai.

2.2 Delhi Sultanate (1206–1526)

Dynasty	Period	Founder
Slave (Mamluk/Ilbari)	1206–1290	Qutbuddin Aibak
Khilji	1290–1320	Jalaluddin Khilji
Tughlaq	1320–1414	Ghiyasuddin Tughlaq
Sayyid	1414–1451	Khizr Khan
Lodi (first Afghan dynasty)	1451–1526	Bahlul Lodi

Slave dynasty

- Qutbuddin Aibak: 'Lakh Bakhsh' (giver of lakhs); started Qutub Minar (completed by Iltutmish; repaired by Firoz Tughlaq); built Quwwat-ul-Islam mosque (Delhi) and Adhai-din-ka-Jhonpra (Ajmer); died falling from a horse playing chaugan (polo) at Lahore.
- Iltutmish: real founder/consolidator; capital Delhi; introduced silver tanka and copper jital; Iqta system; 'Turkan-i-Chahalgani' (Group of Forty); obtained investiture from the Abbasid Caliph.
- Razia Sultan (1236–40): first and only woman ruler of the Delhi Sultanate.
- Balban: theory of kingship (divine right, 'Zil-i-Ilahi' – shadow of God), Sijda and Paibos customs, crushed the Forty, Diwan-i-Arz (military department).

Khilji dynasty

- Alauddin Khilji: market control (price regulation), Dagh (branding horses) and Huliya/Chehra (descriptive rolls of soldiers), standing army paid in cash, Alai Darwaza, Siri fort, Hauz-i-Alai; general Malik Kafur conquered the Deccan; Amir Khusrau ('Parrot of India', invented sitar/tabla per tradition, Khazain-ul-Futuh) at his court; Chittor siege 1303 (Padmavati legend by Jayasi's Padmavat, 1540).

Tughlaq dynasty

- Ghiyasuddin Tughlaq built Tughlaqabad; Muhammad bin Tughlaq: transferred capital from Delhi to Daulatabad (Devagiri) in 1327, token currency of copper/brass, Khurasan and Qarachil expeditions, taxation in the Doab; Ibn Battuta (Moroccan traveller) visited and wrote Rehla; called 'mixture of opposites' and 'wise fool'.
- Firoz Shah Tughlaq: imposed jizya on Brahmins, built canals, towns (Firozabad, Hisar, Jaunpur), brought two Ashokan pillars (Topra and Meerut) to Delhi, Diwan-i-Khairat (charity) and Diwan-i-Bandagan (slaves). Timur invaded Delhi in 1398 (during Nasiruddin Mahmud).

Lodi dynasty

- Sikandar Lodi founded Agra (1504), introduced Gaz-i-Sikandari (measuring yard), wrote poetry as 'Gulrukhi'. Ibrahim Lodi lost the First Battle of Panipat (1526) to Babur.

Qutub Minar – Aibak began, Iltutmish completed | Market reforms – Alauddin | Capital to Daulatabad, token currency – Muhammad bin Tughlaq | Agra founded – Sikandar Lodi (1504)

★ **PYQ-TREND:** Muhammad bin Tughlaq's experiments (capital shift, token currency). Alauddin's market control and Dagh system. Iltutmish's tanka and jital. Razia only woman ruler. Ibn Battuta. Amir Khusrau. Founder of Agra. Balban's theory of kingship (Sijda, Paibos).

✖ **COMMON MISTAKE:** Agra was founded by Sikandar Lodi, not by Akbar. Akbar built the Agra Fort; Fatehpur Sikri was his new capital.

2.3 Vijayanagara and Bahmani Kingdoms

- Vijayanagara founded in 1336 by Harihara I and Bukka I (Sangama dynasty) with guidance of Vidyaranya; capital Hampi (on the Tungabhadra; UNESCO site). Dynasties: Sangama, Saluva, Tuluva, Aravidu.
- Krishnadeva Raya (Tuluva, 1509–29): greatest ruler; wrote Amuktamalyada (Telugu); court had Ashtadiggajas (eight poets) including Tenali Ramakrishna and Allasani Peddana ('Andhra Kavita Pitamaha'); built Vitthala and Hazara Rama temples. Portuguese travellers Domingo Paes and Fernao Nuniz described the city; also Abdur Razzaq (Persian) and Nicolo Conti (Italian, Devaraya I).
- Battle of Talikota (Rakshasi-Tangadi), 1565: Deccan Sultanates defeated Rama Raya; Hampi destroyed.
- Bahmani kingdom: founded 1347 by Alauddin Hasan Bahman Shah (Hasan Gangu); capital Gulbarga, later Bidar; minister Mahmud Gawan (madrasa at Bidar). Split into five Deccan Sultanates: Bijapur (Adil Shahi – Gol Gumbaz), Golconda (Qutb Shahi – Charminar built by Muhammad Quli Qutb Shah, founder of Hyderabad), Ahmednagar (Nizam Shahi), Berar (Imad Shahi), Bidar (Barid Shahi).

Vijayanagara 1336 (Harihara & Bukka) | Talikota 1565 | Bahmani 1347 (Hasan Gangu) | Gol Gumbaz – Bijapur | Charminar – Golconda/Hyderabad

★ **PYQ-TREND:** Founders of Vijayanagara and Bahmani. Krishnadeva Raya's Amuktamalyada and Ashtadiggajas. Battle of Talikota 1565. Gol Gumbaz location. Foreign travellers to Vijayanagara.

2.4 Bhakti and Sufi Movements

Saint	Key facts
Alvars (12, Vishnu) & Nayanars (63, Shiva)	Tamil Bhakti saints, 6th–9th century; Andal – woman Alvar

Saint	Key facts
Adi Shankaracharya	Advaita; four mathas – Sringeri (S), Dwarka (W), Puri (E), Badrinath/Joshimath (N); born Kaladi, Kerala
Ramanuja	Vishishtadvaita (qualified monism)
Madhvacharya	Dvaita (dualism)
Nimbarka	Dvaitadvaita
Vallabhacharya	Shuddhadvaita; Pushti Marga; Krishna worship
Ramananda	Brought Bhakti to the north; disciples Kabir, Ravidas, Dhanna, Sena, Pipa
Kabir	Nirguna; dohas in Bijak; weaver; Hindu–Muslim unity
Guru Nanak	Founder of Sikhism; born Talwandi (Nankana Sahib), 1469
Chaitanya Mahaprabhu	Bengal; Krishna kirtan; Gaudiya Vaishnavism
Mirabai	Rajput princess, devotee of Krishna
Surdas	Sursagar; blind poet of Krishna
Tulsidas	Ramcharitmanas (Awadhi) in Akbar's time
Namdev, Jnaneshwar, Eknath, Tukaram, Ramdas	Maharashtra; Jnaneshwari (Gita commentary); Ramdas – guru of Shivaji, Dasbodh
Shankardev	Assam; Ekasarana dharma; Sattriya dance, Namghar
Basavanna	Lingayat (Virashaiva) movement, Karnataka; Anubhava Mantapa

- Sufi orders (silsilas): Chishti (Moinuddin Chishti – Ajmer; Nizamuddin Auliya – Delhi; Bakhtiyar Kaki; Baba Farid – verses in Guru Granth Sahib; Salim Chishti – Fatehpur Sikri), Suhrawardi (Multan – Bahauddin Zakariya), Naqshbandi (orthodox; Sheikh Ahmad Sirhindi), Qadiri, Firdausi.

★ **PYQ-TREND:** Philosophies of Ramanuja (Vishishtadvaita), Madhva (Dvaita), Shankara (Advaita). Shankara's four mathas. Disciples of Ramananda. Founder of Chishti order in India (Moinuddin Chishti, Ajmer). Shankardev and Sattriya. Basavanna and Lingayats.

2.5 Mughal Empire

Ruler	Period	Key facts
Babur	1526–30	First Battle of Panipat 1526 (Ibrahim Lodi; first use of artillery/'Tulughma'); Khanwa 1527 (Rana Sanga); Chanderi 1528 (Medini Rai); Ghaghra 1529 (Afghans); autobiography Tuzuk-i-Baburi (Baburnama) in Turkish

Ruler	Period	Key facts
Humayun	1530–40, 1555–56	Lost Chausa (1539) and Kannauj/Bilgram (1540) to Sher Shah; returned 1555; died falling from library stairs (Din-panah/Sher Mandal); Humayunnama by sister Gulbadan Begum
Sher Shah Suri (interlude)	1540–45	Rupiya (silver coin), Grand Trunk Road (Sadak-e-Azam), sarais, land survey, Dak chowki; tomb at Sasaram; died at Kalinjar
Akbar	1556–1605	Second Panipat 1556 (Hemu; regent Bairam Khan); Haldighati 1576 (Maharana Pratap; Mughal side led by Man Singh); Mansabdari system; Din-i-Ilahi 1582; abolished jizya 1564 and pilgrim tax; Ibadat Khana at Fatehpur Sikri (1575); Buland Darwaza (Gujarat victory); Todar Mal's Zabti/Dahsala revenue system; Navaratnas
Jahangir	1605–27	Chain of justice; Nur Jahan; Captain William Hawkins (1608) and Sir Thomas Roe (1615) came; executed Guru Arjan Dev (1606); Tuzuk-i-Jahangiri
Shah Jahan	1628–58	Golden age of architecture: Taj Mahal (Agra, for Mumtaz Mahal; architect Ustad Ahmad Lahori), Red Fort Delhi, Jama Masjid Delhi, Moti Masjid (Agra), Peacock Throne; capital Shahjahanabad; imprisoned by Aurangzeb
Aurangzeb	1658–1707	'Zinda Pir'; reimposed jizya (1679); executed Guru Tegh Bahadur (1675); Bibi ka Maqbara (Aurangabad) – 'Taj of the Deccan' (built by his son Azam Shah); died at Ahmednagar

- Akbar's Navaratnas: Birbal (Mahesh Das), Abul Fazl (Akbarnama, Ain-i-Akbari), Faizi (brother of Abul Fazl, poet), Tansen (musician, born Gwalior), Todar Mal (finance), Man Singh (general), Abdul Rahim Khan-i-Khanan (poet, translated Baburnama into Persian), Fakir Aziao-Din, Mulla Do Piaza.
- Later Mughals: Bahadur Shah I; Farrukhsiyar's farman of 1717 gave the East India Company duty-free trade in Bengal; Nadir Shah's invasion (1739, under Muhammad Shah 'Rangila') took the Peacock Throne and Kohinoor; Third Battle of Panipat (1761) – Ahmad Shah Abdali defeated the Marathas; Bahadur Shah Zafar – last Mughal, exiled to Rangoon after 1857.

Panipat I 1526 (Babur v Ibrahim Lodi) | Panipat II 1556 (Akbar v Hemu) | Panipat III 1761 (Abdali v Marathas) | Haldighati 1576 | Khanwa 1527

★ **PYQ-TREND:** Three battles of Panipat with year and combatants – asked very often. Haldighati 1576. Akbar's Din-i-Ilahi, Mansabdari, Ibadat Khana. Sher Shah's Rupiya and GT Road. Shah Jahan's buildings. Bibi ka Maqbara. Thomas Roe at Jahangir's court. Baburnama language (Turkish). Humayunnama by Gulbadan Begum.

✔ **TOPPER TRICK:** Panipat years: 1526 → 1556 (+30) → 1761 (+205). "26–56–61".

2.6 Marathas and Sikhs

- Shivaji (1630–80), born at Shivneri fort; coronation at Raigad 1674 ('Chhatrapati'); Ashtapradhan council (Peshwa – PM, Amatya – finance, Senapati, etc.); revenue taxes Chauth (1/4) and Sardeshmukhi (1/10 extra); killed Afzal Khan (1659); Treaty of Purandar (1665) with Jai Singh; escaped from Agra (1666).
- Peshwas: Balaji Vishwanath (first powerful Peshwa), Baji Rao I (expansion north – 'Hindu Pad Padshahi'), Balaji Baji Rao (Nana Saheb; Panipat III loss), Madhav Rao I, Baji Rao II (Treaty of Bassein 1802 – subsidiary alliance; last Peshwa).
- Anglo-Maratha wars: First (1775–82; Treaty of Salbai), Second (1803–05), Third (1817–18; end of Peshwaship).
- Sikh Gurus: Nanak (1st), Angad (Gurmukhi script), Amar Das (Manji system, langar strengthened), Ram Das (founded Amritsar; land from Akbar), Arjan Dev (compiled Adi Granth 1604, built Harmandir Sahib; executed by Jahangir), Hargobind (Akal Takht; militarisation), Har Rai, Har Krishan (youngest), Tegh Bahadur (executed by Aurangzeb 1675), Gobind Singh (10th; Khalsa founded 1699 at Anandpur Sahib; last human guru; Guru Granth Sahib declared eternal Guru).
- Banda Bahadur – first Sikh political leader after Gobind Singh. Maharaja Ranjit Singh – Sikh empire, capital Lahore; Treaty of Amritsar (1809) with the British; Koh-i-noor. Anglo-Sikh wars 1845–46 and 1848–49 → Punjab annexed by Dalhousie (1849).

Shivaji coronation 1674 Raigad | Khalsa 1699 (Guru Gobind Singh) | Adi Granth 1604 (Guru Arjan Dev) | Amritsar founded – Guru Ram Das

★ **PYQ-TREND:** Shivaji's Ashtapradhan and Chauth–Sardeshmukhi. Founder of Khalsa and year. Guru who compiled Adi Granth; Guru who founded Amritsar; Guru who introduced Gurmukhi. Treaty of Bassein. Maharaja Ranjit Singh's capital.

CHAPTER 3

Modern Indian History

3.1 Arrival of Europeans

Power	Arrival / company	First factory & key facts
Portuguese	Vasco da Gama reached Calicut 1498 (Zamorin)	Francisco de Almeida first viceroy (Blue Water policy); Albuquerque captured Goa from Bijapur (1510); last to leave (Goa liberated 1961)
Dutch	VOC 1602	Masulipatnam 1605; Pulicat; Battle of Bedara (1759) lost to British
English	East India Company, charter by Queen Elizabeth I, 31 Dec 1600	Surat factory 1613 (Jahangir's farman); Madras (Fort St. George, 1639–40); Bombay (dowry from Portugal to Charles II, 1661); Calcutta founded by Job Charnock (1690)
Danes	1616	Tranquebar (Tamil Nadu), Serampore (Bengal)
French	French East India Company 1664 (Colbert)	Surat 1668; Pondicherry 1674 (capital); Dupleix; Carnatic wars lost (Battle of Wandiwash 1760)

- Carnatic wars (British vs French): 1st (1746–48, Treaty of Aix-la-Chapelle), 2nd (1749–54, Treaty of Pondicherry), 3rd (1758–63, Battle of Wandiwash 1760 – Eyre Coote defeated Lally; Treaty of Paris 1763).

Vasco da Gama – Calicut 1498 | EIC – 1600 | Surat factory – 1613 | Wandiwash – 1760 | Goa liberated – 1961

★ **PYQ-TREND:** First Europeans to come and last to leave (Portuguese). Vasco da Gama's arrival at Calicut. Battle of Wandiwash. Founder of Calcutta. Bombay as dowry.

3.2 British Conquest

- Battle of Plassey (23 June 1757): Robert Clive defeated Siraj-ud-Daulah (Nawab of Bengal) through Mir Jafar's treachery. Black Hole tragedy (1756) preceded it.
- Battle of Buxar (1764): British defeated the combined army of Mir Qasim (Bengal), Shuja-ud-Daulah (Awadh) and Shah Alam II (Mughal). Treaty of Allahabad (1765): Company got Diwani (revenue rights) of Bengal, Bihar and Orissa. Clive introduced Dual Government in Bengal (1765–72).
- Anglo-Mysore wars: Hyder Ali and Tipu Sultan. 1st (1767–69, Treaty of Madras), 2nd (1780–84, Treaty of Mangalore), 3rd (1790–92, Treaty of Seringapatam – Cornwallis), 4th (1799 – Tipu killed at Srirangapatna; Wellesley). Tipu planted a 'Tree of Liberty' at Srirangapatna and became a member of the Jacobin Club.

- Policies of expansion: Subsidiary Alliance (Lord Wellesley, 1798; first – Nizam of Hyderabad); Doctrine of Lapse (Lord Dalhousie; first – Satara 1848; also Jhansi, Nagpur, Sambalpur, Udaipur); Awadh annexed in 1856 on grounds of misgovernment.

Plassey 1757 (Clive v Siraj) | Buxar 1764 | Diwani 1765 (Allahabad) | Subsidiary alliance – Wellesley | Doctrine of Lapse – Dalhousie

★ **PYQ-TREND:** Plassey vs Buxar participants and years. Treaty of Allahabad. Subsidiary Alliance (Wellesley) and first state to accept it (Hyderabad). Doctrine of Lapse states. Fourth Anglo-Mysore War and Tipu's death.

3.3 Governors-General and Viceroys (Key Acts and Events)

Person	Remember for
Warren Hastings (first GG of Bengal, 1773)	Regulating Act 1773; Asiatic Society of Bengal (1784, founded by William Jones); Pitt's India Act 1784; impeachment in Britain
Lord Cornwallis	Permanent Settlement 1793 (Bengal, zamindars); Civil Services reforms ('father of Civil Services'); Cornwallis Code
Lord Wellesley	Subsidiary Alliance; Fort William College (1800)
Lord Hastings	Ended Pindaris; Third Anglo-Maratha War; Ryotwari introduced (Munro, Madras 1820)
Lord William Bentinck (first GG of India, 1833)	Abolished Sati (1829, with Raja Ram Mohan Roy); suppressed Thugs; English Education Act 1835 (Macaulay's Minute)
Lord Dalhousie	Doctrine of Lapse; first railway (Bombay–Thane, 16 April 1853); telegraph; Post Office Act 1854; Wood's Despatch 1854 ('Magna Carta of English education'); Widow Remarriage Act drafted (passed July 1856 under Canning)
Lord Canning (last GG, first Viceroy, 1858)	Revolt of 1857; Government of India Act 1858; universities of Calcutta, Bombay, Madras (1857); Indian Councils Act 1861
Lord Mayo	First census 1872 (non-synchronous); assassinated in Andamans
Lord Lytton	Vernacular Press Act 1878; Arms Act 1878; Delhi Durbar 1877 (Victoria 'Kaiser-i-Hind')
Lord Ripon	Father of Local Self-Government (1882); repealed Vernacular Press Act; Ilbert Bill controversy; first regular census 1881; Hunter Commission (education)
Lord Dufferin	INC founded 1885
Lord Curzon	Partition of Bengal (1905); Indian Universities Act 1904; Archaeological Survey reorganised; Ancient Monuments Preservation Act 1904; Police Commission
Lord Minto II	Morley–Minto Reforms 1909 (separate electorates for Muslims); Muslim League formed 1906
Lord Hardinge II	Delhi Durbar 1911; capital shifted from Calcutta to Delhi (announced 1911, from 1912); partition of Bengal annulled 1911

Person	Remember for
Lord Chelmsford	Montagu–Chelmsford Reforms (GoI Act 1919 – dyarchy in provinces); Rowlatt Act; Jallianwala Bagh (13 April 1919); Non-Cooperation
Lord Reading	Chauri Chaura (1922); Moplah rebellion; Swaraj Party
Lord Irwin	Simon Commission (1927); Dandi March; Gandhi–Irwin Pact (1931)
Lord Willingdon	Communal Award (1932), Poona Pact; GoI Act 1935
Lord Linlithgow	Cripps Mission (1942); Quit India; August Offer (1940); longest-serving Viceroy
Lord Wavell	Shimla Conference (1945); Cabinet Mission (1946)
Lord Mountbatten	Last Viceroy; Mountbatten Plan (3 June 1947); first Governor-General of free India
C. Rajagopalachari	Only Indian (and last) Governor-General of India

First GG of Bengal – Warren Hastings | First GG of India – William Bentinck | First Viceroy – Canning | Last Viceroy – Mountbatten | Only Indian GG – C. Rajagopalachari

★ **PYQ-TREND:** Partition of Bengal (Curzon, 1905). Ripon and local self-government. Sati abolition (Bentinck, 1829). First railway 1853 (Dalhousie). Vernacular Press Act (Lytton). Capital shift to Delhi 1911. Permanent Settlement (Cornwallis). Wood's Despatch 1854.

3.4 Land Revenue Systems

System	Introduced by	Region
Permanent Settlement (zamindari)	Cornwallis, 1793	Bengal, Bihar, Orissa, Varanasi
Ryotwari	Thomas Munro & Alexander Read, 1820	Madras, Bombay, Assam
Mahalwari	Holt Mackenzie, 1822; revised by William Bentinck (1833)	North-West Provinces, Punjab, Central India

★ **PYQ-TREND:** Permanent Settlement by Cornwallis; Ryotwari by Munro; Mahalwari by Holt Mackenzie.

3.5 Revolt of 1857

- Immediate cause: greased cartridges of the Enfield rifle. Mangal Pandey attacked officers at Barrackpore (29 March 1857). Outbreak at Meerut on 10 May 1857; rebels marched to Delhi and proclaimed Bahadur Shah Zafar emperor.

Centre	Leader	British officer who suppressed
Delhi	Bahadur Shah Zafar, General Bakht Khan	John Nicholson, Hodson
Kanpur	Nana Saheb, Tantia Tope	Colin Campbell
Lucknow	Begum Hazrat Mahal	Colin Campbell
Jhansi	Rani Lakshmibai	Hugh Rose
Bihar (Jagdishpur)	Kunwar Singh	William Taylor, Vincent Eyre
Faizabad	Maulvi Ahmadullah	–

Centre	Leader	British officer who suppressed
Bareilly	Khan Bahadur Khan	–

- Results: Government of India Act 1858 ended Company rule; Queen Victoria's Proclamation (Allahabad, 1 Nov 1858); post of Secretary of State for India; Viceroy title.
- Books: 'The Indian War of Independence 1857' – V.D. Savarkar; '1857: The Great Rebellion' – Ashok Mehta; S.B. Chaudhuri on civil rebellions.

1857 began at Meerut on 10 May | Mangal Pandey – Barrackpore | GoI Act 1858 ended Company rule

★ **PYQ-TREND:** Leaders at each centre of 1857 (especially Lucknow – Begum Hazrat Mahal, Bihar – Kunwar Singh, Jhansi – Lakshmibai, Kanpur – Nana Saheb). Starting place Meerut. Savarkar called it the first war of independence.

3.6 Socio-Religious Reform Movements

Organisation	Founder & year	Key points
Brahmo Samaj	Raja Ram Mohan Roy, 1828 (Calcutta)	Against sati, idol worship; newspapers Sambad Kaumudi, Mirat-ul-Akhbar; 'father of Indian Renaissance'; died in Bristol
Atmiya Sabha	Ram Mohan Roy, 1815	Precursor of Brahmo Samaj
Tattvabodhini Sabha	Debendranath Tagore, 1839	Revived Brahmo Samaj
Brahmo Samaj of India	Keshab Chandra Sen, 1866	Split
Prarthana Samaj	Atmaram Pandurang, 1867 (Bombay)	M.G. Ranade joined
Arya Samaj	Swami Dayanand Saraswati, 1875 (Bombay)	'Go back to the Vedas'; Satyarth Prakash; Shuddhi movement; DAV schools
Ramakrishna Mission	Swami Vivekananda, 1897 (Belur)	Chicago Parliament of Religions 1893
Theosophical Society	Madame Blavatsky & Col. Olcott, 1875 (New York); HQ Adyar (Madras) 1886	Annie Besant later led it
Satyashodhak Samaj	Jyotiba Phule, 1873	Gulamgiri; anti-caste; first girls' school in Pune with Savitribai Phule (1848)
Young Bengal Movement	Henry Vivian Derozio	Hindu College, Calcutta
Aligarh Movement	Sir Syed Ahmed Khan	Muhammadan Anglo-Oriental College 1875 (became AMU 1920)
Sri Narayana Guru Dharma Paripalana (SNDP) Yogam	Narayana Guru, 1903	Kerala; 'one caste, one religion, one God'
Servants of India Society	Gopal Krishna Gokhale, 1905	–
Self-Respect Movement	E.V. Ramasamy 'Periyar', 1925	Tamil Nadu
Widow remarriage	Ishwar Chandra Vidyasagar	Hindu Widows' Remarriage Act 1856

Brahmo Samaj 1828 – Ram Mohan Roy | Arya Samaj 1875 – Dayanand | Ramakrishna Mission 1897 – Vivekananda | Satyashodhak 1873 – Phule

★ **PYQ-TREND:** Founders of Brahmo Samaj, Arya Samaj, Prarthana Samaj, Satyashodhak Samaj. Satyarth Prakash author. Vivekananda at Chicago 1893. Vidyasagar and widow remarriage. Sir Syed Ahmed Khan and Aligarh.

3.7 Indian National Congress and Early Nationalism

- Precursors: Landholders' Society (1838), British Indian Association (1851), East India Association (Dadabhai Naoroji, London 1866), Indian Association (Surendranath Banerjee & Anand Mohan Bose, 1876), Poona Sarvajanik Sabha (1870), Madras Mahajan Sabha (1884).
- INC founded on 28 December 1885 at Gokuldas Tejpal Sanskrit College, Bombay, by A.O. Hume; 72 delegates; first president W.C. Bonnerjee.

Session	Place & year	President / significance
1st	Bombay 1885	W.C. Bonnerjee
2nd	Calcutta 1886	Dadabhai Naoroji
3rd	Madras 1887	Badruddin Tyabji (first Muslim president)
1888	Allahabad	George Yule (first English president)
1906	Calcutta	Dadabhai Naoroji – 'Swaraj' first demanded
1907	Surat	Rash Behari Ghosh – split (Moderates vs Extremists)
1911	Calcutta	B.N. Dhar – 'Jana Gana Mana' first sung (27 Dec 1911)
1916	Lucknow	A.C. Mazumdar – Lucknow Pact (INC–League); Moderates & Extremists reunited
1917	Calcutta	Annie Besant (first woman president)
1920	Nagpur (and special session at Calcutta)	C. Vijayaraghavachariar – Non-Cooperation ratified
1924	Belgaum	Mahatma Gandhi (only session he presided)
1925	Kanpur	Sarojini Naidu (first Indian woman president)
1929	Lahore	Jawaharlal Nehru – Purna Swaraj resolution; 26 Jan 1930 celebrated as Independence Day
1931	Karachi	Vallabhbhai Patel – resolution on Fundamental Rights & economic policy
1938	Haripura	Subhas Chandra Bose – National Planning Committee
1939	Tripuri	Bose re-elected, resigned; Rajendra Prasad took over
1946	Meerut	J.B. Kripalani – president at independence

- Moderates (1885–1905): Dadabhai Naoroji ('Grand Old Man of India', Drain Theory – 'Poverty and Un-British Rule in India'), Pherozeshah Mehta, Gokhale (Gandhi's political guru),

Surendranath Banerjee. Extremists: Lal-Bal-Pal (Lala Lajpat Rai, Bal Gangadhar Tilak – 'Swaraj is my birthright', Kesari & Maratha newspapers, Ganapati and Shivaji festivals; Bipin Chandra Pal) and Aurobindo Ghosh.

- Swadeshi Movement (1905) against partition of Bengal; 16 Oct 1905 observed as day of mourning; 'Amar Sonar Bangla' by Tagore. Muslim League founded in Dhaka, 1906 (Aga Khan, Nawab Salimullah). Home Rule Leagues (1916): Tilak (April, Belgaum) and Annie Besant (Sept, Madras). Ghadar Party (1913, San Francisco; Lala Hardayal). Komagata Maru incident (1914).

INC 1885 Bombay (Hume; WC Bonnerjee) | Surat split 1907 | Lucknow Pact 1916 | Purna Swaraj – Lahore 1929 | Jana Gana Mana first sung – Calcutta 1911

★ **PYQ-TREND:** First president and venue of INC. Sessions: Surat split, Lucknow Pact, Lahore Purna Swaraj, Belgaum (Gandhi), first woman president (Annie Besant), first Indian woman president (Sarojini Naidu), Haripura (Bose). Drain Theory by Naoroji. Founding of Muslim League in Dhaka. Home Rule Leagues.

3.8 Gandhian Era (1915–1947)

- Gandhi returned from South Africa on 9 January 1915 (now Pravasi Bharatiya Divas). Political guru Gokhale. Founded Sabarmati Ashram (1917, earlier Kochrab 1915). In South Africa: Natal Indian Congress (1894), Phoenix Farm, Tolstoy Farm; newspaper Indian Opinion.

Movement	Year	Key facts
Champaran Satyagraha	1917	Bihar; indigo farmers (tinkathia system); first civil disobedience in India; invited by Rajkumar Shukla
Ahmedabad mill strike	1918	First hunger strike by Gandhi
Kheda Satyagraha	1918	Gujarat; revenue remission; Vallabhbhai Patel joined
Rowlatt Satyagraha	1919	'No vakil, no daleel, no appeal' (Black Act)
Jallianwala Bagh	13 April 1919	Amritsar; General Dyer; Tagore returned knighthood; Hunter Committee; Udham Singh later shot O'Dwyer (1940)
Khilafat & Non-Cooperation	1920–22	Ali brothers; withdrawn after Chauri Chaura (Gorakhpur, 4 Feb 1922)
Swaraj Party	1923	C.R. Das & Motilal Nehru
Simon Commission	1927–28	All-white; 'Simon Go Back'; Lala Lajpat Rai died after lathi charge (1928)
Nehru Report	1928	Motilal Nehru – dominion status; Jinnah's 14 points (1929)
Civil Disobedience / Dandi March	12 March – 6 April 1930	Sabarmati to Dandi (≈ 240 miles, 78 followers); salt law broken; Dharasana (Sarojini Naidu); Khan Abdul Ghaffar Khan's Khudai Khidmatgar (Red Shirts)

Movement	Year	Key facts
Round Table Conferences	1930, 1931, 1932	Gandhi attended only the second (1931)
Gandhi–Irwin Pact	5 March 1931	Suspension of CDM
Communal Award & Poona Pact	1932	Ramsay MacDonald; Gandhi–Ambedkar pact on reserved seats for depressed classes
Individual Satyagraha	1940	First satyagrahi Vinoba Bhave, second Nehru
Cripps Mission	1942	Gandhi: 'post-dated cheque on a crashing bank'
Quit India	8 August 1942	Bombay (Gowalia Tank); 'Do or Die'; Aruna Asaf Ali hoisted flag; parallel governments at Ballia, Satara, Tamluk
INA trials	1945	Red Fort; INA formed by Mohan Singh (1942), led by Subhas Bose ('Delhi Chalo', 'Give me blood…'); Azad Hind government at Singapore (1943)
RIN mutiny	1946	Bombay
Cabinet Mission	1946	Pethick-Lawrence, Cripps, A.V. Alexander; Constituent Assembly
Direct Action Day	16 Aug 1946	Muslim League
Mountbatten Plan & Independence Act	3 June / 18 July 1947	Partition; Radcliffe Line

- Revolutionaries: Khudiram Bose & Prafulla Chaki (Muzaffarpur, 1908); Alipore bomb case (Aurobindo); Kakori conspiracy (1925 – Ram Prasad Bismil, Ashfaqullah Khan); HSRA (1928, Chandrashekhar Azad, Bhagat Singh); Bhagat Singh & B.K. Dutt threw bombs in Central Legislative Assembly (8 April 1929); Bhagat Singh, Rajguru, Sukhdev hanged on 23 March 1931 (Lahore conspiracy, Saunders murder); Chittagong armoury raid (1930, Surya Sen).

Champaran 1917 | Jallianwala 13 Apr 1919 | Chauri Chaura 1922 | Dandi 12 Mar–6 Apr 1930 | Quit India 8 Aug 1942 | Independence 15 Aug 1947

★ **PYQ-TREND:** Champaran as Gandhi's first satyagraha in India. Jallianwala Bagh date and General Dyer. Chauri Chaura and withdrawal of NCM. Dandi March dates. 'Do or Die' and Quit India. Second Round Table – only one Gandhi attended. Poona Pact. First individual satyagrahi. Kakori and Chittagong. Bhagat Singh's execution date. Simon Commission and Lala Lajpat Rai.

✔ **TOPPER TRICK:** Gandhi's movements in order with a 10–12 year rhythm: NCM 1920 → CDM 1930 → QIM 1942.

3.9 Important Newspapers, Books and Slogans

Item	Person
Kesari, Maratha	Bal Gangadhar Tilak
Young India, Harijan, Navjivan	Mahatma Gandhi
Al-Hilal	Maulana Abul Kalam Azad

Item	Person
Comrade	Mohammad Ali
Bande Mataram (paper), Jugantar	Aurobindo Ghosh / Barindra Ghosh
Amrita Bazar Patrika	Sisir Kumar Ghosh & Motilal Ghosh
The Hindu	G. Subramania Iyer (1878)
Bengalee	Surendranath Banerjee
Mook Nayak	B.R. Ambedkar
Hind Swaraj (1909), My Experiments with Truth	Gandhi
Discovery of India, Glimpses of World History	Jawaharlal Nehru
Anandamath (Vande Mataram)	Bankim Chandra Chatterjee
Gitanjali	Rabindranath Tagore (Nobel 1913)
India Wins Freedom	Maulana Azad
'Swaraj is my birthright and I shall have it'	Tilak
'Inquilab Zindabad'	Coined by Maulana Hasrat Mohani; popularised by Bhagat Singh
'Jai Hind', 'Delhi Chalo'	Subhas Chandra Bose
'Do or Die'	Gandhi
'Sarfaroshi ki tamanna'	Bismil Azimabadi (popularised by Ram Prasad Bismil)
'Saare Jahan Se Achha'	Muhammad Iqbal
'Jai Jawan Jai Kisan'	Lal Bahadur Shastri
'Who lives if India dies'	Jawaharlal Nehru

★ **PYQ-TREND:** Newspapers and their founders (Kesari, Al-Hilal, Young India). Authors of Hind Swaraj, Discovery of India, Anandamath. Slogans and their authors (Inquilab Zindabad, Jai Hind).

CHAPTER 4

Art & Culture

4.1 Classical Dances (Sangeet Natak Akademi recognises 8)

Dance	State	Key facts
Bharatanatyam	Tamil Nadu	Oldest; based on Natyashastra; earlier 'Sadir/Dasiattam'; revived by Rukmini Devi Arundale (Kalakshetra)
Kathakali	Kerala	Dance-drama; elaborate make-up (green = noble 'pacha'); male performers traditionally; Ramayana–Mahabharata themes; Kerala Kalamandalam founded by Vallathol
Mohiniyattam	Kerala	'Dance of the enchantress'; solo female; lasya style; white-gold costume
Kuchipudi	Andhra Pradesh	Named after village; Siddhendra Yogi; dancing on brass plate (Tarangam)
Odissi	Odisha	Tribhanga posture; Mahari tradition (temple dancers) and Gotipua (boys); Kelucharan Mohapatra
Kathak	Uttar Pradesh (north India)	Only classical dance of the north; Mughal influence; chakkars; Lucknow, Jaipur, Banaras gharanas; Birju Maharaj
Manipuri	Manipur	Ras Lila themes; Vaishnavism; Jagoi; no ghungroo; Pung cholom (drum dance)
Sattriya	Assam	Introduced by Shankardev (15th century) in sattras; recognised in 2000

- The Ministry of Culture also lists Chhau (Odisha/Jharkhand/West Bengal – masked martial dance; UNESCO ICH 2010) among classical forms; Sangeet Natak Akademi lists 8.
- Natyashastra by Bharat Muni is the source text of Indian dance, drama and music. Nine rasas (navarasa).

Bharatanatyam – TN | Kathakali & Mohiniyattam – Kerala | Kuchipudi – AP | Odissi – Odisha | Kathak – UP | Manipuri – Manipur | Sattriya – Assam

★ **PYQ-TREND:** Classical dance–state matching is one of the most frequent culture questions. Sattriya's founder (Shankardev). Natyashastra author. Kathak as the only northern classical form. Famous dancers and their forms (Birju Maharaj – Kathak, Kelucharan Mohapatra – Odissi, Yamini Krishnamurthy – Bharatanatyam/Kuchipudi, Sonal Mansingh – Odissi).

✔ **TOPPER TRICK:** Two from Kerala ("K & M": Kathakali, Mohiniyattam); Kuchipudi is a village in Krishna district (AP).

4.2 Folk Dances by State

State	Folk dances
Punjab	Bhangra, Giddha (women), Jhumar
Haryana	Phag, Saang, Ghoomar
Rajasthan	Ghoomar (Bhil origin), Kalbelia (snake charmers; UNESCO), Chari, Kathputli, Gair, Terah Taali
Gujarat	Garba, Dandiya Raas, Tippani, Padhar
Maharashtra	Lavani, Tamasha, Koli, Lezim, Povada (ballad)
Goa	Dekhni, Fugdi, Dhalo, Mando
Karnataka	Yakshagana (dance-drama), Dollu Kunitha, Huttari
Kerala	Theyyam, Ottan Thullal, Kaikottikali, Padayani
Tamil Nadu	Kummi, Kolattam, Karagattam, Kavadi, Therukoothu
Andhra Pradesh / Telangana	Burrakatha, Kolattam, Perini (Telangana), Lambadi
Odisha	Chhau, Ghumura, Dalkhai, Gotipua
West Bengal	Gambhira, Jatra, Baul (songs), Purulia Chhau
Bihar	Jat-Jatin, Bidesia, Jhijhiya, Kajari
Jharkhand	Karma, Jhumar, Paika, Chhau (Seraikela)
Uttar Pradesh	Raslila, Nautanki, Charkula, Kajri
Uttarakhand	Langvir Nritya, Chholiya, Barada Nati
Himachal Pradesh	Nati (Guinness record), Dangi, Chamba
Jammu & Kashmir	Rouf, Dumhal, Hafiza
Madhya Pradesh	Matki, Tertali, Gaur Maria, Maanch
Chhattisgarh	Panthi, Raut Nacha, Karma, Saila
Assam	Bihu, Bagurumba (Bodo), Ojapali
Arunachal Pradesh	Ponung, Bardo Chham
Meghalaya	Shad Suk Mynsiem, Nongkrem, Wangala (Garo)
Mizoram	Cheraw (bamboo dance)
Nagaland	Zeliang, Chang Lo
Sikkim	Singhi Chham (snow lion), Tamang Selo
Tripura	Hojagiri, Garia
Manipur	Thang-Ta (martial), Lai Haraoba

★ **PYQ-TREND:** Lavani (Maharashtra), Garba (Gujarat), Bihu (Assam), Cheraw (Mizoram), Yakshagana (Karnataka), Theyyam (Kerala), Ghoomar & Kalbelia (Rajasthan), Hojagiri (Tripura), Nati (Himachal), Raut Nacha & Panthi (Chhattisgarh), Jhumar (Punjab/Jharkhand), Rouf (J&K).

4.3 Music

- Two systems: Hindustani (north; Persian influence; ragas linked to time of day) and Carnatic (south; kriti-based; trinity – Tyagaraja, Muthuswami Dikshitar, Syama Sastri; 72 melakarta ragas; Purandaradasa 'father of Carnatic music').

- Hindustani forms: Dhrupad (oldest, Dagar family), Khayal (Amir Khusrau / popularised by Sadarang), Thumri (Lucknow; light classical), Tappa (Punjab camel riders), Dadra, Ghazal, Qawwali (Amir Khusrau).

Instrument	Maestro
Sitar	Ravi Shankar, Vilayat Khan, Nikhil Banerjee, Anoushka Shankar
Sarod	Amjad Ali Khan, Ali Akbar Khan, Buddhadev Das Gupta
Shehnai	Bismillah Khan (Bharat Ratna 2001)
Tabla	Zakir Hussain, Alla Rakha, Kishan Maharaj
Santoor	Shivkumar Sharma, Bhajan Sopori, Rahul Sharma
Flute (bansuri)	Hariprasad Chaurasia, Pannalal Ghosh
Violin	L. Subramaniam, N. Rajam, M.S. Gopalakrishnan, Lalgudi Jayaraman
Veena	S. Balachander, E. Gayathri
Mohan veena	Vishwa Mohan Bhatt (Grammy 1994)
Rudra veena	Asad Ali Khan
Sarangi	Ram Narayan, Sultan Khan
Mridangam	Palghat Mani Iyer, Umayalpuram Sivaraman
Ghatam	Vikku Vinayakram

- Vocalists: Bhimsen Joshi (Kirana gharana; Bharat Ratna 2009), M.S. Subbulakshmi (first musician to get Bharat Ratna, 1998), Lata Mangeshkar (Bharat Ratna 2001), Kishori Amonkar, Gangubai Hangal, Pandit Jasraj (Mewati), Kumar Gandharva, Girija Devi (thumri, Banaras).
- Instrument classes (Natyashastra): Tata (string – sitar, veena), Sushira (wind – flute, shehnai), Avanaddha (membrane – tabla, mridangam), Ghana (solid – ghatam, cymbals, jaltarang).

★ **PYQ-TREND:** Musician–instrument matching (very frequent): Bismillah Khan – shehnai, Shivkumar Sharma – santoor, Hariprasad Chaurasia – flute, Amjad Ali Khan – sarod, Zakir Hussain – tabla, Ravi Shankar – sitar. Carnatic trinity. First musician with Bharat Ratna (M.S. Subbulakshmi).

4.4 Festivals and Fairs

Festival	State / community	Note
Bihu (Bohag/Rongali, Kati, Magh/Bhogali)	Assam	Agricultural; three Bihus
Onam	Kerala	Harvest; King Mahabali; Vallam Kali (snake boat race)
Pongal	Tamil Nadu	Harvest (January); Jallikattu (bull taming)
Makar Sankranti / Lohri / Uttarayan	Pan-India / Punjab / Gujarat	Sun enters Capricorn; kites in Gujarat
Baisakhi	Punjab	Harvest; Khalsa foundation day
Hornbill Festival	Nagaland	1–10 December, Kisama
Losar	Ladakh, Sikkim, Arunachal	Tibetan New Year

Festival	State / community	Note
Hemis Festival	Ladakh	Guru Padmasambhava
Chhath	Bihar	Sun worship
Durga Puja	West Bengal	UNESCO ICH (2021)
Ganesh Chaturthi	Maharashtra	Public celebration popularised by Tilak (1893)
Ugadi / Gudi Padwa	AP–Karnataka / Maharashtra	New Year
Rath Yatra	Puri, Odisha	Jagannath
Nuakhai	Odisha	New rice
Hareli, Bastar Dussehra	Chhattisgarh	Bastar Dussehra lasts ~75 days
Wangala	Meghalaya (Garo)	Harvest; 100 drums
Sangai Festival	Manipur	Named after state animal (brow-antlered deer)
Chapchar Kut	Mizoram	Spring
Ziro Music Festival, Solung, Losar	Arunachal	–
Thrissur Pooram	Kerala	Elephants, percussion
Pushkar Fair	Rajasthan	Camel fair (Kartik)
Sonepur Mela	Bihar	Cattle (elephant) fair
Kumbh Mela	Prayagraj, Haridwar, Ujjain, Nashik	UNESCO ICH 2017; every 12 years at each site
Hampi Utsav, Mysuru Dasara	Karnataka	–
Kharchi Puja	Tripura	14 gods
Sekrenyi	Nagaland (Angami)	–

- India's UNESCO Intangible Cultural Heritage elements include: Vedic chanting, Ramlila, Kutiyattam (Sanskrit theatre, Kerala), Ramman (Garhwal), Mudiyettu (Kerala), Kalbelia, Chhau, Buddhist chanting of Ladakh, Sankirtana (Manipur), Thatheras of Jandiala Guru (brass craft, Punjab), Yoga (2016), Nowruz (2016), Kumbh Mela (2017), Durga Puja in Kolkata (2021), Garba of Gujarat (2023).

★ **PYQ-TREND:** Bihu–Assam, Onam–Kerala, Pongal–TN, Hornbill–Nagaland, Hemis–Ladakh, Chapchar Kut–Mizoram, Wangala–Meghalaya, Sangai–Manipur, Nuakhai–Odisha. UNESCO ICH (Kumbh Mela, Yoga, Durga Puja, Garba).

4.5 Architecture and Monuments

- Temple styles: Nagara (north – curvilinear shikhara; e.g., Khajuraho, Sun Temple Konark), Dravida (south – pyramidal vimana, gopuram gateways; e.g., Brihadeshwara, Meenakshi), Vesara (mixed – Chalukya & Hoysala; e.g., Hoysaleswara at Halebidu, Chennakesava at Belur). Odisha subtype: Deul (Lingaraja, Jagannath, Konark).
- Konark Sun Temple: Narasimhadeva I (Eastern Ganga, 13th century); chariot with 24 wheels and 7 horses; 'Black Pagoda'. Jagannath temple Puri: 'White Pagoda' (Anantavarman Chodaganga).
- Khajuraho (MP): Chandelas; Kandariya Mahadeva temple. Dilwara temples (Mount Abu): Jain, marble (Vimal Shah and Tejpal). Sanchi Stupa: Ashoka (MP). Ajanta (paintings; Buddhist; Maharashtra) and Ellora (Buddhist, Hindu, Jain; 34 caves). Elephanta (Trimurti Shiva; Mumbai). Meenakshi temple, Madurai (Nayakas). Virupaksha, Hampi. Somnath (Gujarat; first

of 12 jyotirlingas).

- Indo-Islamic: Qutub Minar (red sandstone, 5 storeys), Alai Darwaza (first true arch & dome), Tughlaqabad, Humayun's Tomb (first garden tomb; Hamida Banu Begum; architect Mirak Mirza Ghiyas; precursor of Taj), Fatehpur Sikri (Buland Darwaza – tallest gateway; Panch Mahal; Jodha Bai's palace), Taj Mahal, Red Fort, Jama Masjid, Gol Gumbaz (Bijapur – whispering gallery; Muhammad Adil Shah), Charminar (Hyderabad, 1591), Char Minar's founder Muhammad Quli Qutb Shah.
- Colonial: Victoria Memorial (Kolkata; William Emerson), Gateway of India (Mumbai, 1924; George Wittet; for George V's 1911 visit), India Gate (Delhi; Edwin Lutyens; All-India War Memorial), Rashtrapati Bhavan (Lutyens), Parliament House (old – Lutyens & Herbert Baker; circular), Howrah Bridge (cantilever, 1943), Writers' Building (Kolkata).
- Others: Hawa Mahal (Jaipur; Sawai Pratap Singh; architect Lal Chand Ustad), Jantar Mantar (Sawai Jai Singh II – five observatories: Delhi, Jaipur, Ujjain, Varanasi, Mathura), Golden Temple (Amritsar), Sun Temple Modhera (Gujarat; Solanki), Rani ki Vav (Patan, Gujarat; stepwell; UNESCO; featured on ₹100 note), Kirti Stambh & Vijay Stambh (Chittorgarh; Vijay Stambh built by Rana Kumbha after victory over Mahmud Khilji), Statue of Unity (Kevadia, Gujarat – Sardar Patel, 182 m, tallest statue).

UNESCO World Heritage Sites (selected; India has 43 as of 2024)	State
Ajanta, Ellora, Elephanta Caves, CST (Victoria Terminus), Victorian Gothic & Art Deco Ensembles	Maharashtra
Taj Mahal, Agra Fort, Fatehpur Sikri	UP
Qutub Minar, Red Fort, Humayun's Tomb	Delhi
Sun Temple Konark	Odisha
Mahabalipuram, Great Living Chola Temples (Thanjavur, Gangaikondacholapuram, Darasuram)	Tamil Nadu
Hampi, Pattadakal, Hoysala temples (Belur, Halebidu, Somanathapura – 2023)	Karnataka
Khajuraho, Sanchi, Bhimbetka	MP
Champaner-Pavagadh, Rani ki Vav, Ahmedabad (first Indian World Heritage City, 2017), Dholavira (2021)	Gujarat
Jaipur city (2019), Jantar Mantar Jaipur, Hill Forts of Rajasthan	Rajasthan
Kaziranga, Manas (natural)	Assam
Sundarbans (natural), Santiniketan (2023)	West Bengal
Mahabodhi Temple Bodh Gaya, Nalanda Mahavihara (2016)	Bihar
Ramappa temple (Kakatiya Rudreshwara, 2021)	Telangana
Western Ghats, Great Himalayan NP (HP), Nanda Devi & Valley of Flowers (Uttarakhand), Keoladeo (Rajasthan) – natural	Various
Khangchendzonga NP (mixed – Sikkim)	Sikkim
Mountain Railways of India (Darjeeling, Nilgiri, Kalka–Shimla)	Various
Moidams of Charaideo (Ahom burial mounds, 2024)	Assam

- Natural sites: Kaziranga, Manas, Keoladeo, Sundarbans, Nanda Devi & Valley of Flowers, Western Ghats, Great Himalayan NP. Mixed: Khangchendzonga.

Konark – Narasimhadeva I | Kailasa (Ellora) – Krishna I | Humayun's Tomb – first garden tomb | Gateway of India – 1924 | Rani ki Vav – ₹100 note | Statue of Unity – 182 m

★ **PYQ-TREND:** Monument–builder–city matching. Nagara/Dravida/Vesara examples. Konark 'Black Pagoda'. Rani ki Vav on ₹100 note. Gol Gumbaz whispering gallery. Ahmedabad first Indian World Heritage City. Recently added UNESCO sites (Dholavira, Ramappa, Santiniketan, Hoysala temples, Moidams). Jantar Mantar builder.

✖ **COMMON MISTAKE:** The UNESCO list grows every year. Check the newest additions in current affairs before the exam.

4.6 Paintings, Puppetry and Crafts

Painting	Region
Madhubani (Mithila)	Bihar
Warli	Maharashtra (tribal; white on mud base)
Pattachitra	Odisha, West Bengal
Kalamkari	Andhra Pradesh (Srikalahasti, Machilipatnam)
Tanjore	Tamil Nadu (gold foil)
Kalighat	West Bengal
Phad	Rajasthan (scrolls of Pabuji/Devnarayan)
Pithora	Gujarat, MP (Rathwa/Bhil)
Gond	Madhya Pradesh
Saura	Odisha
Cheriyal scrolls	Telangana
Mysore painting	Karnataka
Kangra, Basohli, Chamba (Pahari)	Himachal, J&K
Thangka	Sikkim, Ladakh, Arunachal (Buddhist scrolls)
Rajput (Mewar, Bundi, Kishangarh – 'Bani Thani')	Rajasthan

- Puppetry: string – Kathputli (Rajasthan), Kundhei (Odisha), Gombeyata (Karnataka), Bommalattam (TN); shadow – Togalu Gombeyata (Karnataka), Tholpavakoothu (Kerala), Ravanchhaya (Odisha), Tholu Bommalata (AP); rod – Putul Nach (Bengal), Yampuri (Bihar); glove – Pavakoothu (Kerala).
- Theatre: Yakshagana (Karnataka), Jatra (Bengal), Nautanki (UP), Tamasha (Maharashtra), Bhavai (Gujarat), Swang (Haryana), Koodiyattam/Kutiyattam (Kerala), Ankia Naat (Assam – Shankardev), Bhand Pather (Kashmir), Therukoothu (TN).
- GI-tagged items (examples): Darjeeling tea (first GI, 2004–05), Banarasi saree, Kanchipuram silk, Pochampally Ikat (Telangana), Chanderi (MP), Mysore silk, Pashmina (Kashmir), Muga silk (Assam), Basmati rice, Alphonso mango (Ratnagiri), Nagpur orange, Tirupati laddu, Rasgulla (Odisha & West Bengal), Madhubani painting, Kolhapuri chappal.

★ **PYQ-TREND:** Madhubani–Bihar, Warli–Maharashtra, Kalamkari–AP, Pattachitra–Odisha, Phad–Rajasthan, Gond–MP, Tanjore–TN. Kathputli–Rajasthan. First GI tag – Darjeeling tea.

4.7 Literature, Languages and Calendars

- 22 languages in the Eighth Schedule (originally 14). Added by: 21st Amendment 1967 – Sindhi; 71st Amendment 1992 – Konkani, Manipuri, Nepali; 92nd Amendment 2003 – Bodo, Dogri, Maithili, Santhali.
- Classical languages: Tamil (first, 2004), Sanskrit (2005), Telugu & Kannada (2008), Malayalam (2013), Odia (2014); in October 2024 Marathi, Pali, Prakrit, Assamese and Bengali were added → 11 in total.
- Sahitya Akademi (1954) – literature, 24 languages; Sangeet Natak Akademi (1952) – performing arts; Lalit Kala Akademi (1954) – fine arts; National School of Drama (1959).
- Scripts: Brahmi (parent of most Indian scripts), Kharosthi (right to left), Gurmukhi (Guru Angad), Ol Chiki (Santhali – Raghunath Murmu), Meetei Mayek (Manipuri).

22 scheduled languages | Classical: Tamil 2004 first; 11 in total after 2024 additions | Sahitya Akademi 1954

★ **PYQ-TREND:** Which amendment added which language (71st – Konkani, Manipuri, Nepali; 92nd – Bodo, Dogri, Maithili, Santhali). First classical language (Tamil). Ol Chiki script for Santhali. Number of scheduled languages (22).

CHAPTER 5

Indian Polity I – Constitution and Rights

5.1 Making of the Constitution

- Constituent Assembly formed under the Cabinet Mission Plan (1946); members indirectly elected by provincial assemblies; total 389 (after partition 299). Idea first proposed by M.N. Roy (1934); INC formally demanded it in 1935.
- First meeting: 9 December 1946 (Dr. Sachchidananda Sinha – temporary president). Dr. Rajendra Prasad elected permanent president (11 Dec 1946); vice-presidents H.C. Mookerjee and V.T. Krishnamachari. Constitutional adviser B.N. Rau.
- Objectives Resolution moved by Nehru on 13 December 1946 (adopted 22 January 1947) – became the Preamble.
- Drafting Committee (29 August 1947): chairman Dr. B.R. Ambedkar ('father/chief architect of the Constitution'); members N. Gopalaswami Ayyangar, Alladi Krishnaswamy Ayyar, K.M. Munshi, Syed Mohammad Saadullah, N. Madhava Rau (replaced B.L. Mitter), T.T. Krishnamachari (replaced D.P. Khaitan).
- Other committees: Union Powers, Union Constitution and States Committees – Nehru; Provincial Constitution Committee and Advisory Committee on Fundamental Rights & Minorities – Sardar Patel; Rules of Procedure and Steering Committees – Rajendra Prasad; Ad hoc Committee on the National Flag – Rajendra Prasad.
- Constitution adopted on 26 November 1949 (Constitution/Samvidhan Divas since 2015); came into force on 26 January 1950 (Republic Day; chosen because Purna Swaraj Day was 26 Jan 1930). Took 2 years, 11 months, 18 days; 11 sessions.
- Original Constitution: 395 Articles, 22 Parts, 8 Schedules. Hand-written by Prem Behari Narain Raizada (calligraphy); illustrated by Nandalal Bose and Beohar Rammanohar Sinha.
- National Anthem adopted 24 January 1950; National Song same day; Rajendra Prasad elected first President on 24 Jan 1950. National Flag adopted 22 July 1947 (designed on Pingali Venkayya's design). Emblem adopted 26 January 1950.

First meeting 9 Dec 1946 | Objectives Resolution 13 Dec 1946 | Adopted 26 Nov 1949 | In force 26 Jan 1950 | Original: 395 Articles, 22 Parts, 8 Schedules

★ **PYQ-TREND:** Chairman of Drafting Committee (Ambedkar), President of Constituent Assembly (Rajendra Prasad), temporary president (Sachchidananda Sinha). Adoption and enforcement dates. Objectives Resolution by Nehru. Constitutional adviser B.N. Rau. Flag adoption date.

5.2 Borrowed Features (Sources)

Source	Features borrowed
Government of India Act 1935	Federal scheme, office of Governor, judiciary, Public Service Commissions, emergency provisions, administrative details (largest source)

Source	Features borrowed
Britain	Parliamentary government, rule of law, legislative procedure, single citizenship, cabinet system, prerogative writs, bicameralism, Speaker
USA	Fundamental Rights, independence of judiciary, judicial review, impeachment of President, removal of SC/HC judges, Vice-President post, Preamble
Ireland	Directive Principles, nomination of Rajya Sabha members, method of election of President
Canada	Federation with a strong Centre, residuary powers with Centre, appointment of Governors by Centre, advisory jurisdiction of Supreme Court
Australia	Concurrent List, freedom of trade and commerce, joint sitting of Parliament
USSR (erstwhile)	Fundamental Duties, ideals of justice (social, economic, political) in Preamble
France	Republic, Liberty, Equality, Fraternity in the Preamble
Germany (Weimar)	Suspension of Fundamental Rights during Emergency
South Africa	Procedure for amendment, election of Rajya Sabha members
Japan	Procedure established by law

✔ **TOPPER TRICK:** "DIR" – Directive principles from IReland; "FD from USSR"; "Emergency suspension of FR from Germany (Weimar)"; "Amendment from South Africa".

★ **PYQ-TREND:** Source of DPSP (Ireland), Fundamental Duties (USSR), Concurrent List (Australia), Fundamental Rights and judicial review (USA), amendment procedure (South Africa), residuary powers (Canada), Emergency suspension (Germany). GoI Act 1935 as the main source.

5.3 Preamble

- 'We, the people of India, having solemnly resolved to constitute India into a Sovereign Socialist Secular Democratic Republic and to secure to all its citizens: Justice – social, economic and political; Liberty of thought, expression, belief, faith and worship; Equality of status and of opportunity; and to promote among them all Fraternity assuring the dignity of the individual and the unity and integrity of the Nation…'
- 42nd Amendment (1976) added 'Socialist', 'Secular' and 'Integrity' – the only amendment to the Preamble so far.
- Berubari Union case (1960): Preamble not part of the Constitution. Kesavananda Bharati case (1973): Preamble is part of the Constitution and can be amended (without altering the basic structure). LIC case (1995): part of the Constitution.
- Preamble is non-justiciable. Called 'identity card of the Constitution' (N.A. Palkhivala), 'political horoscope' (K.M. Munshi), 'soul of the Constitution' (Thakurdas Bhargava).

42nd Amendment 1976 – Socialist, Secular, Integrity | Kesavananda 1973 – Preamble is part, basic structure

★ **PYQ-TREND:** Words added by the 42nd Amendment. Preamble part of the Constitution per Kesavananda Bharati. Source of liberty-equality-fraternity (France). Order of words in Preamble (Sovereign Socialist Secular Democratic Republic).

5.4 Parts and Schedules (Current)

Part	Subject	Articles
I	Union and its territory	1–4
II	Citizenship	5–11
III	Fundamental Rights	12–35
IV	Directive Principles	36–51
IV-A	Fundamental Duties	51A
V	The Union (President, Parliament, SC, CAG)	52–151
VI	The States	152–237
VIII	Union Territories	239–242
IX	Panchayats	243–243O
IX-A	Municipalities	243P–243ZG
IX-B	Co-operative societies	243ZH–243ZT
X	Scheduled and Tribal Areas	244–244A
XI	Centre–State relations	245–263
XII	Finance, property, contracts	264–300A
XIV	Services (UPSC, PSCs)	308–323
XIV-A	Tribunals	323A–323B
XV	Elections	324–329A
XVII	Official language	343–351
XVIII	Emergency provisions	352–360
XX	Amendment	368
XXI	Temporary, transitional and special provisions	369–392

Schedule	Content
1	States and Union Territories
2	Emoluments of President, Governors, Speaker, judges, CAG
3	Forms of oaths and affirmations
4	Allocation of Rajya Sabha seats
5	Administration of Scheduled Areas and Tribes
6	Tribal areas of Assam, Meghalaya, Tripura, Mizoram
7	Union, State and Concurrent Lists
8	Languages (22)
9	Laws protected from judicial review (added by 1st Amendment 1951; land reforms)
10	Anti-defection (added by 52nd Amendment 1985)
11	Panchayats – 29 subjects (73rd Amendment 1992)

Schedule	Content
12	Municipalities – 18 subjects (74th Amendment 1992)

- Seventh Schedule: Union List (originally 97, now 100 entries – defence, foreign affairs, railways, banking, atomic energy), State List (originally 66, now 61 – police, public order, health, agriculture, local government), Concurrent List (originally 47, now 52 – education, forests, marriage, criminal law, electricity). 42nd Amendment moved education, forests, weights & measures, protection of wild animals and birds, and administration of justice to the Concurrent List.

Schedules: 7th – lists | 8th – languages | 9th – 1st Amdt | 10th – anti-defection (52nd) | 11th – Panchayats (29) | 12th – Municipalities (18)

★ **PYQ-TREND:** Schedule contents (especially 7th, 8th, 9th, 10th, 11th). Subjects moved to Concurrent List by the 42nd Amendment. Number of subjects in the 11th and 12th schedules. Part III and IV article ranges.

5.5 Union, Territory and Citizenship

- Article 1: 'India, that is Bharat, shall be a Union of States'. Article 3: Parliament can form new states and alter boundaries (simple majority; President's recommendation; state legislature's view not binding).
- States reorganisation: Andhra State (1953) first on linguistic basis after Potti Sriramulu's death; States Reorganisation Commission – Fazl Ali (1953), Act 1956 (14 states, 6 UTs). Newer states: Gujarat & Maharashtra (1960), Nagaland (1963), Haryana (1966), HP (1971), Manipur, Tripura, Meghalaya (1972), Sikkim (1975, 36th Amendment), Mizoram, Arunachal, Goa (1987), Chhattisgarh, Uttarakhand, Jharkhand (2000), Telangana (2 June 2014 – 29th state). J&K reorganised into two UTs (J&K and Ladakh) on 31 October 2019. Dadra & Nagar Haveli and Daman & Diu merged in 2020. Now 28 states and 8 UTs.
- Citizenship: Articles 5–11; Citizenship Act 1955 – acquisition by birth, descent, registration, naturalisation, incorporation of territory; loss by renunciation, termination, deprivation. Single citizenship (from Britain). Overseas Citizenship of India (OCI). Citizenship (Amendment) Act 2019.

28 states + 8 UTs | Telangana – 29th state, 2014 | J&K and Ladakh UTs – 31 Oct 2019 | Sikkim – 36th Amendment 1975

★ **PYQ-TREND:** Article 1 wording; Article 3 for new states. First linguistic state (Andhra). Fazl Ali Commission. Newest state (Telangana). Number of states and UTs. Sikkim's merger amendment.

5.6 Fundamental Rights (Part III)

Right	Articles
Right to Equality	14–18
Right to Freedom	19–22
Right against Exploitation	23–24
Right to Freedom of Religion	25–28
Cultural and Educational Rights	29–30
Right to Constitutional Remedies	32

- Article 14: equality before law (British) and equal protection of laws (American). 15: no discrimination on religion, race, caste, sex, place of birth. 16: equality of opportunity in public employment. 17: abolition of untouchability (Protection of Civil Rights Act 1955). 18: abolition of titles (except military and academic).
- Article 19: six freedoms – speech & expression, assemble peaceably without arms, form associations/unions/co-operatives, move freely, reside and settle, profession/occupation/trade. (Right to property in 19(1)(f) was removed by the 44th Amendment 1978.)
- Article 20: protection in respect of conviction (no ex-post-facto law, no double jeopardy, no self-incrimination). 21: protection of life and personal liberty (widest; includes privacy – Puttaswamy 2017, clean environment, livelihood, speedy trial, dignity). 21A: right to education for 6–14 years (86th Amendment 2002). 22: protection against arbitrary arrest (produce before magistrate within 24 hours); preventive detention.
- Article 23: prohibition of traffic in human beings and forced labour (begar). 24: no child below 14 in factories, mines or hazardous work.
- Articles 25–28: freedom of conscience and religion; manage religious affairs; no tax for promotion of a religion; no religious instruction in State-funded institutions.
- Articles 29–30: protection of interests of minorities; right of minorities to establish and administer educational institutions.
- Article 32: Right to constitutional remedies – 'heart and soul of the Constitution' (Ambedkar). Writs (SC under Art 32, HC under Art 226 – HC's power wider): Habeas Corpus ('to have the body' – against illegal detention), Mandamus ('we command' – to a public official to perform duty; not against President/Governor), Prohibition (higher court to lower court to stop proceeding), Certiorari ('to be certified' – quash order of lower court/tribunal), Quo Warranto ('by what authority' – against unlawful holding of public office).
- Right to property: now a legal right under Article 300A (44th Amendment 1978).
- FRs suspended during national emergency: Article 19 suspended automatically in war/external aggression (Art 358); others via Presidential order (Art 359), except Articles 20 and 21 (44th Amendment).

Art 14 equality | 17 untouchability | 19 six freedoms | 21 life & liberty | 21A education (86th, 2002) | 32 remedies ('heart & soul') | Property → 300A (44th, 1978)

★ **PYQ-TREND:** Article 32 'heart and soul'. Writs and their meanings (Habeas Corpus, Mandamus, Quo Warranto). Article 17 untouchability, Article 21A education, Article 24 child labour. Right to property removal (44th Amendment) and new Art 300A. Articles 20 and 21 cannot be suspended in emergency. Privacy as FR (2017).

✔ **TOPPER TRICK:** Writs memory: "Habeas = Body; Mandamus = Must do duty; Prohibition = Pre-stop; Certiorari = Correct after decision (quash); Quo Warranto = Which warrant/authority".

5.7 Directive Principles of State Policy (Part IV, Articles 36–51)

- Non-justiciable but 'fundamental in the governance of the country' (Art 37). Idea from the Irish Constitution. 'Novel feature' (Ambedkar).

Article	Directive
38	Social order with justice; minimise inequalities
39	Adequate livelihood, equal pay for equal work, prevent concentration of wealth, protect children

Article	Directive
39A	Equal justice and free legal aid (42nd Amendment)
40	Organise village panchayats (Gandhian)
41	Right to work, education, public assistance
42	Just and humane conditions of work, maternity relief
43	Living wage; promote cottage industries (Gandhian)
43A	Workers' participation in management (42nd)
43B	Co-operative societies (97th, 2011)
44	Uniform Civil Code
45	Early childhood care and education below 6 (changed by 86th Amendment)
46	Educational & economic interests of SCs, STs, weaker sections (Gandhian)
47	Nutrition, public health; prohibition of intoxicating drinks (Gandhian)
48	Organise agriculture & animal husbandry; prohibit cow slaughter (Gandhian)
48A	Protect environment, forests, wildlife (42nd)
49	Protect monuments of national importance
50	Separate judiciary from executive
51	Promote international peace and security

- Classification: socialist, Gandhian, liberal-intellectual. Minerva Mills case (1980): Constitution is founded on balance between FRs and DPSPs.

Art 40 – village panchayats | 44 – Uniform Civil Code | 45 – early childhood care | 48A – environment | 50 – separation of judiciary | 51 – international peace

★ **PYQ-TREND:** Article 44 (UCC), Article 40 (panchayats), Article 48A (environment), Article 50 (separation of judiciary from executive), Article 39A (free legal aid). DPSP source (Ireland). Non-justiciable nature.

5.8 Fundamental Duties (Part IV-A, Article 51A)

- Added by the 42nd Amendment (1976) on the recommendation of the Swaran Singh Committee; originally 10; 11th duty added by the 86th Amendment (2002) – parents/guardians to provide education to children aged 6–14. Idea from the USSR. Non-justiciable.
- Duties include: abide by the Constitution, respect the National Flag and Anthem; cherish ideals of the freedom struggle; uphold sovereignty, unity and integrity; defend the country; promote harmony and renounce practices derogatory to women; preserve composite culture; protect environment including forests, lakes, rivers, wildlife; develop scientific temper, humanism, spirit of inquiry; safeguard public property and abjure violence; strive towards excellence.

Fundamental Duties – 42nd Amdt 1976, Swaran Singh Committee, 10 + 1 (86th, 2002) = 11

★ **PYQ-TREND:** Committee recommending Fundamental Duties (Swaran Singh). Number of duties (11). 11th duty added by 86th Amendment. Article 51A.

CHAPTER 6

Indian Polity II – Institutions

6.1 President (Articles 52–62)

- Art 52: there shall be a President; Art 53: executive power of the Union vested in him; Supreme Commander of the defence forces.
- Election (Art 54–55): indirectly, by an electoral college of elected members of both Houses of Parliament and elected members of state Legislative Assemblies (including Delhi and Puducherry, 70th Amendment). Nominated members and Legislative Councils do not vote. Proportional representation by single transferable vote; secret ballot. Disputes decided by the Supreme Court.
- Qualifications: citizen, 35 years, qualified for Lok Sabha, no office of profit. Term 5 years; eligible for re-election. Oath administered by the Chief Justice of India. Resignation addressed to the Vice-President.
- Impeachment (Art 61): only ground 'violation of the Constitution'; in either House; 14 days' notice signed by 1/4 members; passed by 2/3 of total membership of each House. Nominated members participate; state legislatures do not.
- Powers: appoints PM, ministers, CJI & judges, Governors, CAG, CEC, Attorney General, UPSC chairman; summons and prorogues Parliament, dissolves Lok Sabha; addresses first session each year (Art 87); nominates 12 members to Rajya Sabha (Art 80) – (Anglo-Indian nomination to Lok Sabha ended by the 104th Amendment 2019).
- Ordinance (Art 123): when Parliament is not in session; must be approved within 6 weeks of reassembly.
- Pardoning power (Art 72): pardon, commutation, remission, respite, reprieve; including death sentences and court-martial.
- Veto: absolute, suspensive (return once, except money bills), pocket (no time limit – used by Giani Zail Singh on the Indian Post Office (Amendment) Bill 1986). Money bills: cannot return; Constitutional amendment bills: must assent (24th Amendment).
- Emergency powers: Art 352 (national – war, external aggression, armed rebellion; 'internal disturbance' replaced by 'armed rebellion' by 44th Amendment), Art 356 (President's Rule – failure of constitutional machinery; first in Punjab 1951), Art 360 (financial emergency – never imposed).
- National emergencies proclaimed thrice: 1962 (China war), 1971 (Pakistan war), 1975 (internal disturbance, by Fakhruddin Ali Ahmed on Indira Gandhi's advice).

President	Note
Dr. Rajendra Prasad	First; only one elected twice (1950–62)
Dr. S. Radhakrishnan	Teachers' Day (5 Sept) on his birthday
Dr. Zakir Hussain	First to die in office (1969); first Muslim President
V.V. Giri	First acting President; elected as independent (second-preference count)
Fakhruddin Ali Ahmed	Declared 1975 Emergency; died in office
Neelam Sanjiva Reddy	Elected unopposed; also Lok Sabha Speaker

President	Note
Giani Zail Singh	Pocket veto; first Sikh President
K.R. Narayanan	First Dalit President
A.P.J. Abdul Kalam	'Missile Man'; first scientist President
Pratibha Patil	First woman President
Ram Nath Kovind	14th President
Droupadi Murmu	15th President (2022); first tribal woman President

Art 52 President | Art 61 impeachment | Art 72 pardon | Art 123 ordinance | Art 352/356/360 emergencies | Min age 35

★ **PYQ-TREND:** Electoral college composition (who votes, who doesn't). Oath by CJI; resignation to VP. Impeachment ground. Art 123 ordinance, Art 72 pardon. Pocket veto (Zail Singh). Emergency articles. First woman / Dalit / tribal President. Rajendra Prasad only President elected twice.

6.2 Vice-President, PM and Council of Ministers

- Vice-President (Art 63–71): ex-officio Chairman of Rajya Sabha (Art 64); elected by members (elected + nominated) of both Houses of Parliament; age 35; qualified for Rajya Sabha; term 5 years; removed by Rajya Sabha resolution (effective majority) agreed by Lok Sabha (simple majority), 14 days' notice. First VP: S. Radhakrishnan.
- Prime Minister (Art 74–75): Art 74 – Council of Ministers headed by PM to aid and advise the President (advice binding after 42nd & 44th Amendments; President can return once). Art 75 – PM appointed by President; ministers on PM's advice; collectively responsible to Lok Sabha; size limited to 15% of Lok Sabha strength (91st Amendment 2003).
- Art 78: duty of PM to communicate decisions to the President.
- Prime Ministers: Jawaharlal Nehru (first, longest), Gulzarilal Nanda (acting twice), Lal Bahadur Shastri (Tashkent, 1966), Indira Gandhi (first woman PM), Morarji Desai (first non-Congress PM; only Indian with Bharat Ratna and Nishan-e-Pakistan), Charan Singh (never faced Parliament), V.P. Singh, Chandra Shekhar, P.V. Narasimha Rao (1991 reforms), Atal Bihari Vajpayee, H.D. Deve Gowda, I.K. Gujral, Manmohan Singh, Narendra Modi.
- Attorney General (Art 76): highest law officer; qualified to be SC judge; right to speak in Parliament but not vote; holds office during the pleasure of the President. Solicitor General is not a constitutional post. Advocate General – state (Art 165).

Art 63 VP | Art 64 VP = RS Chairman | Art 74 CoM aids President | Art 75 PM appointment, 15% cap (91st) | Art 76 Attorney General

★ **PYQ-TREND:** VP is ex-officio Chairman of Rajya Sabha. 15% limit on ministry size (91st Amendment). Attorney General can speak but not vote. Art 74 and 75. First non-Congress PM (Morarji Desai).

6.3 Parliament (Articles 79–122)

- Art 79: Parliament = President + Rajya Sabha + Lok Sabha.

Feature	Lok Sabha (House of the People)	Rajya Sabha (Council of States)
Max strength	552 (530 states + 20 UTs + 2 Anglo-Indian earlier; nomination ended 2020)	250 (238 elected + 12 nominated)

Feature	Lok Sabha (House of the People)	Rajya Sabha (Council of States)
Present seats	543 elected	245 (233 + 12)
Election	Direct, FPTP	By elected MLAs (STV); UT members by electoral college
Minimum age	25	30
Term	5 years (can be dissolved)	Permanent; 1/3 retire every 2 years; member's term 6 years
Presiding officer	Speaker (elected by members)	Chairman (VP, ex officio)

- Speaker: elected from members; resigns to the Deputy Speaker; casting vote only in a tie; certifies money bills (decision final); presides over joint sittings (Art 108); removed by effective majority resolution with 14 days' notice. First Speaker G.V. Mavalankar; first woman Speaker Meira Kumar. Pro-tem Speaker administers oaths to new members.
- Sessions: at most 6 months between two sessions (Art 85) → at least two sessions a year; normally Budget, Monsoon and Winter sessions. Quorum: 1/10 of total membership (Art 100).
- Bills: Ordinary bill – either House; deadlock → joint sitting (Art 108; held only 3 times: 1961 Dowry Prohibition, 1978 Banking Service Commission (Repeal), 2002 POTA). Money bill (Art 110) – only in Lok Sabha with President's recommendation; Rajya Sabha can delay 14 days only; no joint sitting. Financial bills (Art 117). Constitutional amendment bills – no joint sitting.
- Budget: 'Annual Financial Statement' (Art 112); Consolidated Fund (Art 266), Contingency Fund (Art 267 – at President's disposal), Public Account. Appropriation Bill (Art 114), Finance Bill. Vote on Account; Cut motions (Policy Cut – reduce to ₹1; Economy Cut; Token Cut – reduce by ₹100).
- Devices: Question Hour (first hour), Zero Hour (Indian innovation, from 1962; not in rules), adjournment motion (LS only), no-confidence motion (LS only; 50 members' support; Art 75 basis), censure motion, calling attention motion (Indian innovation), privilege motion.
- Committees: Public Accounts Committee (22 members – 15 LS + 7 RS; chairman from Opposition by convention since 1967; examines CAG reports; 'twin sister' of Estimates Committee), Estimates Committee (30, all LS; 'continuous economy committee'), Committee on Public Undertakings (22).
- Rajya Sabha's special powers: Art 249 (legislate on State List in national interest by 2/3 resolution), Art 312 (create new All-India Services).

LS max 552, RS max 250 | Ages 25/30 | Art 108 joint sitting | Art 110 money bill | Art 112 budget | Quorum 1/10 | RS 12 nominated (Art 80)

★ **PYQ-TREND:** Who certifies money bill (Speaker). Joint sitting presided by Speaker; held thrice. RS can hold money bill for 14 days. Zero Hour is an Indian innovation. PAC composition and chairman. RS powers under Art 249 and 312. First Speaker G.V. Mavalankar. Minimum ages. Members nominated to RS (12).

✖ **COMMON MISTAKE:** The Vice-President chairs the Rajya Sabha but is not its member, so he cannot vote except in a tie.

6.4 Judiciary

- Supreme Court (Art 124–147): established 28 January 1950; originally CJI + 7 judges; now CJI + 33 (34 total, 2019 Act). Judges appointed by President via the collegium system (Second Judges case 1993, Third 1998; NJAC struck down in 2015). Age limit 65 years. Removal (Art 124(4)) by Parliament on proved misbehaviour or incapacity – special majority in each House (no judge has been removed; V. Ramaswami faced first motion, 1993).
- Jurisdiction: original (Art 131 – Centre–State disputes), writ (Art 32), appellate (Art 132–134), advisory (Art 143 – President seeks opinion; not binding), court of record (Art 129), review (Art 137), complete justice (Art 142). Law declared by SC binding on all courts (Art 141).
- High Courts (Art 214–231): judges retire at 62; Art 226 writs (wider than SC's); common HC for two or more states (Art 231). 25 HCs now; oldest Calcutta (1862), along with Bombay and Madras (1862); Allahabad (1866) is the largest. Telangana and Andhra HCs separated in 2019.
- Lok Adalats: statutory under Legal Services Authorities Act 1987 (NALSA); first held at Junagadh, Gujarat (1982). Public Interest Litigation introduced by Justices P.N. Bhagwati and V.R. Krishna Iyer.
- Landmark cases: Golaknath (1967 – Parliament can't amend FRs), Kesavananda Bharati (1973 – basic structure doctrine; 13-judge bench, largest), Minerva Mills (1980), Maneka Gandhi (1978 – Art 21 expanded, 'procedure established by law' must be fair), Indra Sawhney (1992 – OBC reservation, 50% ceiling, creamy layer), S.R. Bommai (1994 – President's Rule limits, secularism basic structure), Vishaka (1997 – sexual harassment guidelines), K.S. Puttaswamy (2017 – privacy), Shayara Bano (2017 – triple talaq).

SC judges retire 65, HC judges 62 | Art 32 & 226 writs | Art 131 original | Art 143 advisory | Kesavananda 1973 – basic structure | Lok Adalat Act 1987

★ **PYQ-TREND:** Retirement ages of SC and HC judges. Art 143 advisory jurisdiction. Basic structure doctrine case. Oldest High Court. Collegium and NJAC. First Lok Adalat at Junagadh. PIL originators. Removal procedure of judges.

6.5 States, Local Government and Union Territories

- Governor (Art 153–162): appointed by President; holds office during the pleasure of the President; age 35; one person can be Governor of two or more states (7th Amendment 1956). Can reserve bills for President (Art 200–201). Ordinance power (Art 213). Pardon (Art 161) – but not for death sentences.
- Chief Minister appointed by Governor (Art 164). Legislative Councils (Art 169 – created/abolished by Parliament on the Assembly's resolution); currently in 6 states: Andhra Pradesh, Bihar, Karnataka, Maharashtra, Telangana, Uttar Pradesh. Assembly size 60–500 (exceptions: Sikkim 32, Goa 40, Mizoram 40, Puducherry 33).
- Panchayati Raj: Balwant Rai Mehta Committee (1957) recommended three-tier system; first introduced at Nagaur, Rajasthan (2 October 1959) by Nehru; then Andhra Pradesh. Ashok Mehta Committee (1977) – two-tier. L.M. Singhvi – constitutional status. 73rd Amendment (1992; in force 24 April 1993 – National Panchayati Raj Day): Part IX, 11th Schedule (29 subjects), three tiers, 1/3 seats for women, 5-year term, State Election Commission, State Finance Commission (every 5 years), Gram Sabha.
- Urban local bodies: 74th Amendment (1992): Part IX-A, 12th Schedule (18 subjects); Nagar Panchayat, Municipal Council, Municipal Corporation; first municipal corporation – Madras (1688).
- Delhi: National Capital Territory (69th Amendment 1991, Art 239AA). Puducherry also has an assembly. J&K UT has an assembly; Ladakh does not.

Panchayati Raj first – Nagaur, 2 Oct 1959 | 73rd (Panchayat, 29 subjects) & 74th (Municipal, 18 subjects) Amendments 1992 | Legislative Councils in 6 states

★ **PYQ-TREND:** Balwant Rai Mehta Committee and three tiers. First state with Panchayati Raj (Rajasthan, Nagaur). 73rd/74th Amendments details. Women's reservation 1/3 in local bodies. States having Legislative Councils. Governor's pardon power excludes death sentence. Art 239AA Delhi.

6.6 Constitutional and Other Bodies

Body	Article / basis	Key facts
Election Commission	Art 324	Multi-member since 1993 (CEC + 2 ECs); first CEC Sukumar Sen; CEC removed like an SC judge
UPSC	Art 315–323	Chairman & members appointed by President; term 6 years or age 65; annual report to President
Finance Commission	Art 280	Every 5 years; chairman + 4 members; centre–state tax sharing; 1st chaired by K.C. Neogy; 15th by N.K. Singh; 16th by Arvind Panagariya
CAG	Art 148–151	'Guardian of the public purse'; 6 years or 65; removed like SC judge; reports to President → Parliament (PAC)
Attorney General	Art 76	–
National Commission for SCs	Art 338	Separate from STs after 89th Amendment (2003)
National Commission for STs	Art 338A	89th Amendment
National Commission for Backward Classes	Art 338B	102nd Amendment (2018)
GST Council	Art 279A	101st Amendment (2016); chaired by Union Finance Minister
Inter-State Council	Art 263	Sarkaria Commission recommendation; set up 1990
Special Officer for Linguistic Minorities	Art 350B	7th Amendment

- Non-constitutional (statutory/executive) bodies: NITI Aayog (1 January 2015; replaced Planning Commission of 1950; chairman PM; executive resolution), National Human Rights Commission (1993, Protection of Human Rights Act; chairman a retired CJI or SC judge), Central Information Commission (RTI Act 2005), Central Vigilance Commission (Santhanam Committee; statutory 2003), CBI (1963; Delhi Special Police Establishment Act 1946), Lokpal (Act 2013; first Lokpal Justice P.C. Ghose, 2019), National Commission for Women (1992), National Green Tribunal (2010).

Art 148 CAG | Art 280 Finance Commission | Art 315 UPSC | Art 324 Election Commission | Art 279A GST Council | NITI Aayog 1 Jan 2015

★ **PYQ-TREND:** Articles for CAG, Finance Commission, Election Commission, UPSC. First CEC Sukumar Sen. NITI Aayog date and nature (non-constitutional). NHRC year. GST Council article and chairman. CAG called guardian of public purse.

6.7 Important Constitutional Amendments

Amendment	Year	Key change
1st	1951	Ninth Schedule; restrictions on free speech; special provisions for backward classes
7th	1956	Reorganisation of states (A, B, C, D categories abolished)
10th	1961	Dadra & Nagar Haveli
12th	1962	Goa, Daman & Diu
21st	1967	Sindhi language
24th	1971	Parliament can amend any part including FRs; President must assent to amendment bills
26th	1971	Abolished privy purses
31st	1973	Lok Sabha seats raised to 545
36th	1975	Sikkim became a full state
42nd	1976	'Mini Constitution' – Socialist, Secular, Integrity; Fundamental Duties; 5 subjects to Concurrent List; tribunals; CoM advice binding
44th	1978	Right to property removed from FR; 'armed rebellion' for 'internal disturbance'; Arts 20–21 not suspendable
52nd	1985	Anti-defection law (10th Schedule)
61st	1989	Voting age lowered from 21 to 18
69th	1991	Delhi as National Capital Territory
73rd / 74th	1992	Panchayats / Municipalities
86th	2002	Art 21A (education 6–14); Art 45 changed; 11th Fundamental Duty
91st	2003	Ministry size limited to 15%; defection by splits ended
92nd	2003	Bodo, Dogri, Maithili, Santhali
97th	2011	Co-operative societies (Art 19(1)(c), Part IX-B, Art 43B)
99th	2014	NJAC (struck down 2015)
100th	2015	Land Boundary Agreement with Bangladesh
101st	2016	GST
102nd	2018	National Commission for Backward Classes

Amendment	Year	Key change
103rd	2019	10% EWS reservation
104th	2020	SC/ST reservation in legislatures extended to 2030; Anglo-Indian nomination ended
106th	2023	Nari Shakti Vandan Adhiniyam – 1/3 seats for women in Lok Sabha and Assemblies (after delimitation)

- Article 368: amendment procedure – simple majority (outside 368, e.g., new states), special majority (2/3 present and voting + majority of total membership), special majority + ratification by half the states (federal provisions, e.g., GST, election of President, judiciary).

42nd – mini constitution | 44th – property removed | 52nd – anti-defection | 61st – voting age 18 | 73rd/74th – local bodies | 101st – GST | 103rd – EWS | 106th – women's reservation

★ **PYQ-TREND:** 42nd as 'mini constitution'. 61st lowered voting age to 18. 52nd anti-defection. 86th education. 101st GST. 103rd EWS 10%. 106th women's reservation. 24th assent compulsory. 36th Sikkim. Article 368.

6.8 Elections and Political Parties

- Representation of the People Acts 1950 (electoral rolls, seat allocation) and 1951 (conduct of elections). First general election 1951–52 (Oct 1951 – Feb 1952). EVMs first used in Paravur (Kerala) in 1982; VVPAT first in Noksen (Nagaland) 2013. NOTA introduced 2013 (PUCL case). National Voters' Day 25 January (EC founded 25 Jan 1950).
- Delimitation Commission (Art 82, 170); seat allocation frozen on 1971 census until after 2026 (84th Amendment).
- Recognition as national party by Election Commission based on vote share/seats in Lok Sabha or state recognition in 4 states.

First general election 1951–52 | EVM – Paravur 1982 | NOTA 2013 | National Voters' Day 25 Jan

★ **PYQ-TREND:** First use of EVM. NOTA year. National Voters' Day date. Articles 324–329 on elections. Voting age 18 (61st Amendment).

CHAPTER 7

Indian Geography

7.1 India – Location and Size

- Latitudes 8°4'N to 37°6'N; longitudes 68°7'E to 97°25'E. Tropic of Cancer (23.5°N) passes through 8 states: Gujarat, Rajasthan, Madhya Pradesh, Chhattisgarh, Jharkhand, West Bengal, Tripura, Mizoram.
- Area 3.287 million km² – 7th largest country (2.4% of world area); population most populous in the world (overtook China in 2023, UN estimate). N–S extent ≈ 3,214 km; E–W ≈ 2,933 km. Land frontier ≈ 15,200 km; coastline ≈ 7,516.6 km (mainland ≈ 6,100 km).
- Standard Meridian 82°30'E passes through Mirzapur (UP); IST = GMT + 5:30. Time difference between Arunachal and Gujarat ≈ 2 hours.
- Southernmost point: Indira Point (Great Nicobar; earlier Pygmalion Point); mainland southernmost – Kanyakumari (Cape Comorin). Northernmost: Indira Col (Siachen). Easternmost: Kibithu (Arunachal). Westernmost: Guhar Moti (Gujarat).
- Neighbours by length of border: Bangladesh (longest, ≈ 4,096 km), China, Pakistan, Nepal, Myanmar, Bhutan, Afghanistan (≈ 106 km, shortest). States touching Bangladesh: West Bengal (longest), Assam, Meghalaya, Tripura, Mizoram. States bordering Nepal: Uttarakhand, UP, Bihar, West Bengal, Sikkim. States bordering China: Ladakh (UT), HP, Uttarakhand, Sikkim, Arunachal.
- Boundary lines: Radcliffe (India–Pakistan, India–Bangladesh), McMahon (India–China in the east, 1914 Simla), Durand (Pakistan–Afghanistan, 1893). Palk Strait separates India and Sri Lanka (Adam's Bridge/Ram Setu, Gulf of Mannar). Ten Degree Channel separates Andaman and Nicobar; Eight Degree Channel separates Minicoy and Maldives; Nine Degree Channel separates Lakshadweep (Kavaratti) and Minicoy.
- Largest state by area: Rajasthan; smallest: Goa. Most populous: Uttar Pradesh. Largest UT: Ladakh. Smallest UT: Lakshadweep. Longest coastline state: Gujarat (≈ 1,600 km); Andhra Pradesh is second.

Tropic of Cancer – 8 states | Standard meridian 82°30'E (Mirzapur) | Longest border – Bangladesh | Largest state – Rajasthan | Smallest – Goa | Longest coastline – Gujarat

★ **PYQ-TREND:** States crossed by Tropic of Cancer. Standard meridian city. Neighbour with longest/shortest border. Ten Degree Channel. McMahon/Durand/Radcliffe lines. Southernmost point Indira Point. Palk Strait.

7.2 Physiographic Divisions

- The Himalayas: Trans-Himalaya (Karakoram, Ladakh, Zaskar); Greater Himalaya/Himadri (highest peaks, avg 6,000 m; Kangchenjunga 8,586 m – highest peak in India; K2/Godwin Austen 8,611 m in PoK claimed territory), Lesser Himalaya/Himachal (Pir Panjal, Dhauladhar, Mahabharat; hill stations Shimla, Mussoorie, Nainital, Darjeeling), Outer Himalaya/Shiwaliks (youngest; duns like Dehradun; Kosi-Teesta). Purvanchal (Patkai, Naga, Mizo/Lushai hills).
- Important passes: Karakoram (Ladakh–China, highest motorable route region), Zoji La (Srinagar–Leh; Zojila tunnel), Banihal (Jawahar tunnel, Jammu–Srinagar), Khardung La

(Ladakh; among the highest motorable passes), Rohtang (HP – Manali–Lahaul; Atal Tunnel 2020), Shipki La (HP–Tibet; Sutlej enters India), Lipulekh (Uttarakhand; Kailash Mansarovar route), Mana (Uttarakhand), Nathu La (Sikkim–Tibet; trade route reopened 2006), Jelep La (Sikkim–Tibet), Bomdi La (Arunachal), Diphu (Arunachal–Myanmar), Bara-lacha La (HP; Manali–Leh), Palakkad Gap (Kerala–TN, Western Ghats), Thal Ghat & Bhor Ghat (Western Ghats, Maharashtra).

- Northern Plains: Bhabar (pebbles, streams disappear), Terai (marshy), Bhangar (old alluvium; kankar), Khadar (new alluvium; fertile, floodplains).
- Peninsular Plateau: Central Highlands (Malwa, Bundelkhand, Chota Nagpur – 'mineral heartland'), Deccan Plateau. Aravallis – oldest fold mountains; Guru Shikhar (Mount Abu) highest. Vindhyas and Satpuras (Dhupgarh highest peak of Satpura in Pachmarhi). Western Ghats (Sahyadri – continuous; Anamudi 2,695 m – highest peak of South India, in Kerala; Doddabetta – Nilgiris), Eastern Ghats (discontinuous; Mahendragiri; Jindhagada/Arma Konda highest). Nilgiris – meeting point of Western and Eastern Ghats.
- Indian Desert (Thar): mostly in Rajasthan; Luni the only significant river.
- Coastal plains: West – Konkan (Mumbai–Goa), Kanara, Malabar (Kerala; backwaters/kayals, Vembanad lake). East – Northern Circars (Odisha–AP), Coromandel (TN). Chilika lake (Odisha – largest brackish lagoon in India), Pulicat (AP–TN), Kolleru (freshwater, AP).
- Islands: Andaman & Nicobar (572 islands; Barren Island – only active volcano of India; Saddle Peak highest; capital Sri Vijaya Puram, earlier Port Blair) and Lakshadweep (coral; 36 islands; capital Kavaratti).

Highest peak in India – Kangchenjunga | South India – Anamudi | Aravalli – oldest fold mountains (Guru Shikhar) | Barren Island – only active volcano | Chilika – largest brackish lagoon

★ **PYQ-TREND:** Pass–state matching (Zoji La, Nathu La, Shipki La, Rohtang, Bomdi La, Lipulekh). Highest peaks (Kangchenjunga, Anamudi, Guru Shikhar, Dhupgarh). Bhabar/Terai/Bhangar/Khadar. Barren Island volcano. Chilika and Wular lakes. Nilgiris junction of Ghats.

7.3 Drainage – Rivers

River	Origin	Key facts / tributaries
Indus	Near Mansarovar (Bokhar Chu glacier, Tibet)	Enters India in Ladakh; Jhelum, Chenab, Ravi, Beas, Sutlej (Panjnad); Indus Waters Treaty 1960 (India uses Ravi, Beas, Sutlej fully)
Jhelum	Verinag spring	Flows through Wular lake; Srinagar on its banks
Chenab	Bara-lacha La (Chandra + Bhaga)	Largest Indus tributary in India; Chenab bridge (highest railway bridge)
Ravi	Rohtang (Kullu)	Lahore on its banks
Beas	Beas Kund near Rohtang	Joins Sutlej at Harike; Pong dam
Sutlej	Rakas Tal (Tibet)	Enters via Shipki La; Bhakra Nangal dam (Gobind Sagar)
Ganga	Gangotri glacier (Gaumukh) as Bhagirathi; joins Alaknanda at Devprayag	Longest river of India (≈ 2,525 km); national river (2008); Farakka barrage; enters Bangladesh as Padma

River	Origin	Key facts / tributaries
Yamuna	Yamunotri glacier	Longest tributary of Ganga; joins at Prayagraj (Triveni Sangam); Chambal, Betwa, Ken, Sind tributaries
Chambal	Janapao hills (MP, Vindhyas)	Badlands/ravines; Gandhi Sagar, Rana Pratap Sagar, Jawahar Sagar dams; gharial sanctuary
Son	Amarkantak	Right-bank tributary of Ganga
Gandak, Kosi, Ghaghara	Nepal Himalaya	Kosi – 'Sorrow of Bihar'
Damodar	Chota Nagpur	'Sorrow of Bengal'; Damodar Valley Corporation (1948, on TVA model)
Brahmaputra	Chemayungdung glacier (Tibet) as Tsangpo	Enters Arunachal as Siang/Dihang; Majuli (largest river island); Bangladesh – Jamuna; Teesta, Subansiri, Manas tributaries
Narmada	Amarkantak (MP)	West-flowing; rift valley; Dhuandhar falls (Jabalpur); Sardar Sarovar dam; forms estuary
Tapi (Tapti)	Multai (Betul, MP)	West-flowing rift valley; Surat; Ukai dam
Mahi, Sabarmati, Luni	–	West-flowing; Luni ends in Rann of Kutch
Godavari	Trimbakeshwar (Nashik)	Longest peninsular river ('Dakshin Ganga', ≈ 1,465 km); Penganga, Wardha, Wainganga, Indravati, Manjira
Krishna	Mahabaleshwar	Tungabhadra, Bhima, Koyna; Nagarjuna Sagar, Srisailam dams
Kaveri (Cauvery)	Talakaveri (Brahmagiri, Kodagu)	'Dakshin Ganga'; Shivasamudram falls; Krishnaraja Sagar, Mettur dams; Kabini, Hemavati
Mahanadi	Sihawa (Chhattisgarh)	Hirakud dam (one of the longest dams); Odisha delta
Periyar	Sivagiri hills	Longest river of Kerala; Idukki dam
Sharavati	Western Ghats	Jog Falls (Gersoppa) – one of the highest plunge falls

- East-flowing peninsular rivers form deltas (Mahanadi, Godavari, Krishna, Kaveri); west-flowing (Narmada, Tapi) form estuaries.
- Waterfalls: Kunchikal (Karnataka, highest), Jog (Sharavati), Dudhsagar (Mandovi, Goa), Athirappilly (Kerala), Chitrakote ('Niagara of India', Indravati, Chhattisgarh), Hundru (Subarnarekha, Ranchi), Nohkalikai (Meghalaya), Shivasamudram (Kaveri).
- Lakes: Wular (largest freshwater, J&K), Dal (Srinagar), Loktak (Manipur – phumdis, Keibul Lamjao floating national park), Sambhar (largest inland saline, Rajasthan), Pangong Tso (Ladakh), Vembanad (longest lake, Kerala), Chilika (brackish), Kolleru (AP), Hussain Sagar (Hyderabad), Lonar (crater lake, Maharashtra).

Dam	River	State
Tehri (highest dam in India)	Bhagirathi	Uttarakhand
Bhakra Nangal (gravity)	Sutlej	Himachal Pradesh–Punjab
Hirakud	Mahanadi	Odisha
Sardar Sarovar	Narmada	Gujarat
Nagarjuna Sagar	Krishna	Telangana–AP
Mettur	Kaveri	Tamil Nadu
Krishnaraja Sagar	Kaveri	Karnataka
Idukki (arch)	Periyar	Kerala
Indira Sagar	Narmada	MP
Koyna	Koyna	Maharashtra
Rihand (Govind Ballabh Pant Sagar)	Rihand	UP
Tungabhadra	Tungabhadra	Karnataka
Ukai	Tapi	Gujarat
Gandhi Sagar	Chambal	MP
Salal	Chenab	J&K
Farakka barrage	Ganga	West Bengal

Longest river – Ganga | Longest peninsular – Godavari | Highest dam – Tehri | Largest freshwater lake – Wular | Largest saline lake – Sambhar | Largest river island – Majuli

★ **PYQ-TREND:** River origins (Narmada & Son – Amarkantak; Godavari – Trimbakeshwar; Kaveri – Talakaveri; Krishna – Mahabaleshwar). West-flowing rivers through rift valleys (Narmada, Tapi). 'Sorrow of Bihar' (Kosi) and 'Sorrow of Bengal' (Damodar). Dam–river matching (Hirakud, Tehri, Bhakra, Sardar Sarovar). Majuli. Waterfalls (Jog, Chitrakote, Dudhsagar). Loktak floating park. Indus Waters Treaty year.

7.4 Climate

- Tropical monsoon climate. Seasons (IMD): winter (Dec–Feb), summer (Mar–May), south-west monsoon (Jun–Sep), retreating/north-east monsoon (Oct–Nov).
- South-west monsoon onset over Kerala ≈ 1 June; covers India by ≈ 8 July (earlier 15 July). Two branches: Arabian Sea branch and Bay of Bengal branch.
- Mawsynram (Meghalaya) – highest average rainfall in the world (Cherrapunji/Sohra nearby). Least rainfall – Leh, Jaisalmer region.
- Western disturbances bring winter rain to north-west India (good for rabi wheat). Tamil Nadu (Coromandel) gets most rain in winter from the north-east (retreating) monsoon.
- Local winds: Loo (hot dry, north India, summer), Mango showers (Kerala & Karnataka – pre-monsoon, help mango ripening), Cherry blossoms/Coffee showers (Karnataka), Nor'westers / Kalbaisakhi (Bengal–Assam; 'tea showers' in Assam), Bardoli Chheerha (Assam).
- El Niño weakens the Indian monsoon; La Niña strengthens it. Indian Ocean Dipole (IOD) positive phase helps the monsoon.

Monsoon onset Kerala ~1 June | Highest rainfall – Mawsynram | Kalbaisakhi – Bengal | Mango showers – Kerala/Karnataka | NE monsoon – Tamil Nadu winter rain

★ **PYQ-TREND:** Local winds (Loo, Mango showers, Kalbaisakhi). Mawsynram highest rainfall. Tamil Nadu's winter rainfall from NE monsoon. Western disturbances and rabi crops. El Niño effect.

7.5 Soils and Natural Vegetation

Soil	Region	Features
Alluvial	Northern plains, deltas	Most widespread (≈ 43%); rich in potash, poor in nitrogen; khadar & bhangar
Black (regur)	Deccan trap – Maharashtra, Gujarat, MP	Best for cotton; high moisture retention; rich in iron, lime, magnesia; self-ploughing (cracks)
Red & yellow	Eastern & southern Deccan, Odisha, Chhattisgarh	Red due to iron oxide (ferric); yellow when hydrated
Laterite	High rainfall: Western Ghats, Kerala, Karnataka, Odisha hills	Leaching; good for tea, coffee, cashew; used as bricks
Arid (desert)	Rajasthan	Sandy, saline; low humus
Saline & alkaline (usar, reh, kallar)	Dry areas, Punjab–Haryana–UP	Treated with gypsum
Peaty / marshy	Kerala (Kuttanad), Sundarbans	High organic matter
Forest / mountain	Himalayas	Acidic at heights

- ICAR classified 8 major soil groups. Soil erosion control: contour ploughing, terrace farming, shelter belts, strip cropping.
- Forest cover (India State of Forest Report 2021): forest cover ≈ 21.7% of geographical area; forest + tree cover ≈ 24.6%. Largest forest cover by area – Madhya Pradesh; by percentage – Mizoram.
- Forest types: tropical evergreen (Western Ghats, NE; >200 cm rain; ebony, mahogany, rosewood), tropical deciduous/monsoon (most widespread; teak, sal, sandalwood), thorn (Rajasthan; babul, khejri), montane, mangrove/tidal (Sundarbans – sundari tree; largest mangrove forest; Bhitarkanika in Odisha).

Black soil – cotton | Laterite – leaching, tea/coffee/cashew | Alluvial – most widespread | Largest forest area – MP | Highest % forest – Mizoram

★ **PYQ-TREND:** Black soil and cotton (regur). Laterite soil formed by leaching. Red colour due to iron oxide. Gypsum for alkaline soil. Sundarbans mangroves and sundari tree. State with largest forest cover (MP).

7.6 National Parks, Tiger Reserves, Biosphere Reserves

Park / sanctuary	State	Famous for
Jim Corbett (1936, first NP as Hailey NP)	Uttarakhand	Tigers; Project Tiger launched here (1973)
Kaziranga	Assam	One-horned rhinoceros (UNESCO)
Gir	Gujarat	Only home of Asiatic lions
Kanha	Madhya Pradesh	Barasingha (hard-ground swamp deer); Jungle Book inspiration
Bandhavgarh	MP	Highest tiger density; white tigers

Park / sanctuary	State	Famous for
Sundarbans	West Bengal	Royal Bengal tigers, mangroves
Periyar	Kerala	Elephants, tigers
Silent Valley	Kerala	Lion-tailed macaque; tropical evergreen
Eravikulam	Kerala	Nilgiri tahr; Neelakurinji blooms every 12 years
Keoladeo Ghana (Bharatpur)	Rajasthan	Birds; Siberian crane (UNESCO)
Ranthambore	Rajasthan	Tigers
Sariska	Rajasthan	Tigers (relocated)
Dachigam	J&K	Hangul (Kashmir stag)
Hemis	Ladakh	Snow leopard (largest NP of India)
Great Himalayan	HP	UNESCO
Valley of Flowers & Nanda Devi	Uttarakhand	UNESCO
Manas	Assam	Pygmy hog, golden langur; UNESCO
Keibul Lamjao	Manipur	Floating park; Sangai deer
Namdapha	Arunachal	Four big cats (tiger, leopard, snow leopard, clouded leopard)
Kuno	Madhya Pradesh	Cheetahs reintroduced from Namibia (17 Sept 2022) and South Africa
Bandipur & Nagarhole	Karnataka	Tigers, elephants
Dudhwa	UP	Swamp deer, rhino
Simlipal	Odisha	Melanistic tigers
Bhitarkanika	Odisha	Saltwater crocodiles, mangroves
Gahirmatha	Odisha	Olive Ridley turtles (marine sanctuary)
Rann of Kutch (Wild Ass Sanctuary)	Gujarat	Indian wild ass (ghudkhur)
Desert NP	Rajasthan	Great Indian Bustard
Mudumalai	Tamil Nadu	Elephants, tigers
Pench	MP–Maharashtra	Tigers
Tadoba-Andhari	Maharashtra	Tigers
Nagarjunsagar-Srisailam	AP–Telangana	Largest tiger reserve

- Project Tiger 1973 (Jim Corbett first), Project Elephant 1992, Project Crocodile 1975, Project Cheetah 2022. NTCA (National Tiger Conservation Authority) 2006. India has over 55 tiger reserves.
- Biosphere reserves: 18 in India; first – Nilgiri (1986). Largest – Gulf of Kachchh/Rann of Kutch area; 12 are in UNESCO's MAB network (Nilgiri first, 2000). Others: Nanda Devi, Gulf of Mannar, Sundarbans, Nokrek, Pachmarhi, Simlipal, Achanakmar-Amarkantak, Great Nicobar, Agasthyamala, Khangchendzonga, Panna.
- Ramsar sites: India has 80+ (as of 2024); first – Chilika and Keoladeo (1981). Tamil Nadu has the most. Ramsar Convention 1971 (Iran); World Wetlands Day 2 February.

First NP – Jim Corbett | Rhino – Kaziranga | Asiatic lion – Gir | Hangul – Dachigam | Sangai – Keibul Lamjao | Cheetah – Kuno | First biosphere reserve – Nilgiri | First Ramsar – Chilika & Keoladeo

★ **PYQ-TREND:** National park–state–animal matching (very frequent): Kaziranga, Gir, Dachigam, Keibul Lamjao, Silent Valley, Eravikulam, Hemis, Kuno. Project Tiger year and first reserve. Ramsar first sites. First biosphere reserve Nilgiri. Largest tiger reserve.

✖ **COMMON MISTAKE:** Ramsar sites and tiger reserves increase almost every year. Revise the latest count from current affairs.

7.7 Agriculture

- Crop seasons: Kharif (sown June–July with monsoon, harvested Sept–Oct; rice, maize, jowar, bajra, cotton, groundnut, soybean, jute, sugarcane), Rabi (sown Oct–Dec, harvested Mar–Apr; wheat, barley, gram, mustard, peas), Zaid (summer; watermelon, cucumber, muskmelon, fodder).

Crop	Leading state(s)
Rice	West Bengal (largest producer), UP, Punjab
Wheat	Uttar Pradesh, MP, Punjab
Sugarcane	Uttar Pradesh, Maharashtra
Cotton	Gujarat, Maharashtra
Jute	West Bengal ('golden fibre')
Tea	Assam, West Bengal (Darjeeling)
Coffee	Karnataka (≈ 70%), Kerala
Rubber	Kerala
Spices	Kerala ('spice garden of India'); cardamom – Kerala; saffron – J&K (Pampore)
Coconut	Kerala, Karnataka, Tamil Nadu
Tobacco	Gujarat, Andhra Pradesh
Groundnut	Gujarat
Soybean	Madhya Pradesh ('soya state'), Maharashtra
Pulses	Madhya Pradesh, Rajasthan
Millets (bajra)	Rajasthan
Mustard	Rajasthan
Maize	Karnataka, MP

- Revolutions: Green (food grains; M.S. Swaminathan, Norman Borlaug – HYV seeds), White (milk; Verghese Kurien – Operation Flood 1970, Amul, NDDB), Blue (fish), Yellow (oilseeds; Sam Pitroda), Golden (horticulture/honey), Silver (eggs/poultry), Pink (onion, pharma, prawns), Red (meat/tomato), Grey (fertilisers), Brown (leather/cocoa), Round (potato), Black (petroleum).
- India: largest producer of milk, pulses, jute; second in rice, wheat, sugarcane, fruits and vegetables, cotton (top two). International Year of Millets – 2023 (India's proposal).

Kharif – monsoon (rice, cotton) | Rabi – winter (wheat, mustard) | White revolution – Verghese Kurien | Green – Swaminathan | Coffee – Karnataka | Rubber – Kerala | Jute – West Bengal

★ **PYQ-TREND:** Kharif vs rabi crop identification. Leading producer states (coffee, rubber, jute, tea, sugarcane). Revolutions and their products/fathers. Operation Flood. International Year of Millets 2023.

7.8 Minerals, Energy and Industries

Mineral	Leading area / mine
Iron ore	Odisha (largest producer), Chhattisgarh (Bailadila), Jharkhand (Noamundi), Karnataka (Kudremukh)
Coal	Jharkhand (Jharia – largest coking coal), Odisha (Talcher), Chhattisgarh (Korba), West Bengal (Raniganj – first coal mine 1774); Neyveli (lignite, TN)
Petroleum	Mumbai High (offshore, largest), Digboi (Assam – oldest refinery, 1901), Ankleshwar (Gujarat)
Manganese	Odisha, MP, Maharashtra
Mica	Jharkhand (Koderma), Andhra (Nellore), Rajasthan
Bauxite	Odisha (Panchpatmali – NALCO)
Copper	Rajasthan (Khetri), Jharkhand (Singhbhum), MP (Malanjkhand)
Gold	Karnataka (Kolar – closed 2001; Hutti active)
Zinc & lead	Rajasthan (Zawar)
Uranium	Jharkhand (Jaduguda), Andhra (Tummalapalle)
Thorium (monazite)	Kerala beach sands
Diamond	Madhya Pradesh (Panna)
Limestone	MP, Rajasthan

- Iron & steel plants: TISCO Jamshedpur (1907, Tata; Subarnarekha–Kharkai), Bhilai (Chhattisgarh – USSR help), Rourkela (Odisha – West Germany), Durgapur (West Bengal – UK), Bokaro (Jharkhand – USSR), Visakhapatnam (first shore-based integrated plant), Salem, Bhadravati (Vishvesvaraya Iron & Steel).
- Industrial cities: Kanpur ('Manchester of UP'), Ahmedabad ('Manchester of India'), Mumbai (Cottonopolis), Coimbatore ('Manchester of South India'), Bengaluru (Silicon Valley), Ludhiana (hosiery), Firozabad (glass/bangles), Moradabad (brass), Jalandhar (sports goods), Panipat (handloom, 'city of weavers'), Surat (diamond cutting & textiles), Sivakasi (fireworks, matches).
- Nuclear power plants: Tarapur (Maharashtra, first, 1969), Rawatbhata (Rajasthan), Kalpakkam (TN), Narora (UP), Kakrapar (Gujarat), Kaiga (Karnataka), Kudankulam (TN, Russia's help; largest). Bhabha Atomic Research Centre (Trombay).
- Renewable: Bhadla Solar Park (Rajasthan, among the world's largest), Pavagada (Karnataka), Muppandal wind farm (TN). Largest wind power state – Gujarat / Tamil Nadu (close).

Oldest refinery – Digboi | Largest coking coal – Jharia | First nuclear plant – Tarapur | Jaduguda – uranium | Panna – diamonds | TISCO 1907

★ **PYQ-TREND:** Mineral–location matching (Jharia, Kolar, Khetri, Jaduguda, Panna, Mumbai High, Digboi). Steel plants and foreign collaboration (Bhilai–USSR, Rourkela–Germany, Durgapur–UK). Nuclear plants. Industrial city nicknames.

7.9 Transport and Communication

- Railways: first train Bombay (Bori Bunder) – Thane, 16 April 1853 (34 km). 17 zones (+ South Coast Railway announced) – largest zone by route length Northern Railway; headquarters: Northern – New Delhi; Central – Mumbai CST; Western – Churchgate; Eastern – Kolkata; Southern – Chennai; East Central – Hajipur; North Eastern – Gorakhpur; Northeast Frontier – Maligaon (Guwahati); South Central – Secunderabad; South Western – Hubballi; East Coast – Bhubaneswar; North Western – Jaipur; West Central – Jabalpur; North Central – Prayagraj; South East Central – Bilaspur; South Eastern – Kolkata (Garden Reach); Metro Railway Kolkata – Kolkata.
- Longest railway platform: Hubballi (Karnataka, ~1.5 km), then Gorakhpur. Konkan Railway (Roha–Thokur). Vande Bharat (first 2019, Delhi–Varanasi). Dedicated freight corridors. First metro: Kolkata (1984).
- Roads: NH-44 longest national highway (Srinagar–Kanyakumari, ≈ 4,112 km; old NH-7). Golden Quadrilateral (Delhi–Mumbai–Chennai–Kolkata), North-South (Srinagar–Kanyakumari) and East-West (Silchar–Porbandar) corridors. Bharatmala. Atal Tunnel (Rohtang, 9.02 km – longest highway tunnel above 10,000 ft).
- Ports: Kandla/Deendayal (Gujarat – tidal port, largest by cargo among major ports), Mumbai (natural harbour), JNPT/Jawaharlal Nehru (Nhava Sheva – largest container port), Mormugao (Goa – iron ore), New Mangalore, Kochi (natural harbour), Tuticorin/V.O. Chidambaranar, Chennai (oldest artificial harbour), Ennore/Kamarajar (first corporate port), Visakhapatnam (deepest landlocked natural harbour), Paradip (Odisha), Kolkata–Haldia (riverine port). Vadhavan (Maharashtra) under construction.
- Inland waterways: NW-1 Ganga (Prayagraj–Haldia, longest), NW-2 Brahmaputra (Sadiya–Dhubri), NW-3 West Coast Canal (Kerala). 111 national waterways declared (2016 Act).

First train 16 Apr 1853 | NH-44 longest | NW-1 Ganga | JNPT – largest container port | Kandla – tidal | Visakhapatnam – deepest landlocked harbour

★ **PYQ-TREND:** First train date and route. Railway zone headquarters. Longest NH. Port characteristics (Kandla tidal, Visakhapatnam deepest, Kolkata riverine, Mormugao iron ore). National waterways. Atal tunnel.

CHAPTER 8

World & Physical Geography

8.1 Universe and Solar System

- Universe origin: Big Bang theory (Georges Lemaître; expansion – Edwin Hubble). Milky Way – our galaxy (spiral); nearest large galaxy – Andromeda. Light year = distance light travels in one year (≈ 9.46 × 10^{12} km) – a unit of distance. Nearest star after the Sun: Proxima Centauri (≈ 4.24 light years). Brightest star in the night sky: Sirius (Dog Star).
- Sun: ≈ 73% hydrogen, ≈ 25% helium; energy by nuclear fusion; light takes ≈ 8 min 20 s to reach Earth; surface (photosphere) ≈ 5,500–6,000 °C; corona visible during a total solar eclipse; sunspots are cooler regions.

Planet	Remember for
Mercury	Smallest, closest to Sun, fastest orbit (88 days), no moon
Venus	Hottest planet; brightest; rotates clockwise (east to west); 'Earth's twin'; morning/evening star; no moon
Earth	Only planet known with life; densest planet; 'Blue Planet'; 1 moon
Mars	'Red Planet' (iron oxide); Olympus Mons – highest volcano in the solar system; moons Phobos & Deimos
Jupiter	Largest; Great Red Spot; Ganymede – largest moon in solar system; fastest rotation (~10 h)
Saturn	Prominent rings; least dense (would float on water); Titan – moon with thick atmosphere; most moons (count keeps rising)
Uranus	Rotates on its side ('rolling planet'); greenish (methane); discovered by William Herschel (1781); also rotates east to west
Neptune	Farthest planet; strongest winds; discovered by mathematical prediction (1846)

- Inner (terrestrial) planets: Mercury, Venus, Earth, Mars. Outer (Jovian/gas giants): Jupiter, Saturn, Uranus, Neptune. Asteroid belt between Mars and Jupiter. Pluto reclassified as a dwarf planet (IAU, 2006).
- Moon: ≈ 3,84,400 km away; light takes ≈ 1.3 s; same side always faces Earth (synchronous rotation); no atmosphere; first man on Moon – Neil Armstrong (Apollo 11, 20 July 1969). Solar eclipse: Moon between Sun and Earth (on new moon day). Lunar eclipse: Earth between Sun and Moon (full moon). Tides due to Moon's gravity (spring tides at full/new moon; neap tides at quarter moon).

Largest planet – Jupiter | Smallest – Mercury | Hottest – Venus | Red – Mars | Largest moon – Ganymede | Dwarf planet – Pluto (2006) | Sunlight to Earth ≈ 8 min 20 s

★ **PYQ-TREND:** Hottest planet Venus (not Mercury). Planets rotating clockwise (Venus, Uranus). Largest moon Ganymede. Red planet. Pluto's reclassification. Light year as distance. Brightest star Sirius. Solar vs lunar eclipse positions.

✖ **COMMON MISTAKE:** Light year is a unit of distance, not time. Mercury is closest to the Sun but Venus is the hottest because of its thick CO_2 greenhouse atmosphere.

8.2 The Earth – Motions, Latitudes and Time

- Shape: oblate spheroid/geoid. Equatorial radius ≈ 6,378 km; polar ≈ 6,357 km. Rotation (west to east, ≈ 23 h 56 min) → day and night, Coriolis force. Revolution ≈ 365¼ days → seasons (due to 23.5° tilt of axis), leap year.
- Perihelion (closest to Sun, ≈ 3 January); Aphelion (farthest, ≈ 4 July). Equinoxes (equal day and night): 21 March (vernal) and 23 September (autumnal). Solstices: 21 June (summer, longest day in N. hemisphere; Sun over Tropic of Cancer) and 22 December (winter, shortest day; Sun over Tropic of Capricorn).
- Latitudes: Equator 0°, Tropic of Cancer 23.5°N, Tropic of Capricorn 23.5°S, Arctic Circle 66.5°N, Antarctic Circle 66.5°S. Longitude: Prime Meridian 0° (Greenwich); 1° longitude = 4 minutes of time; 15° = 1 hour. International Date Line ≈ 180° (zigzag; crossing westward adds a day).
- Equator passes through 13 countries including Ecuador, Colombia, Brazil, Gabon, Congo, DR Congo, Uganda, Kenya, Somalia, Indonesia, Maldives (waters), Kiribati. Tropic of Cancer passes through India, Mexico, Egypt, Saudi Arabia, Bangladesh, China, Myanmar etc. Tropic of Capricorn through Brazil, Argentina, Chile, South Africa, Australia, Madagascar, Mozambique etc. Both tropics and the Equator cross Africa; only the Equator and Capricorn cross South America; Brazil is crossed by both Equator and Tropic of Capricorn.

1° longitude = 4 min | 21 June – longest day (N) | 22 Dec – shortest day (N) | Equinoxes 21 Mar & 23 Sep | Perihelion ~3 Jan

★ **PYQ-TREND:** Longest day 21 June. Equinox dates. Time-longitude relation. Countries on the Equator and tropics. Seasons due to axial tilt. International Date Line.

8.3 Earth's Interior, Rocks, Landforms

- Layers: crust (SIAL – silica + aluminium, continental; SIMA – silica + magnesium, oceanic), mantle (Moho discontinuity between crust and mantle; asthenosphere – source of magma), core (NIFE – nickel + iron; outer liquid, inner solid; Gutenberg discontinuity between mantle and core). Most abundant element in crust: oxygen (≈ 46.6%), then silicon, aluminium, iron. In Earth as a whole: iron.
- Rocks: igneous (primary; granite – intrusive, basalt – extrusive), sedimentary (layers, fossils; sandstone, limestone, shale, coal), metamorphic (limestone → marble, sandstone → quartzite, shale → slate, granite → gneiss, coal → graphite/diamond under pressure).
- Plate tectonics (continental drift – Alfred Wegener 1912; Pangaea, Panthalassa; Laurasia & Gondwana). Fold mountains (Himalayas, Alps, Andes, Rockies – young), block mountains (Vosges, Black Forest), volcanic mountains (Fujiyama, Kilimanjaro, Vesuvius).
- Volcanoes: Pacific 'Ring of Fire' (≈ 75% of active volcanoes); Mauna Loa (Hawaii – largest active), Ojos del Salado (highest volcano), Stromboli ('lighthouse of the Mediterranean'), Kilimanjaro (Tanzania – highest in Africa, dormant), Krakatoa, Vesuvius (destroyed Pompeii).

- Earthquakes: focus/hypocentre (point of origin), epicentre (on the surface above it). Seismic waves: P (primary – fastest, longitudinal, through solids & liquids), S (secondary – transverse, only solids – their absence shows the outer core is liquid), L/surface waves (most destructive). Measured on the Richter scale (magnitude, logarithmic) and Mercalli scale (intensity). Seismograph records them. Tsunami – seismic sea wave (Japanese word).

Moho – crust/mantle | Gutenberg – mantle/core | Crust – oxygen most abundant | Limestone → marble | P waves fastest; S waves not through liquid | Richter scale

★ **PYQ-TREND:** Most abundant element in crust (oxygen) and metal (aluminium). SIAL/SIMA/NIFE. Metamorphic rock examples (marble, slate, quartzite). P and S waves. Ring of Fire. Wegener's continental drift. Richter scale.

8.4 Atmosphere

- Composition (dry air by volume): nitrogen 78.08%, oxygen 20.95%, argon 0.93%, CO_2 ≈ 0.04%. Water vapour variable.

Layer	Height (approx.)	Features
Troposphere	0–8 km (poles) to 18 km (equator)	All weather; temperature falls ~6.5 °C per km (lapse rate); tropopause
Stratosphere	Up to ~50 km	Ozone layer (15–35 km) absorbs UV; jet aircraft fly in lower part (calm)
Mesosphere	Up to ~80–85 km	Coldest layer; meteors burn up
Thermosphere (incl. ionosphere)	80–400+ km	Reflects radio waves; aurora; temperature rises
Exosphere	Outermost	Merges with space; satellites

- Ozone depletion by CFCs; ozone hole over Antarctica; measured in Dobson units; Montreal Protocol (1987); International Ozone Day 16 September (Vienna Convention 1985).
- Pressure belts and winds: Equatorial low (doldrums), sub-tropical highs (horse latitudes, 30°), sub-polar lows (60°), polar highs. Planetary winds: trade winds (sub-tropical high → equator; easterlies), westerlies (30–60°; 'Roaring Forties', 'Furious Fifties', 'Shrieking Sixties' in the Southern Hemisphere), polar easterlies. Coriolis force deflects winds to the right in the Northern Hemisphere (Ferrel's law).
- Local winds: Chinook (Rockies – 'snow eater', warm), Foehn (Alps – warm, dry), Sirocco (Sahara → Italy, hot), Harmattan (West Africa – 'the doctor', dry), Mistral (France – cold), Bora (Adriatic – cold), Khamsin (Egypt), Santa Ana (California), Zonda (Argentina), Loo (India), Brickfielder (Australia), Pampero (Argentina), Blizzard (polar).
- Cyclones: tropical cyclones are called hurricanes (Atlantic, Caribbean), typhoons (China Sea/Pacific), cyclones (Indian Ocean), willy-willies (Australia), tornadoes (USA – small, most violent). Eye of a cyclone is calm. Anticlockwise in the Northern Hemisphere.
- Clouds: cirrus (high, feathery), cumulus (fair weather, puffy), stratus (layered, low), nimbus/cumulonimbus (rain, thunderstorms).

N_2 78% | O_2 21% | Ar 0.93% | CO_2 0.04% | Ozone – stratosphere | Weather – troposphere | Meteors burn – mesosphere | Radio reflection – ionosphere

★ **PYQ-TREND:** Layer with ozone (stratosphere) and weather (troposphere). Ionosphere reflects radio waves. Composition percentages. Local winds and their regions (Chinook, Foehn, Sirocco, Harmattan, Mistral). Cyclone names by region. Roaring Forties. Ozone Day.

8.5 Hydrosphere – Oceans

- About 71% of Earth's surface is water; ≈ 97% of it is in oceans. Five oceans: Pacific (largest, deepest – Mariana Trench, Challenger Deep ≈ 10,994 m), Atlantic (second largest; S-shaped; busiest for trade; Puerto Rico Trench), Indian (only ocean named after a country; Java/Sunda Trench), Southern (Antarctic), Arctic (smallest, shallowest).
- Ocean currents: warm – Gulf Stream (keeps NW Europe warm), Kuroshio (Japan), Brazil, Agulhas, East Australian; cold – Labrador, Humboldt/Peru (rich fishing; El Niño reverses it), California, Canaries, Benguela, Oyashio, West Australian. Meeting of warm & cold currents creates fishing grounds (Grand Banks off Newfoundland – Gulf Stream + Labrador).
- Salinity: average ≈ 35‰ (parts per thousand). Highest: Lake Van (Turkey) and Dead Sea (≈ 340‰ – lowest point on land, ≈ −430 m); Great Salt Lake (USA). Among seas: Red Sea highest; Baltic Sea low.
- Coral reefs: Great Barrier Reef (Australia – largest living structure). Atolls (Lakshadweep, Maldives).

Largest & deepest ocean – Pacific | Deepest point – Mariana Trench (Challenger Deep) | Smallest ocean – Arctic | Gulf Stream – warm | Humboldt – cold | Dead Sea – lowest land point

★ **PYQ-TREND:** Deepest trench. Warm vs cold currents. Gulf Stream effect on Europe. Grand Banks fishing ground. Ocean named after a country (Indian). Saltiest water bodies. Great Barrier Reef.

8.6 Continents and Countries – Key Facts

Continent	Facts
Asia	Largest & most populous; Mount Everest (8,849 m, Nepal–China; height revised 2020); Dead Sea; Caspian Sea (largest lake, saline); Lake Baikal (deepest, oldest freshwater lake, Siberia); Yangtze (longest in Asia); Gobi desert
Africa	Second largest; Nile (longest river in the world ≈ 6,650 km); Sahara (largest hot desert); Kilimanjaro; Lake Victoria (largest in Africa, source of White Nile); Victoria Falls (Zambezi); crossed by Equator and both tropics; 54 countries
North America	Lake Superior (largest freshwater lake by area); Mississippi–Missouri; Rockies; Niagara Falls (USA–Canada); Death Valley; Greenland (largest island – Denmark)
South America	Amazon (largest by volume/discharge; 'lungs of the world'); Andes (longest mountain range); Angel Falls (Venezuela – highest waterfall); Atacama (driest desert); Lake Titicaca (highest navigable lake)
Europe	Volga (longest in Europe); Alps (Mont Blanc); Danube flows through most countries (10); Vatican City (smallest country)

Continent	Facts
Australia	Smallest continent; Great Barrier Reef; Murray–Darling; 'island continent'; Great Artesian Basin
Antarctica	Coldest, highest average elevation; Vinson Massif; India's stations Dakshin Gangotri (1983, first), Maitri (1989), Bharati (2012)

- Largest countries by area: Russia, Canada, USA/China, Brazil, Australia, India (7th). Most populous: India, China, USA, Indonesia, Pakistan. Smallest: Vatican City; Monaco second.
- Deserts: Sahara (Africa – largest hot), Arabian, Gobi (Mongolia–China), Kalahari (Botswana), Namib (oldest), Atacama (Chile – driest), Thar (India–Pakistan), Great Victoria (Australia), Patagonian (Argentina), Antarctica (largest cold desert).
- Grasslands: Prairies (N. America), Pampas (Argentina), Steppes (Eurasia), Veld (South Africa), Downs (Australia), Savanna (Africa), Llanos (Venezuela), Campos (Brazil), Selvas (Amazon rainforest).
- Important straits: Strait of Gibraltar (Atlantic–Mediterranean; Spain–Morocco), Bosphorus & Dardanelles (Turkey; Black Sea–Mediterranean), Strait of Hormuz (Persian Gulf–Gulf of Oman; oil route), Bab-el-Mandeb (Red Sea–Gulf of Aden), Strait of Malacca (Indonesia–Malaysia; Indian–Pacific Oceans), Palk Strait (India–Sri Lanka), Bering Strait (Asia–North America; Russia–USA), English Channel/Strait of Dover (UK–France), Cook Strait (NZ islands), Bass Strait (Australia–Tasmania), Florida Strait, Sunda Strait (Java–Sumatra), Korea Strait, Magellan Strait (S. America).
- Canals: Suez (1869; Mediterranean–Red Sea; Egypt), Panama (1914; Atlantic–Pacific), Kiel (North Sea–Baltic; Germany).
- Lines: Mason-Dixon (historical US), 38th Parallel (North–South Korea), 49th Parallel (USA–Canada), 17th Parallel (old N–S Vietnam), Durand (Pak–Afghan), McMahon (India–China), Radcliffe, Oder-Neisse (Germany–Poland), Hindenburg (Germany–Poland WWI), Maginot (France–Germany), Siegfried (Germany fortification).

Longest river – Nile | Largest by volume – Amazon | Largest lake – Caspian | Deepest lake – Baikal | Highest waterfall – Angel | Driest – Atacama | Largest hot desert – Sahara | Largest island – Greenland

★ **PYQ-TREND:** Strait–countries matching (Gibraltar, Hormuz, Malacca, Bering, Dover). Boundary lines (38th parallel, 49th parallel, Durand). Grasslands names. Highest waterfall and longest river. Suez/Panama. Indian Antarctic stations. Lake Baikal deepest. Everest revised height.

8.7 Countries, Capitals, Parliaments and Currencies (Neighbours & Key)

Country	Capital	Parliament	Currency
Nepal	Kathmandu	Federal Parliament	Nepalese Rupee
Bhutan	Thimphu	Parliament (Tshogdu/National Assembly)	Ngultrum
Bangladesh	Dhaka	Jatiya Sangsad	Taka
Pakistan	Islamabad	Majlis-e-Shoora	Pakistani Rupee
Sri Lanka	Sri Jayawardenepura Kotte (legislative), Colombo (commercial)	Parliament	Sri Lankan Rupee

Country	Capital	Parliament	Currency
Myanmar	Naypyidaw	Pyidaungsu Hluttaw	Kyat
Afghanistan	Kabul	–	Afghani
Maldives	Malé	People's Majlis	Rufiyaa
China	Beijing	National People's Congress	Renminbi (Yuan)
Japan	Tokyo	Diet	Yen
Russia	Moscow	Federal Assembly (Duma)	Ruble
USA	Washington DC	Congress	Dollar
UK	London	Parliament	Pound Sterling
Israel	Jerusalem (claimed)	Knesset	Shekel
Iran	Tehran	Majlis	Rial
Germany	Berlin	Bundestag	Euro
Norway	Oslo	Storting	Krone
Denmark	Copenhagen	Folketing	Krone
Sweden	Stockholm	Riksdag	Krona
Spain	Madrid	Cortes Generales	Euro
Switzerland	Bern	Federal Assembly	Swiss Franc
Netherlands	Amsterdam (seat of govt – The Hague)	States General	Euro
Turkey (Türkiye)	Ankara	Grand National Assembly	Lira
South Africa	Pretoria (exec.), Cape Town (leg.), Bloemfontein (jud.)	Parliament	Rand
Brazil	Brasília	National Congress	Real
Australia	Canberra	Parliament	Australian Dollar
Saudi Arabia	Riyadh	Majlis al-Shura	Riyal
UAE	Abu Dhabi	Federal National Council	Dirham
Indonesia	Jakarta (moving to Nusantara)	People's Consultative Assembly	Rupiah
South Korea	Seoul	National Assembly	Won
Egypt	Cairo	House of Representatives	Egyptian Pound
Iceland	Reykjavik	Althing (oldest parliament)	Krona

★ **PYQ-TREND:** Parliament names (Diet – Japan, Knesset – Israel, Majlis – Iran/Maldives, Jatiya Sangsad – Bangladesh, Storting – Norway, Riksdag – Sweden, Folketing – Denmark, Bundestag – Germany, Duma – Russia). Currencies of neighbours (Ngultrum, Taka, Kyat, Rufiyaa). Capitals of Myanmar (Naypyidaw) and Sri Lanka. Three capitals of South Africa.

✔ **TOPPER TRICK:** Neighbours' currency chain: "Nepal Rupee – Bhutan Ngultrum – Bangladesh Taka – Myanmar Kyat – Maldives Rufiyaa – Afghanistan Afghani".

CHAPTER 9

Indian Economy

9.1 Basic Concepts

- Economics: Adam Smith ('father of economics'; Wealth of Nations, 1776); J.M. Keynes (macroeconomics; General Theory, 1936); Alfred Marshall (welfare, Principles of Economics). Micro vs macroeconomics (Ragnar Frisch coined terms).
- Sectors: Primary (agriculture, mining, fishing), Secondary (manufacturing, construction, electricity), Tertiary (services). India's GDP: services ≈ 55%, industry ≈ 25–30%, agriculture ≈ 17–18%; agriculture employs ≈ 45% of workers.
- Economic systems: capitalist (market), socialist (state), mixed (India).
- Demand and supply: Law of demand – price ↑, quantity demanded ↓ (inverse). Exceptions: Giffen goods (inferior staples), Veblen goods (luxury/prestige). Law of diminishing marginal utility. Elasticity of demand. Market types: perfect competition, monopoly (single seller), duopoly (two), oligopoly (few sellers), monopsony (single buyer), monopolistic competition (many sellers, differentiated products).

Adam Smith – Wealth of Nations 1776 | Keynes – General Theory 1936 | Giffen & Veblen – exceptions to law of demand | Monopsony – one buyer

★ **PYQ-TREND:** Father of economics. Market forms (monopsony, oligopoly, duopoly). Giffen goods. Law of demand. Sector classification of activities.

9.2 National Income

- GDP: market value of final goods and services produced within a country's borders in a year. GNP = GDP + net factor income from abroad. NNP = GNP − depreciation. National Income = NNP at factor cost.
- GDP at market price − indirect taxes + subsidies = GDP at factor cost. Real GDP (constant prices; base year 2011–12) vs nominal GDP (current prices). GDP deflator = nominal/real × 100.
- Per capita income = national income / population. First estimate of national income: Dadabhai Naoroji (1867–68, 'Poverty and Un-British Rule'); first scientific estimate – V.K.R.V. Rao (1931). After independence: National Income Committee (P.C. Mahalanobis, 1949). Now National Statistics Office (NSO, merged CSO & NSSO in 2019) under MoSPI.
- Measurement methods: product/value-added, income, expenditure.
- Human Development Index (UNDP since 1990): life expectancy, education, per capita income (GNI); devised by Mahbub ul Haq with Amartya Sen.
- Inflation: rise in general price level. Demand-pull vs cost-push. Measured by WPI (Office of Economic Adviser, DPIIT; base 2011–12) and CPI (NSO; base 2012; RBI targets CPI inflation 4% ± 2%). Stagflation = inflation + stagnation; deflation = falling prices; disinflation = falling rate of inflation; hyperinflation; skewflation.

GNP = GDP + NFIA | NNP = GNP − depreciation | Base year 2011–12 | Inflation target CPI 4% ± 2% | First NI estimate – Dadabhai Naoroji

★ **PYQ-TREND:** GDP vs GNP definitions. First person to estimate national income (Naoroji). HDI components and founder (Mahbub ul Haq). CPI and WPI agencies. Inflation target. Stagflation meaning.

9.3 Planning in India

- Planning Commission set up on 15 March 1950 (executive resolution; chairman PM). National Development Council (1952). Replaced by NITI Aayog on 1 January 2015 (National Institution for Transforming India; 'think tank'; bottom-up approach; Governing Council includes CMs).
- Earlier plans: Bombay Plan (1944, industrialists), Gandhian Plan (S.N. Agarwal), People's Plan (M.N. Roy), M. Visvesvaraya's Planned Economy for India (1934), National Planning Committee (INC, 1938, Nehru).

Plan	Period	Model / focus
First	1951–56	Harrod–Domar; agriculture, irrigation; successful
Second	1956–61	P.C. Mahalanobis; heavy industry; steel plants
Third	1961–66	Self-reliance; failed (China & Pakistan wars, drought)
Plan holiday	1966–69	Three annual plans; Green Revolution began
Fourth	1969–74	Growth with stability; Gadgil formula; bank nationalisation 1969
Fifth	1974–79 (ended 1978)	Garibi Hatao; poverty removal; D.P. Dhar
Rolling Plan	1978–80	Janata govt (Gunnar Myrdal concept)
Sixth	1980–85	Poverty eradication; IRDP
Seventh	1985–90	Food, work, productivity
Annual plans	1990–92	Economic crisis
Eighth	1992–97	LPG reforms; Rao–Manmohan model; human development
Ninth	1997–2002	Growth with social justice
Tenth	2002–07	8% growth target
Eleventh	2007–12	Faster and more inclusive growth
Twelfth (last)	2012–17	Faster, more inclusive and sustainable growth

Planning Commission 15 Mar 1950 | NITI Aayog 1 Jan 2015 | 2nd Plan – Mahalanobis | Plan holiday 1966–69 | 12th Plan – last

★ **PYQ-TREND:** Second Plan's Mahalanobis model. First Plan – Harrod-Domar, agriculture. Plan holiday years. Rolling plan concept. NITI Aayog date and chairman. Bombay Plan.

9.4 Money, Banking and RBI

- Reserve Bank of India: established 1 April 1935 (Hilton Young Commission / RBI Act 1934); nationalised 1 January 1949; HQ Mumbai (initially Calcutta); first Governor Sir Osborne

Smith; first Indian Governor C.D. Deshmukh. Functions: monetary policy, issue of currency (except ₹1 notes and coins – Government of India/Finance Ministry; RBI distributes), banker to the government, banker's bank, lender of last resort, custodian of forex reserves.

- Monetary Policy Committee (2016): 6 members (3 RBI incl. Governor with casting vote, 3 government nominees); meets at least four times a year.
- Instruments: quantitative – Repo rate (RBI lends to banks), Reverse repo, CRR (cash reserve ratio, kept with RBI), SLR (statutory liquidity ratio, kept by banks themselves in gold, cash, approved securities), Bank rate (long-term lending), Open Market Operations, MSF (Marginal Standing Facility), SDF (Standing Deposit Facility, 2022 – floor of LAF corridor); qualitative – margin requirements, moral suasion, credit rationing, direct action.
- To control inflation, RBI raises repo/CRR/SLR (contractionary/dear money). To boost growth, it lowers them (cheap money).
- Money supply: M0 (reserve money), M1 (narrow money = currency + demand deposits + other deposits with RBI), M2, M3 (broad money = M1 + time deposits; most commonly used), M4.
- Banking history: first bank – Bank of Hindostan (1770); oldest existing – State Bank of India (originally Bank of Calcutta 1806 → Imperial Bank 1921 → SBI 1 July 1955); first purely Indian bank – Punjab National Bank (1894/95, Lala Lajpat Rai among founders). Bank nationalisation: 14 banks (19 July 1969), 6 more (1980). Lead Bank Scheme 1969. Regional Rural Banks (2 Oct 1975; Narasimham Committee). NABARD (12 July 1982; agriculture & rural credit). SIDBI (1990; MSMEs). EXIM Bank (1982). NHB (1988; housing).
- Payments: NPCI (2008) runs UPI (2016), RuPay, IMPS, NACH, FASTag (NETC), BHIM app. Payments banks (Nachiket Mor committee; e.g., India Post Payments Bank). Small finance banks. Mudra Yojana (2015; Shishu/Kishore/Tarun loans). Jan Dhan Yojana (28 Aug 2014; financial inclusion).
- Committees: Narasimham (banking reforms, 1991 & 1998), Raja Chelliah (tax reforms), Tarapore (capital account convertibility), Malhotra (insurance reforms), Kelkar (fiscal/tax), Urjit Patel (monetary policy framework, inflation targeting), Rangarajan (poverty), Tendulkar (poverty line), Y.V. Reddy (14th FC), Abid Hussain (small industries), Bimal Jalan (RBI reserves).
- Other regulators: SEBI (1988; statutory 1992; securities market; HQ Mumbai), IRDAI (insurance, Hyderabad), PFRDA (pensions), FSSAI (food safety), TRAI (telecom), CCI (competition).
- Stock exchanges: BSE (1875, oldest in Asia; Sensex – 30 stocks), NSE (1992; Nifty – 50 stocks).

RBI 1 Apr 1935, nationalised 1949 | ₹1 note issued by GoI | MPC – 6 members | SBI 1 Jul 1955 | 14 banks nationalised 19 Jul 1969 | NABARD 1982 | Sensex 30, Nifty 50

★ **PYQ-TREND:** RBI establishment and nationalisation years. Who issues ₹1 note (Finance Ministry). Repo/CRR/SLR definitions and their effect on inflation. Bank nationalisation dates. NABARD functions. SEBI as market regulator. Sensex/Nifty composition. Narasimham committee. UPI and NPCI. First Indian bank (PNB). M3 as broad money.

✖ **COMMON MISTAKE:** Repo rate is the rate at which RBI lends to commercial banks; reverse repo is the rate at which RBI borrows from them. Don't swap them.

9.5 Public Finance – Budget and Taxes

- Union Budget: Annual Financial Statement (Art 112); presented on 1 February since 2017 (railway budget merged with the general budget in 2017, ending a practice since 1924 – Acworth Committee). First budget of independent India – R.K. Shanmukham Chetty (26 Nov 1947). First budget in British India – James Wilson (1860). Most budgets – Morarji Desai (10). First woman to present – Indira Gandhi (1970); first full-time woman FM – Nirmala Sitharaman.
- Receipts: revenue receipts (tax + non-tax) and capital receipts (borrowings, disinvestment, loan recovery). Expenditure: revenue (salaries, interest, subsidies) and capital (asset creation).

Revenue deficit = revenue expenditure − revenue receipts
Fiscal deficit = total expenditure − (revenue receipts + non-debt capital receipts) = total borrowings
Primary deficit = fiscal deficit − interest payments
Effective revenue deficit = revenue deficit − grants for creation of capital assets

- FRBM Act 2003 (fiscal discipline). Outcome budget (2005), gender budget (2005–06). Zero-based budgeting.
- Taxes: direct (burden cannot be shifted – income tax, corporation tax, wealth tax (abolished 2015)) and indirect (burden shifted – GST, customs, excise on some goods). Progressive (rate rises with income), regressive, proportional.
- GST: launched 1 July 2017 (101st Amendment 2016); 'One Nation One Tax'; destination-based; CGST, SGST, IGST (inter-state), UTGST; slabs 0, 5, 12, 18, 28% (with cess) – rationalisation announced in 2025 to mainly 5% and 18% plus a 40% special rate; check the latest; GST Council (Art 279A) chaired by Union FM; GSTN for IT backbone; first state to ratify – Assam. Petroleum crude, diesel, petrol, ATF, natural gas and alcohol for human consumption are outside GST for now.
- Laffer curve: relation between tax rates and revenue. Fiscal policy (government – tax & spending) vs monetary policy (RBI – money & interest).

GST 1 July 2017 (101st Amdt) | Budget date 1 Feb since 2017 | First budget of free India – R.K. Shanmukham Chetty | FRBM 2003

★ **PYQ-TREND:** Fiscal deficit definition (= borrowings). Primary deficit. Direct vs indirect tax examples. GST launch date, amendment, first state to ratify. Rail budget merger 2017. First budget presenter. Laffer curve. FRBM Act.

9.6 Economic Reforms, Trade and International Organisations

- 1991 New Economic Policy: LPG – Liberalisation, Privatisation, Globalisation (PM P.V. Narasimha Rao; FM Manmohan Singh); balance of payments crisis; gold pledged; rupee devalued; licence raj dismantled; MRTP diluted (replaced by Competition Act 2002); FEMA 1999 replaced FERA 1973.
- Balance of payments: current account (trade in goods & services, remittances) + capital account (FDI, FPI, loans). India is the largest recipient of remittances.
- International bodies: IMF & World Bank (Bretton Woods Conference 1944; HQ Washington DC); World Bank group includes IBRD, IDA (soft loans), IFC, MIGA, ICSID. WTO (1 Jan 1995; replaced GATT 1947; HQ Geneva). AIIB (Beijing), New Development Bank (BRICS; Shanghai), ADB (Manila, 1966). UNCTAD (Geneva). OPEC (Vienna). OECD (Paris).
- Special Drawing Rights (SDR): IMF reserve asset; basket of USD, Euro, Yuan, Yen, Pound.
- Make in India (25 Sept 2014), Startup India (2016), Digital India (2015), Atmanirbhar Bharat (2020), PLI schemes (2020), Insolvency & Bankruptcy Code (2016), demonetisation of ₹500

and ₹1000 notes (8 Nov 2016).

LPG 1991 | WTO 1 Jan 1995 (Geneva) | IMF & World Bank – Bretton Woods 1944 | FEMA 1999 | IBC 2016 | Demonetisation 8 Nov 2016

★ **PYQ-TREND:** 1991 reforms (LPG) and leaders. WTO formation and HQ. Bretton Woods institutions. FEMA replaced FERA. IDA as soft-loan window. HQs of ADB, AIIB, NDB. SDR basket.

9.7 Poverty, Employment and Government Schemes

- Poverty line committees: Alagh (1979), Lakdawala (1993), Tendulkar (2009 – 37.2% poverty, 2004–05), Rangarajan (2014). Multidimensional Poverty Index (NITI Aayog/UNDP-OPHI; health, education, standard of living).
- Unemployment types: disguised (agriculture – marginal productivity zero), seasonal, structural, frictional, cyclical, educated. Periodic Labour Force Survey (PLFS) measures employment.

Scheme	Year	Purpose
MGNREGA	2005 (Act); launched 2 Feb 2006 (Anantapur)	100 days of guaranteed wage employment for rural households
Pradhan Mantri Jan Dhan Yojana	2014	Bank accounts for all
Swachh Bharat Mission	2 Oct 2014	Sanitation, ODF
Beti Bachao Beti Padhao	2015 (Panipat)	Child sex ratio & girls' education
Pradhan Mantri Ujjwala Yojana	1 May 2016 (Ballia)	LPG connections to BPL women
PM-KISAN	2019	₹6,000 per year income support to farmers in 3 instalments
Ayushman Bharat – PMJAY	23 Sept 2018 (Ranchi)	Health cover ₹5 lakh per family per year
Atal Pension Yojana	2015	Pension for unorganised sector
PM Jeevan Jyoti / Suraksha Bima	2015	Life (₹2 lakh) / accident insurance
Pradhan Mantri Awas Yojana	2015	Housing for all
Saubhagya	2017	Household electrification
Jal Jeevan Mission	2019	Tap water to every rural household (Har Ghar Jal)
PM Garib Kalyan Anna Yojana	2020	Free food grains
Stand-Up India	2016	Loans to SC/ST & women entrepreneurs
Sukanya Samriddhi Yojana	2015	Savings for girl child
SVAMITVA	2020	Property cards in rural areas via drones
PM Vishwakarma	2023	Support to artisans
National Food Security Act	2013	Subsidised grains to ~2/3 population
One Nation One Ration Card	2019–20	Portability of PDS

MGNREGA – 100 days | PM-KISAN ₹6,000/yr | Ayushman Bharat ₹5 lakh | Jan Dhan 2014 | Ujjwala 2016 | Swachh Bharat 2 Oct 2014

★ **PYQ-TREND:** Scheme–purpose matching (MGNREGA days, Ujjwala LPG, PM-KISAN amount, Ayushman Bharat cover, Jal Jeevan). Tendulkar committee on poverty. Disguised unemployment in agriculture. Launch places (Beti Bachao – Panipat).

✖ **COMMON MISTAKE:** Scheme amounts and coverage get revised. Note the version given in the official notification year you are preparing for.

9.8 Census and Demography (Census 2011)

- Census: first non-synchronous 1872 (Mayo); first synchronous 1881 (Ripon); 2011 – 15th census, 7th after independence. 2021 census postponed (to be conducted with caste enumeration; check updates).

Indicator (2011)	Value
Population	121.09 crore (≈ 17.5% of world)
Decadal growth	17.7%
Sex ratio	943 females per 1000 males (highest Kerala 1084; lowest Haryana 879; UT lowest Daman & Diu 618)
Child sex ratio (0–6)	919
Literacy	74.04% (male 82.14%, female 65.46%); highest Kerala (94%), lowest Bihar (61.8%)
Density	382 per km² (highest Bihar 1106; lowest Arunachal 17)
Most populous state	Uttar Pradesh; least – Sikkim
Urban population	31.1%

Census 2011: 121 crore | sex ratio 943 | literacy 74.04% | density 382 | Kerala highest sex ratio & literacy | Bihar highest density

★ **PYQ-TREND:** Census 2011 sex ratio, literacy, density, highest/lowest states. First synchronous census 1881. Most and least populous states.

CHAPTER 10

General Science – Physics

10.1 Units and Measurements

- Seven SI base units: metre (length), kilogram (mass), second (time), ampere (current), kelvin (temperature), mole (amount of substance), candela (luminous intensity). Supplementary: radian, steradian. SI redefined on 20 May 2019 using fundamental constants (kilogram via Planck's constant).

Quantity	SI unit	Note
Force	newton (N)	$kg·m/s^2$
Work / energy	joule (J)	1 kWh = 3.6×10^6 J; 1 calorie = 4.186 J; 1 eV = 1.6×10^{-19} J
Power	watt (W)	1 hp = 746 W
Pressure	pascal (Pa)	1 atm = 1.013×10^5 Pa; bar = 10^5 Pa; torr
Frequency	hertz (Hz)	–
Charge	coulomb (C)	–
Potential difference	volt (V)	–
Resistance	ohm (Ω)	–
Capacitance	farad (F)	–
Inductance	henry (H)	–
Magnetic flux	weber (Wb)	–
Magnetic field	tesla (T)	1 T = 10^4 gauss
Luminous flux	lumen	Illuminance – lux
Radioactivity	becquerel (Bq)	curie = 3.7×10^{10} Bq
Absorbed radiation dose	gray (Gy)	sievert (Sv) for equivalent dose
Power of lens	dioptre (D)	P = 1/f (m)
Sound intensity level	decibel (dB)	–
Viscosity	Pa·s (poise in CGS)	–

- Large/small distance units: light year (≈ 9.46×10^{15} m), parsec (3.26 light years), astronomical unit (Sun–Earth ≈ 1.5×10^{11} m), angstrom (10^{-10} m), fermi (10^{-15} m), nautical mile (1,852 m – sea and air navigation), knot (speed, nautical mile per hour), fathom (depth of water, 6 ft).
- Temperature: °F = (9/5)°C + 32; K = °C + 273.15; −40 °C = −40 °F; normal body temperature 37 °C = 98.6 °F = 310 K. Absolute zero 0 K = −273.15 °C.
- Scalars: mass, speed, distance, time, work, energy, power, temperature, pressure, current (treated as scalar). Vectors: displacement, velocity, acceleration, force, momentum, torque, weight, electric & magnetic field.

7 base SI units | 1 kWh = 3.6 MJ | 1 hp = 746 W | 1 nautical mile = 1,852 m | −40 °C = −40 °F | Body temp 98.6 °F

★ **PYQ-TREND:** SI units of physical quantities (pressure – pascal, power – watt, magnetic field – tesla, radioactivity – becquerel, luminous intensity – candela). Light year/parsec as distance. Nautical mile and knot usage. Temperature conversion −40. Scalar vs vector identification. Unit of power of lens.

10.2 Motion, Force and Gravitation

- Newton's laws: 1st – inertia (Galileo's concept; e.g., passengers jerk forward when a bus stops); 2nd – F = ma (rate of change of momentum); 3rd – action-reaction (rocket propulsion, recoil of gun, swimming, walking).
- Momentum p = mv; conservation of momentum (rocket, gun recoil). Impulse = force × time (catching a ball by moving hands back increases time, reducing force; airbags, shock absorbers).
- Friction: static > kinetic > rolling. Lubricants and ball bearings reduce friction.
- Circular motion: centripetal force mv^2/r towards the centre (centrifugal is a pseudo force; cream separator, washing machine dryer). Banking of roads and bending of cyclist.
- Gravitation: $F = Gm_1m_2/r^2$; $G = 6.67 \times 10^{-11}$ N·m²/kg² (universal constant, measured by Cavendish). g = 9.8 m/s²; g is maximum at poles, minimum at the equator; decreases with height and depth; zero at Earth's centre. g on Moon ≈ 1/6 of Earth's (weight 1/6; mass unchanged).
- Weight in a lift: increases when accelerating up, decreases when accelerating down, zero in free fall (weightlessness in an orbiting satellite).
- Escape velocity from Earth ≈ 11.2 km/s (independent of mass of the body); orbital velocity near Earth ≈ 7.9 km/s. Geostationary satellite: ≈ 35,786 km above the equator (≈ 36,000 km), period 24 h, moves west to east. Kepler's laws (elliptical orbits; $T^2 \propto r^3$).
- Simple pendulum: $T = 2\pi\sqrt{(l/g)}$ – independent of mass; longer at the Moon; increases with length; a pendulum clock runs slow in summer (length increases) and at hills.
- Work = F·s cos θ (zero if force ⊥ displacement, e.g., coolie carrying load horizontally, satellite in circular orbit). Energy forms: KE = ½mv² (doubling speed → 4× KE), PE = mgh. Law of conservation of energy.

F = ma | g = 9.8 m/s² (max at poles) | Escape velocity 11.2 km/s | Geostationary ≈ 36,000 km, 24 h | $T = 2\pi\sqrt{(l/g)}$ | KE ∝ v²

★ **PYQ-TREND:** Examples of Newton's laws (rocket – 3rd law; inertia examples). g at poles vs equator. Weightlessness in satellites. Escape velocity value and its independence from mass. Pendulum period independent of mass. Work done zero when force ⊥ displacement. Doubling velocity quadruples KE.

10.3 Properties of Matter and Fluids

- Elasticity: Hooke's law (stress ∝ strain). Steel is more elastic than rubber (higher Young's modulus).
- Pressure = force/area (sharp knives, wide straps of bags, broad tyres for tractors). Pressure in liquids increases with depth (dams thicker at the base). Atmospheric pressure decreases with altitude (nose bleeding at heights; pressure cooker needed at hills because water boils below 100 °C). Barometer (Torricelli) measures atmospheric pressure; sudden fall indicates storm; gradual fall – rain; rise – fair weather.
- Pascal's law – hydraulic lift, hydraulic brakes, hydraulic press.

- Archimedes' principle: buoyant force = weight of liquid displaced. Floatation: a body floats if its density is less than the liquid's. Ships float due to large displaced volume; ice floats (density 0.92); a person floats more easily in sea water (higher density) and in the Dead Sea. Hydrometer/lactometer (milk purity) use floatation.
- Surface tension: spherical drops, rise of oil in a lamp wick (capillarity), insects walking on water; detergents/soap reduce surface tension; hot soup tastes better (lower surface tension spreads it). Capillarity – blotting paper, rise of water in plants (partly), towel drying.
- Viscosity: liquids' viscosity decreases with temperature; gases' increases. Honey more viscous than water.
- Bernoulli's principle: faster flow → lower pressure (aeroplane lift, roof blown off in storms, atomiser/sprayer, spinning ball swing).

Pascal – hydraulic lift/brakes | Archimedes – buoyancy, ships | Lactometer – milk | Bernoulli – aeroplane lift | Water boils below 100 °C at hills

★ **PYQ-TREND:** Pascal's law applications. Archimedes' principle and floatation. Why it's easier to swim in sea water. Surface tension examples (soap reduces it). Bernoulli examples (aeroplane, roof blown off). Pressure cooker cooks faster (raises boiling point). Barometer reading and weather.

10.4 Heat and Thermodynamics

- Heat is energy in transit (joule; calorie). Temperature measures hotness (thermometer: mercury used because of uniform expansion, high boiling point, shines/opaque, doesn't wet glass; clinical thermometer range 35–42 °C/94–108 °F; pyrometer for very high temperatures; thermocouple; bolometer).
- Specific heat: water has a very high specific heat (4.186 kJ/kg·K) – used as a coolant in radiators, moderates coastal climates, hot water bottles. Land heats and cools faster than sea → land & sea breezes (sea breeze by day, land breeze at night).
- Latent heat: of fusion of ice 336 J/g (80 cal/g); of vaporisation of water 2,260 J/g (540 cal/g). Steam at 100 °C burns more than water at 100 °C (extra latent heat). Ice at 0 °C cools more than water at 0 °C.
- Anomalous expansion of water: maximum density at 4 °C – lakes freeze from the top; aquatic life survives below. Water expands on freezing (pipes burst in winter).
- Heat transfer: conduction (solids; metals good conductors; wood, air poor – woollen clothes trap air, double walls), convection (fluids; sea breeze, monsoon, boiling water), radiation (no medium; Sun's heat; black surfaces absorb and emit best; white reflects – white clothes in summer). Thermos flask reduces all three (vacuum stops conduction & convection, silvered walls reduce radiation).
- Boiling point rises with pressure (pressure cooker) and with dissolved impurities (salt); melting point of ice falls with pressure (regelation – skating) and with impurities (salt on icy roads).
- Evaporation causes cooling (earthen pot/surahi, sweating, desert cooler); faster with higher temperature, larger surface area, wind, lower humidity.
- Thermodynamics laws: zeroth (thermal equilibrium – temperature), first (energy conservation), second (heat flows from hot to cold; no 100% efficient engine), third (absolute zero unattainable). Refrigerator transfers heat from cold to hot using work.

Max density of water at 4 °C | Latent heat of ice 80 cal/g; steam 540 cal/g | Sea breeze by day, land breeze at night | Thermos – vacuum + silvered walls

★ **PYQ-TREND:** Anomalous expansion of water (4 °C). Steam burns more than boiling water (latent heat). Pressure cooker principle. Salt raises boiling point and lowers melting point. Sea/land breeze timing. Earthen pot cooling by evaporation. Thermos flask design. Why mercury in thermometers. White clothes in summer. Woollen clothes keep warm (trapped air).

10.5 Sound

- Sound: mechanical longitudinal wave; needs a medium (cannot travel in vacuum – no sound on the Moon). Speed: solids > liquids > gases; in air ≈ 330–343 m/s (increases with temperature and humidity; independent of pressure); in water ≈ 1,480 m/s; in steel ≈ 5,000–6,000 m/s.
- Audible range: 20 Hz–20 kHz. Infrasonic (< 20 Hz – earthquakes, elephants, whales). Ultrasonic (> 20 kHz – bats (echolocation), dolphins, dogs hear up to ~45 kHz; used in SONAR, ultrasound scanning, cleaning, detecting flaws, killing bacteria).
- Characteristics: pitch (depends on frequency; women/children have higher pitch), loudness (amplitude; decibel), quality/timbre (waveform).
- Echo: reflected sound heard separately; minimum distance ≈ 17 m (persistence of hearing 0.1 s). Reverberation – multiple reflections (reduced by soft absorbing materials in halls). SONAR (sound navigation and ranging) uses ultrasonic echoes to find ocean depth and submarines.
- Doppler effect: apparent change in frequency due to relative motion (siren pitch rising as an ambulance approaches; used in radar speed guns, astronomy – red shift).
- Resonance: forced vibration at natural frequency (soldiers break step on bridges; tuning a radio). Beats: interference of slightly different frequencies.
- Supersonic speed: faster than sound (Mach > 1); sonic boom.

Sound needs medium | v: solid > liquid > gas | Air ≈ 343 m/s | Audible 20 Hz–20 kHz | Echo min distance ≈ 17 m | Bats – ultrasonic

★ **PYQ-TREND:** Sound cannot travel in vacuum. Speed of sound highest in solids. Audible range. Bats and dolphins use ultrasound. SONAR principle. Doppler effect example. Pitch depends on frequency. Decibel unit. Soldiers break step on bridges (resonance).

10.6 Light and Optics

- Light: electromagnetic transverse wave; speed in vacuum ≈ 3×10^8 m/s (fastest); slower in glass/water. Dual nature (wave–particle). Sunlight takes ≈ 8 min 20 s to reach Earth.
- Reflection: angle of incidence = angle of reflection. Plane mirror image: virtual, erect, same size, laterally inverted (AMBULANCE written in reverse). Concave mirror: converging; used in shaving mirrors, dentist's mirror, torch/headlight reflectors, solar furnaces, satellite dishes. Convex mirror: diverging, wider field of view, virtual diminished image – rear-view mirrors of vehicles, street lights.
- Refraction: bending of light when it changes medium. Examples: pool appears shallower, stick appears bent in water, twinkling of stars (atmospheric refraction), sun visible ≈ 2 min before actual sunrise and after sunset, mirage (in deserts and on hot roads – total internal reflection).
- Total internal reflection (light from denser to rarer, angle > critical angle): sparkling diamond (critical angle ≈ 24°), optical fibres (communication, endoscopy), mirage, looming.
- Dispersion: white light splits into VIBGYOR through a prism (Newton); violet deviates most, red least. Rainbow – dispersion + refraction + total internal reflection in raindrops; seen

opposite to the Sun.

- Scattering (Rayleigh: $\propto 1/\lambda^4$): sky looks blue (blue scattered most); sun reddish at sunrise/sunset; danger signals are red (least scattered, travels farthest); clouds white (large particles scatter all colours); sky appears black to astronauts. Tyndall effect – scattering by colloids. Raman effect – change in wavelength on scattering (C.V. Raman, Nobel 1930; National Science Day 28 February).
- Lenses: convex (converging) and concave (diverging). Power P = 1/f (dioptre); convex +ve, concave −ve.

Eye defect	Cause	Correction
Myopia (short-sightedness)	Eyeball elongated; image before retina; can't see far	Concave lens
Hypermetropia (long-sightedness)	Eyeball short; image behind retina; can't see near	Convex lens
Presbyopia	Ageing; weak ciliary muscles	Bifocal lens
Astigmatism	Irregular cornea	Cylindrical lens
Cataract	Lens becomes opaque	Surgery (lens replacement)

- Human eye: image formed on the retina (real and inverted); rods (dim light, black & white; rhodopsin needs vitamin A), cones (colour; bright light); least distance of distinct vision 25 cm; persistence of vision ≈ 1/16 s (basis of motion pictures). Colour blindness – inability to distinguish red & green (X-linked).
- Optical instruments: microscope (convex lenses), telescope (astronomical – convex; Galileo), periscope (two plane mirrors/prisms – submarines), kaleidoscope (multiple reflections from inclined mirrors), binoculars (prisms), spectrometer.
- Primary colours of light: red, green, blue (RGB); secondary: yellow, cyan, magenta. Red + green + blue = white. An object appears in the colour it reflects (a red rose in green light looks black).

c = 3×10^8 m/s | Rear-view mirror – convex | Shaving/dentist mirror – concave | Myopia – concave lens | Hypermetropia – convex | Blue sky – scattering | Diamond sparkle – TIR | Raman effect – 28 Feb

★ **PYQ-TREND:** Mirror used in vehicles (convex), shaving (concave). Correction lenses for eye defects. Blue sky and red sunset (scattering). Twinkling of stars (refraction). Diamond's brilliance and optical fibres (total internal reflection). Rainbow formation. Mirage. Raman effect and National Science Day. Lateral inversion. Danger lights red. Power of lens in dioptre.

10.7 Electricity and Magnetism

- Current I = Q/t (ampere). Ohm's law V = IR. Resistance R = ρL/A (increases with length, decreases with area; metals' resistance rises with temperature; semiconductors' falls). Conductors (silver best, copper used for wires), insulators, semiconductors (silicon, germanium), superconductors (zero resistance at very low temperature).
- Series: $R = R_1 + R_2$ (same current; one bulb fails → all go off; decorative lights). Parallel: $1/R = 1/R_1 + 1/R_2$ (same voltage; household wiring – each appliance gets full voltage and can be switched independently).
- Power $P = VI = I^2R = V^2/R$. Commercial unit: kilowatt-hour (1 unit = 1 kWh). Heating effect (Joule's law $H = I^2Rt$): electric heater, iron, bulb (tungsten filament – high melting point ≈ 3,422 °C; bulb filled with argon/nitrogen), fuse (low melting point alloy of tin and lead – connected in series; protects against overload/short circuit). Heating elements use nichrome (high resistivity, doesn't oxidise).

- Household supply in India: 220–230 V AC, 50 Hz (USA: 120 V, 60 Hz). Wires: live (red/brown), neutral (black/blue), earth (green). Earthing protects against shocks. MCB (miniature circuit breaker) replaces fuse.
- Electric cell: chemical → electrical energy. Dry cell (Leclanché; zinc container, carbon rod; 1.5 V). Lead–acid battery (car; rechargeable). Lithium-ion (phones, EVs – Nobel 2019). Solar cell: light → electricity (silicon).
- Magnetism: like poles repel; magnetic field lines from N to S outside. Earth behaves like a giant magnet (geographic north near magnetic south pole). Compass. Electromagnet (soft iron core) – cranes, electric bell. Ferromagnetic: iron, cobalt, nickel. Oersted discovered magnetic effect of current.
- Electromagnetic induction (Faraday): changing magnetic flux induces current – generator/dynamo (mechanical → electrical), transformer (changes AC voltage; works only on AC; step-up at power stations, step-down near homes), induction cooktop. Electric motor: electrical → mechanical (Fleming's left-hand rule). Generator – Fleming's right-hand rule.
- AC vs DC: AC can be transformed and transmitted efficiently over long distances at high voltage. Devices: rectifier (AC → DC), inverter (DC → AC), galvanometer (detects small current), ammeter (current; in series; low resistance), voltmeter (voltage; in parallel; high resistance), rheostat (variable resistance).

V = IR | P = VI | 1 unit = 1 kWh | India 230 V, 50 Hz AC | Fuse – low melting point, in series | Filament – tungsten | Heater – nichrome | Transformer – AC only

★ **PYQ-TREND:** Household wiring is parallel. Fuse wire properties. Tungsten in bulbs; nichrome in heaters. Commercial unit of energy (kWh). Transformer works on AC only and principle of mutual induction. Generator vs motor energy conversion. Ammeter in series, voltmeter in parallel. Frequency of AC in India (50 Hz). Silver best conductor.

10.8 Modern Physics and Electronics

- Atom: electron (J.J. Thomson, 1897; cathode rays), proton (Goldstein/Rutherford), neutron (James Chadwick, 1932), nucleus (Rutherford's gold-foil experiment, 1911). Bohr's model of orbits (1913).
- Radioactivity: discovered by Henri Becquerel (1896); Marie & Pierre Curie (polonium, radium). Alpha (helium nucleus, least penetrating), beta (electrons), gamma (electromagnetic, most penetrating, no charge). Half-life. Carbon-14 dating (age of fossils/archaeological samples); uranium/potassium-argon dating (rocks). Geiger–Müller counter detects radiation.
- Nuclear fission (heavy nucleus splits – U-235; atom bomb; nuclear reactors; chain reaction controlled by cadmium/boron rods; moderators – heavy water, graphite slow neutrons). Nuclear fusion (light nuclei combine – Sun's energy, hydrogen bomb; very high temperature needed).
- Einstein: $E = mc^2$ (mass-energy equivalence); photoelectric effect (Nobel 1921) – basis of solar cells, photocells.
- Electromagnetic spectrum (increasing wavelength): gamma rays < X-rays < ultraviolet < visible < infrared < microwaves < radio waves. Uses: gamma – cancer treatment, sterilisation; X-rays (Röntgen 1895) – bone imaging; UV – sterilising water, vitamin D synthesis, causes sunburn (ozone absorbs); IR – night vision, TV remote, heat therapy; microwaves – ovens, radar, satellite communication; radio waves – broadcasting.
- Semiconductors: p-type (doped with trivalent – boron, gallium; holes), n-type (pentavalent – phosphorus, arsenic; electrons). Diode (rectifier), transistor (amplifier/switch; Bardeen,

Brattain, Shockley 1947), LED, integrated circuit (Jack Kilby). Logic gates: NAND and NOR are universal gates.

- LASER: Light Amplification by Stimulated Emission of Radiation (monochromatic, coherent). RADAR: Radio Detection And Ranging (radio waves/microwaves). MRI uses strong magnetic fields and radio waves (no ionising radiation). CT scan uses X-rays.

Electron – Thomson | Neutron – Chadwick | Nucleus – Rutherford | Radioactivity – Becquerel | Fission – reactors/atom bomb | Fusion – Sun/H-bomb | X-rays – Röntgen | $E = mc^2$

★ **PYQ-TREND:** Discoverers of electron, proton, neutron, X-rays, radioactivity. Fission vs fusion. Energy source of the Sun (fusion). Carbon dating. Order of electromagnetic spectrum. TV remote uses infrared. Microwave oven principle. LASER full form. Moderator and control rods. Universal gates. Photoelectric effect and Einstein's Nobel.

10.9 Instruments and Inventions (Physics)

Instrument	Measures / use
Ammeter	Electric current
Altimeter	Altitude (aircraft)
Anemometer	Wind speed
Audiometer	Sound intensity / hearing
Barometer	Atmospheric pressure
Bolometer	Heat radiation
Calorimeter	Quantity of heat
Cardiograph (ECG)	Heart activity
Chronometer	Accurate time on ships (longitude)
Dynamo	Converts mechanical to electrical energy
Galvanometer	Small electric currents
Hydrometer	Specific gravity/density of liquids
Hygrometer	Relative humidity
Lactometer	Purity (density) of milk
Manometer	Gas pressure
Odometer	Distance travelled by vehicle
Pyrometer	Very high temperatures
Seismograph	Earthquakes
Sphygmomanometer	Blood pressure
Tachometer	Speed of rotation (rpm)
Ondometer	Frequency of electromagnetic waves
Hypsometer	Boiling point of water / altitude
Rain gauge (udometer)	Rainfall
Spectrometer	Spectra / wavelength
Stethoscope	Heart and lung sounds (Laennec)
Fathometer	Ocean depth (sonar)
Saccharimeter	Sugar concentration

Hygrometer – humidity | Anemometer – wind | Sphygmomanometer – BP | Lactometer – milk | Seismograph – earthquakes | Fathometer – sea depth | Pyrometer – high temperature

★ **PYQ-TREND:** Instrument–quantity matching is asked in almost every SSC paper (hygrometer, anemometer, altimeter, pyrometer, tachometer, sphygmomanometer, lactometer, fathometer).

CHAPTER 11

General Science – Chemistry

11.1 Matter and Atomic Structure

- States: solid, liquid, gas, plasma (fourth state – Sun, lightning, neon signs), Bose–Einstein condensate (fifth; S.N. Bose & Einstein; near absolute zero).
- Sublimation (solid → gas directly): camphor, naphthalene (mothballs), iodine, dry ice (solid CO_2), ammonium chloride. Deposition is the reverse.
- Pure substances: elements and compounds. Mixtures: homogeneous (solutions – air, brass, salt water) and heterogeneous (sand + water, milk is colloidal). Colloids show Tyndall effect: milk, fog, smoke, blood, butter, jelly, ink. Emulsion: milk, butter (liquid in liquid). Aerosol: fog, mist (liquid in gas), smoke (solid in gas). Foam: shaving cream. Gel: jelly, cheese.
- Separation: distillation (liquids with different boiling points; fractional distillation for petroleum and liquid air), chromatography (dyes, pigments), centrifugation (cream from milk, blood tests), sublimation, crystallisation (salt from sea water; purification of alum/copper sulphate), evaporation, filtration, decantation, separating funnel (oil and water), magnetic separation.
- Atomic number (Z) = number of protons; mass number (A) = protons + neutrons. Isotopes: same Z, different A (protium, deuterium, tritium; C-12/C-14; U-235/U-238). Isobars: same A, different Z (Ar-40/Ca-40). Isotones: same number of neutrons. Isoelectronic species: same number of electrons.
- Dalton's atomic theory (1808). Avogadro number 6.022×10^{23} per mole. Molar volume of gas at STP 22.4 L.
- Valency and electronic configuration (K, L, M, N shells hold 2, 8, 18, 32 = $2n^2$).

Plasma – 4th state | BEC – 5th | Sublimation – camphor, naphthalene, iodine, dry ice | Isotopes – same Z | Avogadro 6.022×10^{23} | $2n^2$ rule

★ **PYQ-TREND:** Examples of sublimation. Colloids and Tyndall effect (milk, fog). Separation techniques (cream from milk – centrifugation; petroleum – fractional distillation; dyes – chromatography). Isotopes vs isobars. Avogadro number. Plasma state.

11.2 Periodic Table

- Dobereiner's triads, Newlands' law of octaves, Mendeleev's periodic table (atomic mass; left gaps for gallium & germanium – eka-aluminium, eka-silicon), Moseley's modern periodic law (atomic number). Modern table: 7 periods, 18 groups; 118 elements (Oganesson, 118, heaviest).
- Groups: 1 alkali metals (Li, Na, K, Rb, Cs, Fr – very reactive, stored in kerosene), 2 alkaline earth metals (Be, Mg, Ca, Sr, Ba, Ra), 13 boron family, 14 carbon family, 15 nitrogen family (pnictogens), 16 oxygen family (chalcogens), 17 halogens (F, Cl, Br, I, At), 18 noble gases (He, Ne, Ar, Kr, Xe, Rn; inert; complete octet).
- Trends: across a period – atomic size decreases, electronegativity and ionisation energy increase; down a group – size increases, electronegativity decreases, metallic character increases.

- Extremes: most electronegative – fluorine; most reactive non-metal – fluorine; most reactive metal – caesium/francium; lightest element – hydrogen; lightest metal – lithium; heaviest naturally occurring – uranium; densest – osmium; most abundant in universe – hydrogen; in Earth's crust – oxygen (metal – aluminium); in atmosphere – nitrogen; in human body – oxygen (by mass; carbon second); hardest natural substance – diamond; liquid metal – mercury (also caesium, gallium, francium near room temperature); liquid non-metal – bromine; best conductor – silver; poorest conductor metal – lead/bismuth; highest melting metal – tungsten; lowest melting metal – mercury; metalloids – B, Si, Ge, As, Sb, Te.
- Noble gas uses: helium – balloons, airships, deep-sea diving (with oxygen; avoids bends), MRI cooling; neon – advertising signs (red glow); argon – bulbs, welding; krypton & xenon – flash lamps, headlights; radon – radiotherapy (radioactive).

Modern periodic law – Moseley (atomic number) | 18 groups, 7 periods | Most electronegative – F | Liquid metal – Hg | Liquid non-metal – Br | Densest – Os | Highest melting – W

★ **PYQ-TREND:** Father of the periodic table (Mendeleev) and modern periodic law (Moseley). Group names (halogens, alkali, noble gases). Extremes (most electronegative, liquid non-metal, lightest metal, best conductor). Noble gas uses (helium for diving & balloons, neon for signs). Periodic trends.

11.3 Chemical Reactions, Acids, Bases and Salts

- Reaction types: combination, decomposition (thermal, electrolytic, photolytic – silver chloride darkens in light, used in photography), displacement (more reactive metal displaces less reactive; iron nail in copper sulphate turns brown), double displacement (precipitation), redox (oxidation = loss of electrons/gain of oxygen; reduction = gain of electrons/loss of oxygen). Exothermic (respiration, burning, neutralisation, quicklime + water) and endothermic (photosynthesis, dissolving ammonium chloride/nitrate).
- Corrosion (rusting of iron needs oxygen and water: $Fe_2O_3 \cdot xH_2O$); prevented by galvanisation (zinc coating), painting, oiling, alloying (stainless steel), electroplating, cathodic protection. Rancidity of fats (oxidation) prevented by antioxidants, nitrogen flushing in chip packets, refrigeration.
- Catalysts: speed up reactions without being consumed – iron (Haber process for ammonia), V_2O_5 (Contact process for H_2SO_4), nickel (hydrogenation of oils to vanaspati ghee), platinum (Ostwald process for nitric acid; catalytic converters), enzymes (biocatalysts).
- Acids: sour, turn blue litmus red, pH < 7, release H^+. Bases: bitter, soapy, turn red litmus blue, pH > 7. Indicators: litmus (from lichens), phenolphthalein (colourless in acid, pink in base), methyl orange (red in acid, yellow in base), turmeric (red in base).
- pH scale (Sørensen): 0–14; pure water 7; human blood ≈ 7.4 (slightly basic); gastric juice ≈ 1.5–2 (HCl); lemon juice ≈ 2.2; vinegar ≈ 3; milk ≈ 6.5; tomato ≈ 4.2; saliva ≈ 6.5–7.5; sea water ≈ 8; milk of magnesia ≈ 10; acid rain pH < 5.6; tooth decay begins below pH 5.5.

Natural acid	Source
Citric acid	Lemon, orange
Lactic acid	Curd, sour milk; muscles during exercise (cramps)
Acetic acid	Vinegar
Tartaric acid	Tamarind, grapes
Oxalic acid	Tomato, spinach (kidney stones – calcium oxalate)
Malic acid	Apple

Natural acid	Source
Formic (methanoic) acid	Ant sting, nettle sting
Ascorbic acid	Vitamin C (citrus, amla)
Hydrochloric acid	Stomach (gastric juice)
Uric acid	Urine (gout)
Butyric acid	Rancid butter
Stearic acid	Fats

- Neutralisation: acid + base → salt + water. Antacids (milk of magnesia – $Mg(OH)_2$, baking soda) relieve acidity. Ant sting treated with baking soda; wasp sting (alkaline) with vinegar. Acidic soil treated with quicklime/slaked lime; basic soil with organic matter/gypsum.
- Aqua regia: 3 parts conc. HCl + 1 part conc. HNO_3 – dissolves gold and platinum. 'King of chemicals' / oil of vitriol: sulphuric acid (H_2SO_4).

Common name	Chemical name / formula	Use
Common salt	Sodium chloride, NaCl	Food, chlor-alkali
Baking soda	Sodium bicarbonate, $NaHCO_3$	Baking, antacid, fire extinguisher
Washing soda	Sodium carbonate decahydrate, $Na_2CO_3 \cdot 10H_2O$	Removing hardness of water, glass
Caustic soda	Sodium hydroxide, NaOH	Soap making
Bleaching powder	Calcium oxychloride, $CaOCl_2$	Disinfecting water
Plaster of Paris	Calcium sulphate hemihydrate, $CaSO_4 \cdot \frac{1}{2}H_2O$	Fractures, statues (sets with water into gypsum)
Gypsum	$CaSO_4 \cdot 2H_2O$	Cement (slows setting), fertiliser
Quicklime	Calcium oxide, CaO	Cement, whitewash (with water → slaked lime)
Slaked lime	Calcium hydroxide, $Ca(OH)_2$	Whitewash, soil treatment
Limestone / marble / chalk	Calcium carbonate, $CaCO_3$	Cement, building
Blue vitriol	Copper sulphate, $CuSO_4 \cdot 5H_2O$	Fungicide (Bordeaux mixture)
Green vitriol	Ferrous sulphate, $FeSO_4 \cdot 7H_2O$	Ink, iron supplement
White vitriol	Zinc sulphate	Eye lotion
Epsom salt	Magnesium sulphate, $MgSO_4 \cdot 7H_2O$	Laxative
Glauber's salt	Sodium sulphate decahydrate	Laxative
Potash alum (fitkari)	$K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24H_2O$	Water purification, after-shave (antiseptic)
Laughing gas	Nitrous oxide, N_2O	Anaesthetic
Dry ice	Solid CO_2	Refrigerant, stage fog
Marsh gas	Methane, CH_4	Biogas, CNG
Heavy water	Deuterium oxide, D_2O	Moderator in nuclear reactors
Sal ammoniac	Ammonium chloride, NH_4Cl	Dry cells
Muriatic acid	Hydrochloric acid	–
Lunar caustic	Silver nitrate, $AgNO_3$	Indelible (election) ink
Calomel	Mercurous chloride	Medicine
Mohr's salt	Ferrous ammonium sulphate	Lab reagent

Common name	Chemical name / formula	Use
Chile saltpetre / saltpetre	$NaNO_3$ / KNO_3	Fertiliser / gunpowder
Sugar of lead	Lead acetate	–
Rust	Hydrated ferric oxide, $Fe_2O_3 \cdot xH_2O$	–
Hypo	Sodium thiosulphate, $Na_2S_2O_3$	Photography (fixer)
Galena / Cinnabar / Bauxite / Haematite / Magnetite	PbS / HgS / $Al_2O_3 \cdot 2H_2O$ / Fe_2O_3 / Fe_3O_4	Ores

Baking soda $NaHCO_3$ | Washing soda $Na_2CO_3 \cdot 10H_2O$ | Bleaching powder $CaOCl_2$ | POP $CaSO_4 \cdot ½H_2O$ | Gypsum $CaSO_4 \cdot 2H_2O$ | Laughing gas N_2O | Indelible ink – $AgNO_3$ | Aqua regia 3 HCl : 1 HNO_3

★ **PYQ-TREND:** Common name–formula matching (very frequent). Natural acids (ant sting – formic, tomato – oxalic, curd – lactic, tamarind – tartaric, vinegar – acetic). pH of blood, gastric juice, pure water. Indicator colours (phenolphthalein pink in base). Aqua regia composition. Rusting conditions and galvanisation. Catalysts in Haber/Contact processes. Hydrogenation with nickel. Antacids.

✔ **TOPPER TRICK:** "POP is half-hydrated, gypsum is double-hydrated": $CaSO_4 \cdot ½H_2O$ vs $CaSO_4 \cdot 2H_2O$.

11.4 Metals, Non-metals and Alloys

- Metals: lustrous, malleable (gold most), ductile (gold, silver), good conductors (silver > copper > gold > aluminium), sonorous. Exceptions: mercury liquid; sodium & potassium soft (cut with knife) and stored in kerosene; gallium & caesium melt in the hand; iodine (non-metal) is lustrous; graphite (non-metal) conducts electricity; diamond (non-metal) hardest.
- Reactivity series: K > Na > Ca > Mg > Al > Zn > Fe > Pb > H > Cu > Hg > Ag > Au > Pt.
- Extraction: ore → concentration (froth flotation for sulphide ores) → roasting (sulphide ores in air) / calcination (carbonate ores without air) → reduction (carbon; electrolysis for highly reactive metals like Na, Al – Hall–Héroult) → refining (electrolytic). Thermite reaction (Al + Fe_2O_3) – welding rail tracks.
- Ores: iron – haematite (Fe_2O_3), magnetite (Fe_3O_4 – richest, magnetic), siderite; aluminium – bauxite; copper – chalcopyrite, cuprite; zinc – zinc blende, calamine; lead – galena; mercury – cinnabar; tin – cassiterite; sodium – rock salt; magnesium – magnesite, dolomite, carnallite; calcium – limestone; uranium – pitchblende; thorium – monazite.
- Iron forms: pig iron/cast iron (≈ 2–4% C, brittle), wrought iron (purest, < 0.25% C), steel (0.1–1.5% C). Stainless steel (Fe + Cr + Ni; chromium prevents rust).

Alloy	Composition	Use
Brass	Cu + Zn	Utensils, fittings
Bronze	Cu + Sn	Statues, medals, bells
German silver	Cu + Zn + Ni	Utensils
Gun metal	Cu + Sn + Zn	Gears, guns
Bell metal	Cu + Sn	Bells
Solder	Pb + Sn	Joining electrical wires (low melting point)
Duralumin	Al + Cu + Mg + Mn	Aircraft

Alloy	Composition	Use
Magnalium	Al + Mg	Aircraft parts, scientific instruments
Stainless steel	Fe + Cr + Ni + C	Utensils, surgical tools
Nichrome	Ni + Cr	Heating elements
Amalgam	Mercury + any metal	Dental fillings (silver–tin amalgam)
Invar	Fe + Ni	Clock pendulums, measuring tapes
Alnico	Al + Ni + Co + Fe	Magnets
Type metal	Pb + Sb + Sn	Printing
22-carat gold	22 parts gold + 2 parts Cu/Ag	Jewellery (24 carat = pure)

- Non-metals: hydrogen (lightest; 'future fuel'; Cavendish), oxygen (Priestley/Scheele; supports combustion), nitrogen (Rutherford; inert; food packaging; liquid N_2 for preservation), carbon allotropes – diamond (each C bonded to 4, hardest, non-conductor; used in cutting & jewellery), graphite (layers, soft, conductor; pencils, lubricant, electrodes; pencil 'lead' is graphite + clay), fullerene (C-60, buckminsterfullerene, football shape), graphene (single layer; Nobel 2010); sulphur (vulcanisation of rubber – Goodyear; gunpowder); phosphorus (white – stored in water, red – safety matches); chlorine (bleaching, water disinfection); iodine (antiseptic tincture; iodised salt prevents goitre).

Most malleable/ductile – gold | Best conductor – silver | Na & K stored in kerosene | White phosphorus in water | Pencil 'lead' = graphite | Diamond hardest | Brass Cu+Zn | Bronze Cu+Sn | Solder Pb+Sn | 24 carat pure gold

★ **PYQ-TREND:** Alloy compositions (brass, bronze, solder, German silver, duralumin, stainless steel). Ores of metals (bauxite – Al, galena – Pb, cinnabar – Hg, haematite – Fe). Allotropes of carbon and their properties. Metals stored in kerosene. Reactivity series. Froth flotation. Pure gold carat. Vulcanisation with sulphur. Thermite welding.

11.5 Carbon Compounds and Everyday Chemistry

- Carbon forms millions of compounds due to catenation and tetravalency. Hydrocarbons: alkanes (single bonds, C_nH_{2n+2} – methane, ethane, propane, butane), alkenes (double bond – ethene ripens fruits), alkynes (triple – ethyne/acetylene for welding & artificial ripening with calcium carbide – banned).
- Fuels: LPG – mainly butane + propane (ethyl mercaptan added for smell); CNG – mainly methane; biogas – methane (≈ 55–75%) + CO_2; coal gas – H_2 + CH_4 + CO; water gas – CO + H_2; producer gas – CO + N_2; natural gas – methane. Calorific value highest – hydrogen. Octane number (petrol anti-knock quality), cetane number (diesel). Ethanol blending in petrol (E20 target).
- Alcohols: ethanol (in beverages, spirit; 'rectified spirit' 95%; 'absolute alcohol' 100%; denatured with methanol – poisonous, causes blindness). Fermentation of sugar by yeast produces ethanol and CO_2.
- Soaps (sodium/potassium salts of fatty acids; saponification – fat + NaOH) don't lather well in hard water (Ca^{2+}/Mg^{2+} form scum). Detergents work in hard water. Temporary hardness (bicarbonates) removed by boiling or Clark's method (lime); permanent hardness (chlorides, sulphates) by washing soda or ion exchange (permutit/zeolite).

- Polymers: natural (rubber – isoprene, cellulose, starch, proteins, silk, wool), synthetic: polythene (packaging), PVC (pipes), Teflon/PTFE (non-stick; most chemically resistant), Bakelite (thermosetting – switches, handles), nylon (first synthetic fibre – parachutes, ropes; Wallace Carothers), polyester/terylene/dacron (clothes; PET bottles), acrylic (artificial wool), rayon (artificial silk from cellulose), melamine (crockery; thermoset), neoprene (synthetic rubber), Kevlar (bulletproof vests). Thermoplastics soften on heating (polythene, PVC); thermosets don't (Bakelite, melamine).
- Glass: silica (sand) + soda + lime. Types: soda-lime (windows), Pyrex/borosilicate (heat resistant, lab ware), flint/lead crystal (lenses, prisms), photochromatic (sunglasses – silver halide), coloured glass – cobalt oxide (blue), chromium oxide (green), manganese dioxide (violet), ferrous oxide (green), cuprous oxide (red), gold chloride (ruby red).
- Cement: Portland cement (Joseph Aspdin, 1824) – limestone + clay; gypsum added to slow setting. Concrete = cement + sand + gravel + water.
- Fertilisers: urea (46% N – highest), ammonium sulphate, DAP, NPK. Pesticides: DDT (banned in agriculture; biomagnification), BHC.
- Environmental chemistry: greenhouse gases – CO_2, methane, N_2O, CFCs, water vapour; CFCs deplete ozone (Montreal Protocol); acid rain – SO_2 and NOx (damaged Taj Mahal – 'marble cancer'); photochemical smog (ozone, PAN); Bhopal gas tragedy (1984) – methyl isocyanate (MIC) from Union Carbide; Minamata disease – mercury; Itai-itai – cadmium; blue baby syndrome – nitrates in water; fluorosis – excess fluoride; arsenic poisoning – groundwater in Bengal; carbon monoxide binds haemoglobin (≈ 200× stronger than O_2) – poisonous.
- Fire extinguishers: CO_2 (electrical fires), soda-acid, foam (oil fires – don't use water on oil/electrical fires).
- Food chemistry: artificial sweeteners (saccharin, aspartame – unstable when heated, sucralose); preservatives (sodium benzoate, salt, sugar, vinegar); MSG (Ajinomoto, flavour enhancer); baking powder = baking soda + tartaric acid.
- Medicines: analgesics (aspirin, paracetamol), antipyretics (paracetamol – fever), antibiotics (penicillin – Alexander Fleming, 1928), antiseptics (dettol, savlon, tincture iodine – on living tissue) vs disinfectants (phenol – on non-living objects), tranquilisers, antacids, antihistamines.

LPG – butane + propane | CNG – methane | Urea 46% N | Bhopal 1984 – methyl isocyanate | Minamata – Hg | Itai-itai – Cd | Teflon – non-stick | Nylon – first synthetic fibre | Pyrex – borosilicate

★ **PYQ-TREND:** Main constituents of LPG, CNG, biogas. Bhopal gas (MIC). Diseases from pollutants (Minamata – mercury, Itai-itai – cadmium, blue baby – nitrate). Polymers and uses (Teflon, Bakelite, PVC, nylon, rayon). Hard water and its removal. Saponification. CO poisoning. Ethene ripens fruits; calcium carbide banned. Acid rain & Taj Mahal. Penicillin discovery. Fire extinguisher types. Glass colouring oxides.

CHAPTER 12

General Science – Biology

12.1 Cell – The Unit of Life

- Cell discovered by Robert Hooke (1665, cork – dead cells; book Micrographia). Living cells first seen by Anton van Leeuwenhoek (father of microbiology). Nucleus discovered by Robert Brown (1831). Cell theory: Schleiden (plants) & Schwann (animals), 1838–39; Rudolf Virchow – 'omnis cellula e cellula' (cells arise from pre-existing cells).
- Prokaryotes (no true nucleus/membrane-bound organelles; bacteria, cyanobacteria) vs eukaryotes (true nucleus). Largest cell: ostrich egg; smallest: Mycoplasma (PPLO); longest human cell: neuron; largest human cell: ovum; smallest human cell: sperm (by volume); RBCs lack nuclei in mammals.

Organelle	Function / fact
Cell membrane (plasma membrane)	Selectively permeable; fluid mosaic model (Singer & Nicolson)
Cell wall	Plants (cellulose), fungi (chitin), bacteria (peptidoglycan); absent in animal cells
Nucleus	Control centre; contains DNA/chromosomes; nucleolus makes ribosomes
Mitochondria	'Powerhouse of the cell'; ATP via aerobic respiration; own DNA & ribosomes (semi-autonomous); maternal inheritance
Chloroplast	Photosynthesis; chlorophyll (Mg at centre); own DNA; only in plants & algae
Ribosome	Protein synthesis; non-membranous; 70S (prokaryotes) / 80S (eukaryotes)
Endoplasmic reticulum	Rough (protein synthesis, with ribosomes), smooth (lipid synthesis, detoxification)
Golgi apparatus	Packaging & secretion ('post office'); forms lysosomes, acrosome of sperm; Camillo Golgi
Lysosome	'Suicide bags' – digestive enzymes (Christian de Duve)
Vacuole	Storage; large central vacuole in plant cells (tonoplast)
Centrosome	Cell division (spindle) in animal cells
Plastids	Chloroplast (green), chromoplast (colour – tomato, carrot), leucoplast (storage – starch)

- Cell division: mitosis (somatic cells; two identical diploid daughter cells; growth & repair) and meiosis (germ cells; four haploid gametes; reduction division; crossing over → variation).
- DNA: double helix (Watson & Crick, 1953, with Rosalind Franklin's X-ray data); bases A–T, G–C; sugar deoxyribose. RNA: single strand, uracil instead of thymine, ribose sugar. Gene – unit of heredity (term by Johannsen). Human chromosomes: 46 (23 pairs; 22 autosomes + XX/XY).

Cell – Robert Hooke 1665 | Nucleus – Robert Brown | Powerhouse – mitochondria | Suicide bag – lysosome | Protein factory – ribosome | DNA double helix – Watson & Crick 1953 | 46 chromosomes

★ **PYQ-TREND:** Discoverer of cell and nucleus. Powerhouse (mitochondria), suicide bags (lysosomes), protein synthesis (ribosomes). Organelles with own DNA. Cell wall material. Mitosis vs meiosis. Watson & Crick. Number of human chromosomes. Largest/longest human cell.

12.2 Classification of Living Beings

- Carl Linnaeus – father of taxonomy; binomial nomenclature (genus + species, e.g., Homo sapiens, Panthera tigris, Mangifera indica). Hierarchy: Kingdom → Phylum/Division → Class → Order → Family → Genus → Species.
- Whittaker's five kingdoms (1969): Monera (bacteria, cyanobacteria – prokaryotes), Protista (unicellular eukaryotes – Amoeba, Paramecium, Euglena, Plasmodium), Fungi (heterotrophs; chitin walls; yeast, mushroom, Penicillium), Plantae, Animalia. (Carl Woese's three domains: Bacteria, Archaea, Eukarya.)
- Viruses: acellular, crystallisable, living only inside host cells (link between living and non-living); DNA or RNA; first discovered – Tobacco Mosaic Virus (Ivanovsky 1892; named by Beijerinck). Viroids (RNA only), prions (proteins – mad cow disease).
- Plant kingdom: thallophyta (algae), bryophyta ('amphibians of the plant kingdom' – mosses, liverworts), pteridophyta (ferns – first vascular plants), gymnosperms (naked seeds – pine, cycas, deodar; Sequoia – tallest tree), angiosperms (flowering; seeds in fruits; monocots – one cotyledon, parallel veins, fibrous roots – rice, wheat, maize, banana, coconut, grasses; dicots – two cotyledons, reticulate veins, tap roots – pea, mango, neem).
- Animal kingdom: Porifera (sponges, pores), Coelenterata/Cnidaria (Hydra, jellyfish, corals), Platyhelminthes (flatworms – tapeworm, liver fluke), Nematoda (roundworms – Ascaris, filarial worm), Annelida (segmented – earthworm, leech; earthworm 'farmer's friend'), Arthropoda (largest phylum – insects, spiders, crabs, prawns; jointed legs; open circulation), Mollusca (snail, octopus, Pila; octopus/squid – 'devil fish'), Echinodermata (starfish, sea urchin – all marine; water vascular system), Chordata (notochord).
- Vertebrate classes: Pisces (gills; 2-chambered heart; cold-blooded), Amphibia (frog, toad, salamander; 3-chambered heart; live on land and water), Reptilia (lizards, snakes, crocodiles – 4-chambered heart in crocodiles; cold-blooded), Aves (birds; warm-blooded; hollow bones; 4-chambered heart), Mammalia (mammary glands, hair, external ears; 4-chambered heart; warm-blooded; egg-laying mammals – platypus, echidna).
- Odd facts: whale, dolphin, bat are mammals; seahorse, shark are fish; starfish, jellyfish, silverfish, cuttlefish are not fish; penguins are birds; kiwi, ostrich, emu – flightless birds; ostrich lays the largest egg; hummingbird – smallest bird; blue whale – largest animal.

Father of taxonomy – Linnaeus | Five kingdoms – Whittaker 1969 | TMV – first virus | Largest phylum – Arthropoda | Bryophytes – amphibians of plant kingdom | Platypus – egg-laying mammal

★ **PYQ-TREND:** Binomial nomenclature and Linnaeus. Five-kingdom classification. Scientific names of common organisms (Homo sapiens, Panthera leo/tigris, Mangifera indica, Oryza sativa, Triticum aestivum, Azadirachta indica – neem, Ficus benghalensis – banyan). Viruses as a link. Animals wrongly called fish. Heart chambers by class. Bryophytes/pteridophytes characteristics.

12.3 Human Nutrition – Food and Vitamins

- Nutrients: carbohydrates (main energy source; 4 kcal/g), proteins (body building; 4 kcal/g; amino acids – 20 types, 9 essential), fats (concentrated energy; 9 kcal/g), vitamins, minerals,

water, roughage (fibre – prevents constipation).

Vitamin	Chemical name	Deficiency disease	Sources
A (fat-soluble)	Retinol	Night blindness, xerophthalmia	Carrot, milk, liver, fish oil, mango, papaya
D (fat-soluble)	Calciferol	Rickets (children), osteomalacia (adults)	Sunlight (skin), fish oil, egg yolk
E (fat-soluble)	Tocopherol	Infertility/reproductive issues ('anti-sterility vitamin')	Vegetable oils, nuts, green leaves
K (fat-soluble)	Phylloquinone	Delayed blood clotting, haemorrhage	Green leafy vegetables; made by gut bacteria
B1	Thiamine	Beriberi	Whole grains, yeast, pulses
B2	Riboflavin	Ariboflavinosis – cracked lips/tongue (cheilosis)	Milk, eggs, green vegetables
B3	Niacin (nicotinic acid)	Pellagra (3 Ds – dermatitis, diarrhoea, dementia)	Meat, groundnut, cereals
B5	Pantothenic acid	Burning feet syndrome	Widespread
B6	Pyridoxine	Anaemia, convulsions	Cereals, meat
B7	Biotin (vitamin H)	Dermatitis, hair loss	Egg yolk, liver
B9	Folic acid	Megaloblastic anaemia, neural tube defects	Leafy vegetables
B12	Cobalamin (contains cobalt)	Pernicious anaemia	Animal products only (meat, milk, egg)
C (water-soluble)	Ascorbic acid	Scurvy (bleeding gums)	Citrus fruits, amla (richest), guava, tomato

- Fat-soluble vitamins: A, D, E, K (stored in liver/fat). Water-soluble: B-complex and C (not stored; excreted; vitamin C destroyed by heating).

Mineral	Role	Deficiency
Iron	Haemoglobin	Anaemia
Iodine	Thyroxine	Goitre; cretinism in children
Calcium	Bones, teeth, clotting	Rickets, osteoporosis
Phosphorus	Bones, ATP	Weak bones
Sodium & potassium	Nerve impulses, fluid balance	Muscle cramps
Fluorine	Tooth enamel	Dental caries (excess → fluorosis)
Zinc	Enzymes, immunity	Growth retardation
Magnesium	Chlorophyll, enzymes	–
Cobalt	Vitamin B12	Anaemia

- Protein–energy malnutrition: kwashiorkor (protein deficiency – swollen belly, oedema), marasmus (protein + calorie deficiency – wasting).

A – night blindness | B1 – beriberi | B3 – pellagra | B12 – pernicious anaemia (cobalt) | C – scurvy | D – rickets | K – clotting | Iodine – goitre | Fat-soluble: A, D, E, K

★ **PYQ-TREND:** Vitamin–deficiency disease matching is one of the most frequent GA questions. Chemical names (retinol, calciferol, ascorbic acid, tocopherol). Fat- vs water-soluble vitamins. Vitamin made in skin by sunlight (D). Vitamin by gut bacteria (K). Mineral in B12 (cobalt). Iodine and goitre. Kwashiorkor vs marasmus. Richest source of vitamin C (amla).

12.4 Human Body Systems

Digestive system

- Mouth (saliva – salivary amylase/ptyalin digests starch), oesophagus, stomach (HCl kills germs & activates pepsin; pepsin – proteins; renin – milk protein in infants), small intestine (longest part ≈ 6–7 m; duodenum receives bile & pancreatic juice; villi absorb nutrients), large intestine (absorbs water; caecum & vestigial appendix), rectum, anus.
- Liver: largest gland (≈ 1.5 kg); secretes bile (stored in gall bladder; emulsifies fats; no enzymes); stores glycogen; detoxification; makes urea. Pancreas: both exocrine (pancreatic juice – trypsin, lipase, amylase) and endocrine (islets of Langerhans – insulin (β cells, lowers sugar), glucagon (α cells, raises sugar)).
- Digestion end products: carbohydrates → glucose; proteins → amino acids; fats → fatty acids & glycerol.

Respiratory system

- Nose → pharynx → larynx (voice box) → trachea → bronchi → bronchioles → alveoli (gas exchange). Diaphragm helps breathing. Normal breathing rate ≈ 12–18 per minute (adult). Tidal volume ≈ 500 mL; vital capacity ≈ 3.5–4.5 L.
- Aerobic respiration: glucose + O_2 → CO_2 + H_2O + energy (38 ATP theoretically); anaerobic in muscles → lactic acid (fatigue/cramps); in yeast → ethanol + CO_2 (fermentation). Glycolysis in cytoplasm; Krebs cycle in mitochondria.

Circulatory system

- Discovered by William Harvey (1628). Heart: 4 chambers (2 atria, 2 ventricles); left ventricle thickest wall (pumps to whole body); pacemaker – SA node; heartbeat ≈ 72/min; normal BP 120/80 mm Hg (measured by sphygmomanometer). Arteries carry blood away from heart (oxygenated except pulmonary artery); veins carry towards heart (deoxygenated except pulmonary vein). Aorta – largest artery; vena cava – largest vein. Capillaries – exchange. First heart transplant – Christiaan Barnard (South Africa, 1967).
- Blood (≈ 5–6 litres in adults; ≈ 7% of body weight; pH 7.4): plasma (≈ 55%; 90% water) and cells: RBCs (erythrocytes – haemoglobin carries O_2; no nucleus in mammals; lifespan ≈ 120 days; made in red bone marrow; destroyed in spleen – 'graveyard of RBCs'), WBCs (leucocytes – immunity; neutrophils most abundant, lymphocytes make antibodies), platelets (thrombocytes – clotting; fall in dengue). Clotting needs vitamin K, calcium, fibrinogen → fibrin. Haemophilia – bleeder's disease (X-linked). Anticoagulant: heparin.
- Blood groups (Karl Landsteiner, 1900): A, B, AB, O. Universal donor O (negative), universal recipient AB (positive). Rh factor (Landsteiner & Wiener, Rhesus monkey); Rh incompatibility – erythroblastosis fetalis. ABO is an example of multiple alleles and codominance.
- Lymphatic system: lymph nodes, spleen, thymus (T-cell maturation), tonsils.

Excretory system

- Kidneys (≈ 1 million nephrons each – functional unit; Bowman's capsule, glomerulus). Urine: water, urea, uric acid, creatinine, salts; yellow colour from urochrome. Dialysis for kidney failure. Kidney stones – calcium oxalate. Other excretory organs: skin (sweat), lungs (CO_2),

liver (urea formation – ornithine cycle).

Nervous system

- Brain (≈ 1.35 kg; protected by skull & meninges; cerebrospinal fluid): cerebrum (largest; intelligence, memory, thinking, voluntary actions), cerebellum (balance, posture, coordination – affected by alcohol), medulla oblongata (involuntary actions – heartbeat, breathing, vomiting, sneezing), hypothalamus (body temperature, hunger, thirst, controls pituitary), thalamus (relay). Spinal cord – reflex actions. Neuron – structural & functional unit; synapse – junction. 12 pairs of cranial nerves, 31 pairs of spinal nerves.

Endocrine system (hormones)

Gland	Hormone	Function / disorder
Pituitary ('master gland'; at base of brain)	Growth hormone, TSH, ACTH, FSH, LH, prolactin, ADH (vasopressin), oxytocin	GH excess – gigantism/acromegaly; deficiency – dwarfism; ADH deficiency – diabetes insipidus
Thyroid (neck; largest endocrine gland)	Thyroxine (needs iodine), calcitonin	Goitre, cretinism (children), myxoedema (adults), hyperthyroidism – Graves' disease
Parathyroid	Parathormone	Blood calcium (raises)
Adrenal (on kidneys)	Adrenaline ('fight or flight', emergency hormone), cortisol, aldosterone	Addison's disease (cortex deficiency)
Pancreas (islets)	Insulin, glucagon	Diabetes mellitus (insulin deficiency)
Thymus	Thymosin	Immunity (T cells); shrinks in adults
Pineal	Melatonin	Sleep–wake cycle (biological clock)
Testes / ovaries	Testosterone / oestrogen, progesterone	Secondary sexual characters; pregnancy

Skeletal and muscular system

- Adult human: 206 bones (newborn ≈ 270–300). Largest/longest bone – femur (thigh); smallest – stapes (in middle ear); strongest – femur; hardest substance – tooth enamel. Skull 22 bones; vertebral column 33 vertebrae (26 in adults after fusion); ribs 12 pairs (floating ribs – last 2 pairs). Joints: ball-and-socket (shoulder, hip), hinge (knee, elbow), pivot (neck – atlas & axis), gliding (wrist). Largest muscle – gluteus maximus; smallest – stapedius.
- Sense organs: eye (cornea – transplanted; iris gives colour; retina – rods & cones; blind spot – no receptors; yellow spot – sharpest vision), ear (hearing & balance; ossicles – malleus, incus, stapes; cochlea – hearing; semicircular canals – balance), tongue, nose, skin (largest organ).
- Teeth: 32 in adults (dental formula 2123), 20 milk teeth. Enamel hardest substance in body.

Largest gland – liver | Master gland – pituitary | Largest organ – skin | 206 bones; femur longest; stapes smallest | RBC life 120 days | Universal donor O, recipient AB | BP 120/80 | Insulin – β cells | Graveyard of RBCs – spleen

★ **PYQ-TREND:** Largest gland, master gland, largest organ. Blood groups – universal donor/recipient and Landsteiner. Hormone–gland–disorder matching (insulin – diabetes, thyroxine – goitre, ADH – diabetes insipidus, adrenaline – emergency). Function of cerebellum and medulla. Number of bones; smallest and largest bones. RBC lifespan, site of formation and destruction. Bile has no enzymes. HCl in stomach. Heart chambers and pacemaker. Nephron as kidney unit. Haemophilia.

12.5 Diseases, Immunity and Health

Disease	Pathogen	Notes
Tuberculosis	Bacterium – Mycobacterium tuberculosis (Robert Koch, 1882)	Lungs; BCG vaccine; DOTS; World TB Day 24 March
Cholera	Vibrio cholerae (bacterium)	Water-borne; ORS treatment
Typhoid	Salmonella typhi (bacterium)	Widal test; intestine
Tetanus	Clostridium tetani	Lockjaw; DPT vaccine
Diphtheria	Corynebacterium diphtheriae	Throat; DPT
Whooping cough (pertussis)	Bordetella pertussis	DPT
Leprosy	Mycobacterium leprae (Hansen)	Skin & nerves; MDT; Anti-Leprosy Day 30 Jan
Plague	Yersinia pestis	Rat flea; 'Black Death'
Pneumonia	Streptococcus pneumoniae (also viral)	Lungs (alveoli)
Anthrax	Bacillus anthracis	Cattle
Leptospirosis, botulism, gonorrhoea, syphilis	Bacteria	–
Common cold, influenza	Viruses (rhinovirus, influenza virus)	–
Polio	Poliovirus	Paralysis; OPV (Sabin) and IPV (Salk); India declared polio-free 2014
Measles, mumps, rubella	Viruses	MMR vaccine
Chickenpox	Varicella-zoster virus	–
Smallpox	Variola virus	First vaccine – Edward Jenner (1796); eradicated 1980
Rabies	Rabies virus (dog bite)	Hydrophobia; vaccine by Louis Pasteur (1885)
Hepatitis	Hepatitis A–E viruses	Liver; jaundice
AIDS	HIV (retrovirus)	Attacks helper T-cells (CD4); ELISA test; World AIDS Day 1 Dec
Dengue	Dengue virus	Aedes aegypti mosquito (day-biting); falling platelets; 'break-bone fever'
Chikungunya, Zika, yellow fever	Viruses	Aedes mosquito
Japanese encephalitis	Virus	Culex mosquito
COVID-19	SARS-CoV-2 (coronavirus)	Pandemic declared by WHO March 2020
Ebola, Nipah, mpox	Viruses	Nipah – fruit bats (Kerala outbreaks)

Disease	Pathogen	Notes
Malaria	Protozoan – Plasmodium (vivax, falciparum – most severe)	Female Anopheles mosquito; Ronald Ross (Nobel 1902, discovered in Secunderabad); quinine/chloroquine/artemisinin
Amoebic dysentery	Entamoeba histolytica (protozoan)	–
Kala-azar	Leishmania donovani (protozoan)	Sandfly
Sleeping sickness	Trypanosoma	Tsetse fly
Filariasis (elephantiasis)	Wuchereria bancrofti (worm)	Culex mosquito
Ringworm, athlete's foot, dandruff	Fungi	–

- Non-communicable/deficiency/genetic: diabetes, hypertension, cancer (carcinoma – epithelial, sarcoma – connective tissue, leukaemia – blood 'blood cancer', lymphoma), haemophilia, colour blindness, sickle-cell anaemia (abnormal haemoglobin; resistance to malaria), Down syndrome (trisomy 21), Turner's syndrome (XO), Klinefelter's (XXY), thalassaemia. Occupational: silicosis (stone/glass workers), asbestosis, byssinosis (cotton dust), black lung/anthracosis (coal miners).
- Organs affected: jaundice/hepatitis – liver; pneumonia/TB/asthma/bronchitis – lungs; glaucoma/trachoma/conjunctivitis/cataract – eyes; pyorrhoea/gingivitis – gums; arthritis/gout – joints; meningitis/encephalitis – brain; eczema/psoriasis – skin; goitre – thyroid; diabetes – pancreas.
- Immunity: innate vs acquired; active (vaccines, infection) vs passive (antibodies from mother via colostrum/placenta, antiserum). Antibodies (immunoglobulins: IgG – crosses placenta, most abundant; IgA – colostrum; IgM – first/largest; IgE – allergy). Vaccine: weakened/killed pathogen; Edward Jenner – father of immunology. Louis Pasteur – pasteurisation, rabies & anthrax vaccines, germ theory. Antibiotic – penicillin (Fleming, 1928).
- Universal Immunisation Programme (1985); Mission Indradhanush (2014). India's COVID vaccines: Covishield (Serum Institute/Oxford–AstraZeneca), Covaxin (Bharat Biotech – indigenous). Vaccination drive began 16 January 2021.
- Tests: Widal – typhoid; ELISA & Western Blot – HIV; Mantoux & sputum test – TB; Schick test – diphtheria; Dick test – scarlet fever; Tourniquet test – dengue; PCR/RT-PCR – COVID.

Malaria – Plasmodium via female Anopheles | Dengue – Aedes | Filaria – Culex | TB – Koch | Smallpox vaccine – Jenner | Rabies vaccine – Pasteur | Penicillin – Fleming 1928 | Widal – typhoid | ELISA – HIV

★ **PYQ-TREND:** Disease–pathogen–vector matching (malaria, dengue, kala-azar, sleeping sickness, filaria). Bacterial vs viral diseases. Tests (Widal, ELISA, Mantoux). Vaccine pioneers (Jenner, Pasteur, Salk). Polio-free India 2014. HIV attacks T-helper cells. Occupational diseases (silicosis, byssinosis). Genetic disorders (Down syndrome trisomy 21). Ronald Ross and malaria.

✖ **COMMON MISTAKE:** Only the female Anopheles mosquito transmits malaria (males feed on nectar). Dengue's Aedes bites by day.

12.6 Plant Biology

- Photosynthesis: $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ (light, chlorophyll). Oxygen released comes from water (photolysis). Light reaction in grana (thylakoids), dark reaction (Calvin cycle) in stroma. Red and blue light most effective; green least (reflected). C_3 plants (wheat, rice) vs

C_4 plants (maize, sugarcane – more efficient). CAM plants (cactus, pineapple – stomata open at night).

- Transpiration: loss of water vapour through stomata (guard cells); cools the plant, aids water ascent. Xylem conducts water & minerals (dead tissue mostly – vessels, tracheids); phloem conducts food (sieve tubes, companion cells; living). Annual rings in trees (dendrochronology – estimating age).
- Plant hormones: auxins (growth, apical dominance, phototropism; IAA; rooting of cuttings), gibberellins (stem elongation, bolting, seedless fruits; first found in rice 'foolish seedling' disease), cytokinins (cell division; delay ageing), abscisic acid ('stress hormone'; dormancy; stomatal closure), ethylene (only gaseous hormone; fruit ripening).
- Tropisms: phototropism (light – shoots +ve), geotropism (gravity – roots +ve), hydrotropism (water), thigmotropism (touch – tendrils). Nastic movements: Mimosa pudica (touch-me-not) folds leaves – seismonasty.
- Reproduction: vegetative propagation – potato (stem tuber), ginger (rhizome), onion (bulb), sweet potato & dahlia (root tuber), Bryophyllum (leaf buds), grafting (mango, rose), layering, tissue culture (micropropagation). Pollination by wind (anemophily – grasses, maize), insects (entomophily), water (hydrophily – Vallisneria), birds (ornithophily). Double fertilisation unique to angiosperms (Nawaschin).
- Edible parts: potato – stem; carrot, radish, beetroot, sweet potato – root; onion, garlic – bulb (leaves/stem); ginger, turmeric – rhizome (stem); cauliflower – inflorescence; cabbage, lettuce, spinach – leaves; saffron – stigma; clove – flower bud; black pepper – fruit; coconut water – liquid endosperm; apple – thalamus (false fruit); mango – mesocarp; banana – parthenocarpic fruit.
- Nitrogen fixation: Rhizobium in root nodules of legumes (peas, beans, groundnut); Azotobacter (free-living); cyanobacteria (Anabaena, Nostoc – paddy fields); Azolla (fern with Anabaena) – biofertiliser.
- Economic plants: rubber (Hevea brasiliensis), quinine (Cinchona bark – malaria), morphine (opium poppy), cocaine (coca), tea (Camellia sinensis), coffee (Coffea arabica), neem (Azadirachta indica – natural pesticide), Bt cotton (Bacillus thuringiensis gene – pest resistant; India's only approved GM crop for cultivation), golden rice (vitamin A).

O_2 in photosynthesis from water | Ethylene – ripening (gaseous) | ABA – stress | Gibberellin – stem elongation | Potato – stem | Ginger – rhizome | Saffron – stigma | Clove – flower bud | Rhizobium – legumes

★ **PYQ-TREND:** Edible parts of plants (potato stem, carrot root, ginger rhizome, saffron stigma, clove bud, cauliflower inflorescence, apple thalamus). Plant hormones and functions (ethylene ripening, auxin phototropism, ABA stress). Source of oxygen in photosynthesis. Xylem vs phloem. Rhizobium and legumes. Mimosa movement. Bt cotton. Annual rings for age. Insectivorous plants (Nepenthes – pitcher plant, Drosera, Venus flytrap) grow in nitrogen-poor soil.

12.7 Genetics, Evolution, Ecology

- Gregor Mendel – father of genetics; experiments on garden pea (Pisum sativum), 7 traits; laws of dominance, segregation, independent assortment. Monohybrid ratio 3:1 (phenotype), 1:2:1 (genotype); dihybrid 9:3:3:1. Rediscovered in 1900 (de Vries, Correns, Tschermak). T.H. Morgan – fruit fly (Drosophila), linkage. Sex of child determined by father's chromosome (X or Y). Human Genome Project completed in 2003.
- Biotechnology: recombinant DNA (Cohen & Boyer); first cloned mammal – Dolly the sheep (1996, Ian Wilmut); DNA fingerprinting (Alec Jeffreys; in India – Lalji Singh, 'father of Indian

DNA fingerprinting'); CRISPR-Cas9 gene editing (Nobel 2020 – Charpentier & Doudna); first transgenic animal; insulin by genetically engineered bacteria; stem cells; Flavr Savr tomato (first GM food).

- Evolution: Lamarck (inheritance of acquired characters; 'use and disuse'), Darwin (natural selection; 'Origin of Species' 1859; 'survival of the fittest' – phrase by Herbert Spencer), Hugo de Vries (mutation theory). Homologous organs (same origin, different function – forelimbs of man, bat, whale; divergent evolution) vs analogous organs (different origin, same function – wings of bird & insect; convergent evolution). Vestigial organs: appendix, wisdom teeth, coccyx, nictitating membrane. Archaeopteryx – connecting link between reptiles and birds.
- Ecology: ecosystem (biotic + abiotic; term by A.G. Tansley), food chain (producers → herbivores → carnivores → decomposers); 10% law of energy transfer (Lindeman); pyramid of energy always upright; pyramid of numbers inverted in a tree ecosystem; biomass pyramid inverted in the ocean. Biomagnification (DDT, mercury increase along the food chain). Keystone species. Biodiversity hotspots in India: Himalaya, Indo-Burma, Western Ghats & Sri Lanka, Sundaland (Nicobar).
- IUCN Red List: critically endangered, endangered, vulnerable. Great Indian Bustard – critically endangered (state bird of Rajasthan). Ex-situ (zoos, seed banks, botanical gardens) vs in-situ (national parks, sanctuaries, biosphere reserves) conservation. Environment Protection Act 1986; Wildlife Protection Act 1972; Forest Conservation Act 1980; Biological Diversity Act 2002. Chipko movement (1973, Uttarakhand – Sunderlal Bahuguna, Gaura Devi; earlier Bishnoi sacrifice – Amrita Devi, 1730); Appiko movement (Karnataka); Narmada Bachao Andolan (Medha Patkar); Silent Valley movement (Kerala).

Father of genetics – Mendel (pea) | 3:1 and 9:3:3:1 | Dolly – 1996 | Origin of Species – Darwin 1859 | 10% law – Lindeman | Ecosystem – Tansley | Chipko – 1973

★ **PYQ-TREND:** Mendel's ratios and pea plant. Darwin vs Lamarck. Homologous vs analogous organs. Vestigial organs. 10% law of energy transfer. Biomagnification. Biodiversity hotspots of India. Chipko and Bishnoi movements. Dolly the sheep. DNA fingerprinting in India (Lalji Singh). Sex determination by father.

CHAPTER 13

Static GK I – Sports

13.1 Olympics

- Ancient Olympics: Olympia, Greece, 776 BCE. Modern Olympics: founded by Pierre de Coubertin (IOC, 1894; HQ Lausanne, Switzerland); first Games Athens 1896. Motto: 'Citius, Altius, Fortius' (Faster, Higher, Stronger) – 'Communiter' (Together) added in 2021. Five rings (blue, yellow, black, green, red) represent five continents. Olympic Day 23 June.

Year	Host	India's highlight
1900	Paris	Norman Pritchard – 2 silver (athletics)
1928	Amsterdam	First hockey gold (Dhyan Chand)
1952	Helsinki	K.D. Jadhav – bronze (wrestling), first individual medal of independent India
1980	Moscow	Hockey gold (8th and last hockey gold)
1996	Atlanta	Leander Paes – bronze (tennis)
2000	Sydney	Karnam Malleswari – bronze (weightlifting), first Indian woman medallist
2004	Athens	Rajyavardhan Rathore – silver (shooting, double trap)
2008	Beijing	Abhinav Bindra – gold (10 m air rifle), first individual gold
2012	London	6 medals (Sushil Kumar silver, Vijay Kumar silver, Saina, Mary Kom, Gagan Narang, Yogeshwar Dutt)
2016	Rio	P.V. Sindhu – silver (badminton), Sakshi Malik – bronze (wrestling)
2020 (held 2021)	Tokyo	7 medals – best haul then; Neeraj Chopra – gold (javelin, first athletics gold); Mirabai Chanu silver; hockey bronze
2024	Paris	6 medals (Neeraj silver; Manu Bhaker 2 bronze – first Indian with two medals at one Games; Aman Sehrawat bronze; Swapnil Kusale bronze; hockey bronze)
2028	Los Angeles	Upcoming (cricket included)
2032	Brisbane	Upcoming

- India's 8 Olympic hockey golds: 1928, 1932, 1936, 1948, 1952, 1956, 1964, 1980. Dhyan Chand – 'Wizard of Hockey'; his birthday 29 August = National Sports Day; Khel Ratna

renamed Major Dhyan Chand Khel Ratna (2021).

- Asian Games: first held in New Delhi, 1951 (also 1982). Commonwealth Games: Delhi 2010. Paralympics: Tokyo 2020 – Avani Lekhara first Indian woman Paralympic gold; Paris 2024 – India's best (29 medals).

Modern Olympics – Athens 1896 (Coubertin) | IOC HQ – Lausanne | First individual gold – Abhinav Bindra 2008 | First athletics gold – Neeraj Chopra 2021 | Hockey golds – 8 | National Sports Day – 29 Aug

★ **PYQ-TREND:** First Indian individual gold (Bindra), first woman medallist (Malleswari), first individual medal (K.D. Jadhav). Neeraj Chopra's javelin gold. Olympic ring colours. Coubertin. First Asian Games host (Delhi 1951). National Sports Day and Dhyan Chand. Number of hockey golds.

13.2 Games – Terms, Players and Measurements

Sport	Terms
Cricket	Googly, doosra, yorker, LBW, maiden over, hat-trick, silly point, gully, mid-on, cover drive, duck, stumped, follow-on, Duckworth–Lewis–Stern
Football	Penalty, corner kick, off-side, dribble, header, free kick, hat-trick, red card
Hockey	Bully, penalty corner, short corner, push-in, dribble, scoop, striking circle
Tennis	Ace, deuce, love, advantage, volley, smash, let, break point, grand slam
Badminton	Shuttlecock, smash, let, drop, deuce, love
Golf	Birdie (1 under par), eagle (2 under), albatross, bogey (1 over), caddie, hole-in-one, tee, putt, links
Chess	Checkmate, castling, stalemate, gambit, en passant, grandmaster
Boxing	Upper cut, jab, knockout, hook, punch
Polo	Chukker, mallet
Bridge	Grand slam, no trump, rubber
Wrestling	Half nelson, full nelson, take down
Kabaddi	Raid, cant, lona
Shooting	Bull's eye, plinker
Baseball	Home run, pitcher, strike
Billiards / snooker	Cannon, cue, pot, break

Sport	Field / numbers
Cricket	11 players; pitch 22 yards (20.12 m); ball 5.5–5.75 oz; stumps 28 inches high
Football	11 players; 90 minutes (two halves of 45)
Hockey	11 players; 60 minutes (four quarters of 15)
Basketball	5 players
Volleyball	6 players
Kabaddi	7 players
Kho-kho	9 players (12 in a team)

Sport	Field / numbers
Polo	4 players
Water polo	7 players
Netball	7 players
Rugby (union)	15 players
Baseball	9 players
Badminton/tennis	Singles 1, doubles 2

- Origins: hockey, polo (modern) – India/Persia; chess – India (Chaturanga); badminton – India (Poona); kabaddi – India; cricket – England; football – England (modern); basketball – USA (James Naismith); volleyball – USA (William Morgan); judo – Japan; table tennis – England; baseball – USA; carrom – India; snooker – India (Jabalpur, British officers); tennis – England (modern lawn tennis).
- National games: India – no officially declared national game (hockey popularly considered); USA – baseball; Canada – ice hockey (winter) & lacrosse (summer); Japan – sumo/judo; China – table tennis; Pakistan – hockey; Bangladesh – kabaddi; Sri Lanka – volleyball; Spain – bullfighting; Brazil – football; Australia – cricket; Afghanistan – buzkashi; Russia – chess (unofficial); Scotland – rugby.

Cricket pitch 22 yards | Kho-kho 9, Kabaddi 7, Polo 4, Volleyball 6, Basketball 5 | Chess & badminton – India origin | Basketball – Naismith

★ **PYQ-TREND:** Number of players per team (kabaddi 7, kho-kho 9, polo 4, volleyball 6, netball 7). Terms belonging to sports (googly – cricket; bully – hockey; chukker – polo; birdie – golf; deuce – tennis; castling – chess). Origin of games. National games of countries. India has no official national game.

13.3 Trophies, Cups and Stadiums

Trophy / cup	Sport
Ranji Trophy, Duleep Trophy, Irani Cup, Deodhar Trophy, Vijay Hazare, Syed Mushtaq Ali, C.K. Nayudu, Vizzy Trophy	Cricket (domestic)
Ashes	Cricket (England–Australia)
Border–Gavaskar Trophy	Cricket (India–Australia)
Anderson–Tendulkar Trophy (earlier Pataudi Trophy)	Cricket (India–England Tests)
Durand Cup (oldest football tournament in Asia, 1888), Santosh Trophy, Federation Cup, Subroto Cup (schools), Rovers Cup, IFA Shield	Football
Beighton Cup (oldest hockey tournament), Rangaswami Cup, Aga Khan Cup, Nehru Trophy, Dhyan Chand Trophy	Hockey
Davis Cup (men), Billie Jean King Cup (women; formerly Fed Cup)	Tennis
Thomas Cup (men), Uber Cup (women), Sudirman Cup (mixed)	Badminton
Swaythling Cup (men), Corbillon Cup (women)	Table tennis
Ryder Cup, Walker Cup	Golf
Stanley Cup	Ice hockey

Trophy / cup	Sport
America's Cup	Yachting
Westchester Cup	Polo
Grand Prix	Motor racing (Formula 1)
FIFA World Cup (every 4 years), UEFA Champions League, Copa América	Football

- Grand Slams (tennis): Australian Open (hard court, January), French Open/Roland Garros (clay, May–June), Wimbledon (grass, oldest – 1877, June–July), US Open (hard court, Aug–Sept).
- Cricket World Cups: first ODI World Cup 1975 (England; won by West Indies). India won ODI World Cup in 1983 (Kapil Dev, Lord's) and 2011 (M.S. Dhoni, Mumbai); T20 World Cup 2007 (Dhoni) and 2024 (Rohit Sharma, Barbados); Champions Trophy 2002 (shared), 2013, 2025. 2023 ODI World Cup in India – won by Australia.
- FIFA: first World Cup 1930 (Uruguay – hosted & won). Brazil has most titles (5). 2022 Qatar – won by Argentina (Messi). 2026 – USA, Canada, Mexico (48 teams).

Stadium	City
Eden Gardens	Kolkata (cricket)
Narendra Modi Stadium (Motera)	Ahmedabad – largest cricket stadium in the world
Wankhede, Brabourne	Mumbai
M. Chinnaswamy	Bengaluru
M.A. Chidambaram (Chepauk)	Chennai
Arun Jaitley (Feroz Shah Kotla)	Delhi
Jawaharlal Nehru Stadium	Delhi (also Kochi, Chennai)
Salt Lake (Yuva Bharati Krirangan)	Kolkata – football
Green Park	Kanpur
Barabati	Cuttack
Rajiv Gandhi International	Hyderabad
Kalinga Stadium	Bhubaneswar – hockey
Birsa Munda International Hockey Stadium	Rourkela – largest hockey stadium
Major Dhyan Chand National Stadium	Delhi – hockey
Lord's	London – 'Mecca of cricket'
Maracanã	Rio de Janeiro – football

Ranji – cricket | Durand – football (1888) | Beighton – hockey | Davis – tennis | Thomas & Uber – badminton | Wimbledon – grass, oldest | Largest cricket stadium – Narendra Modi Stadium

★ **PYQ-TREND:** Trophy–sport matching (Ranji, Durand, Santosh, Beighton, Thomas, Uber, Swaythling, Davis, Ryder). Grand Slam surfaces. India's World Cup wins and captains. Stadium–city matching (Eden Gardens, Wankhede, Kalinga, Chepauk). Oldest football tournament in Asia (Durand).

✖ **COMMON MISTAKE:** Sports winners change every season. Recheck the last 12 months' winners (IPL, Wimbledon, Thomas Cup, World Cups) from current affairs.

13.4 Famous Indian Sportspersons and Nicknames

Player	Sport / identity
Sachin Tendulkar	Cricket – 'Master Blaster', 100 international centuries; Bharat Ratna 2014 (youngest & first sportsperson)
Sunil Gavaskar	Cricket – 'Little Master'; first to 10,000 Test runs
Kapil Dev	Cricket – 1983 World Cup captain; 'Haryana Hurricane'
M.S. Dhoni	Cricket – 'Captain Cool'; three ICC trophies
Virat Kohli	Cricket – 'Run Machine'
Rahul Dravid	Cricket – 'The Wall'
Sourav Ganguly	Cricket – 'Dada', 'Prince of Kolkata'
Anil Kumble	Cricket – 10 wickets in a Test innings (1999, vs Pakistan, Delhi)
Milkha Singh	Athletics – 'Flying Sikh'
P.T. Usha	Athletics – 'Payyoli Express', 'Golden Girl'; 4th at 1984 Olympics (400 m hurdles)
Hima Das	Athletics – 'Dhing Express'
Dutee Chand	Athletics – sprinter
Neeraj Chopra	Javelin – Olympic gold 2021, silver 2024
Dhyan Chand	Hockey – 'The Wizard'
Dhanraj Pillay	Hockey
Viswanathan Anand	Chess – India's first Grandmaster (1988); 'Lightning Kid'; 'Tiger of Madras'
D. Gukesh	Chess – youngest undisputed World Champion (2024, aged 18)
Koneru Humpy	Chess – first Indian woman to earn the Grandmaster title (2002); World Rapid Champion
Prakash Padukone	Badminton – first Indian to win All England (1980)
Pullela Gopichand	Badminton – All England 2001; coach
Saina Nehwal	Badminton – Olympic bronze 2012
P.V. Sindhu	Badminton – Olympic silver 2016, bronze 2020; World Champion 2019
Mary Kom	Boxing – 'Magnificent Mary'; six-time world champion
Vijender Singh	Boxing – first Indian Olympic boxing medal (2008)
Leander Paes, Mahesh Bhupathi, Sania Mirza, Rohan Bopanna	Tennis
Pankaj Advani	Billiards & snooker – multiple world titles
Geet Sethi	Billiards
Abhinav Bindra, Gagan Narang, Manu Bhaker	Shooting
Sushil Kumar, Yogeshwar Dutt, Bajrang Punia, Vinesh Phogat, Sakshi Malik	Wrestling
Mirabai Chanu, Karnam Malleswari	Weightlifting
Dipa Karmakar	Gymnastics – Produnova vault, 4th at Rio 2016
Aditi Ashok	Golf
Arjun Atwal, Jeev Milkha Singh	Golf

Player	Sport / identity
Narain Karthikeyan	First Indian F1 driver
Bachendri Pal	First Indian woman to climb Everest (1984)
Tenzing Norgay & Edmund Hillary	First to climb Everest (29 May 1953)
Arunima Sinha	First woman amputee to climb Everest (2013)
Mihir Sen	First Indian to swim the English Channel (1958)
Arati Saha	First Asian woman to swim the English Channel (1959)

- National sports awards: Major Dhyan Chand Khel Ratna (highest; first awardee Viswanathan Anand, 1991–92), Arjuna Award (1961; bronze statuette of Arjuna), Dronacharya Award (coaches, 1985), Arjuna Award (Lifetime) for veterans, Rashtriya Khel Protsahan Puraskar, Maulana Abul Kalam Azad (MAKA) Trophy (university). Khelo India (2018).

Flying Sikh – Milkha Singh | Payyoli Express – PT Usha | Wizard – Dhyan Chand | First Indian GM – Anand | First Indian woman on Everest – Bachendri Pal 1984 | Khel Ratna first – Anand

★ **PYQ-TREND:** Nicknames (Flying Sikh, Payyoli Express, Dhing Express, Master Blaster, Little Master, The Wall). Sport of famous players (Pankaj Advani – billiards, Dipa Karmakar – gymnastics, Mary Kom – boxing, Mirabai – weightlifting). First Indian woman on Everest. First Grandmaster. Arjuna/Dronacharya award categories. D. Gukesh world champion.

CHAPTER 14

Static GK II – Books, Awards, Days, Firsts, Organisations

14.1 National Symbols of India

Symbol	Fact
National Flag	Tiranga; ratio 3:2; saffron (courage), white (peace, truth), green (faith, fertility); Ashoka Chakra navy blue with 24 spokes; designed by Pingali Venkayya; adopted 22 July 1947; Flag Code 2002 (amended 2022 – polyester/machine-made allowed, 'Har Ghar Tiranga')
National Emblem	Lion Capital of Ashoka at Sarnath; adopted 26 Jan 1950; 'Satyameva Jayate' (Mundaka Upanishad) in Devanagari; 4 lions (3 visible), bull (left), horse (right), wheel
National Anthem	Jana Gana Mana – Rabindranath Tagore; adopted 24 Jan 1950; first sung 27 Dec 1911 (Calcutta INC); playing time ≈ 52 seconds (short version ≈ 20 s); originally in Bengali (Bharoto Bhagyo Bidhata)
National Song	Vande Mataram – Bankim Chandra Chatterjee, from Anandamath (1882); first sung by Tagore at 1896 Calcutta INC session; equal status with anthem
National Calendar	Saka calendar (from 22 March 1957); Chaitra first month
National Animal	Tiger (Panthera tigris), since 1973 (earlier lion)
National Heritage Animal	Elephant (2010)
National Bird	Indian Peacock (Pavo cristatus), 1963
National Flower	Lotus (Nelumbo nucifera)
National Tree	Banyan (Ficus benghalensis)
National Fruit	Mango
National Aquatic Animal	Ganges river dolphin (2009)
National River	Ganga (2008)
National Reptile	King cobra
National Currency symbol	₹ – designed by D. Udaya Kumar (2010)
National Micro-organism	Lactobacillus delbrueckii (2012)
Pledge	Written by P.V. Subba Rao (1962, Telugu)

Flag 3:2, 24 spokes, 22 July 1947 | Anthem 24 Jan 1950 (52 s) | Emblem – Sarnath, 26 Jan 1950 | National calendar – Saka (22 Mar 1957) | Tiger 1973 | Dolphin 2009 | ₹ symbol 2010

★ **PYQ-TREND:** Flag ratio and spokes, designer Pingali Venkayya. Anthem duration (52 s) and first singing. Source of 'Satyameva Jayate'. National aquatic animal (dolphin), heritage animal (elephant). ₹ symbol designer. Vande Mataram source (Anandamath).

14.2 Important Days

Date	Day
9 January	Pravasi Bharatiya Divas (Gandhi's return 1915)
12 January	National Youth Day (Vivekananda's birthday)
15 January	Army Day (K.M. Cariappa took over 1949)
23 January	Parakram Divas (Subhas Chandra Bose)
24 January	National Girl Child Day
25 January	National Voters' Day; National Tourism Day
26 January	Republic Day; International Customs Day
30 January	Martyrs' Day (Shaheed Diwas); Anti-Leprosy Day
4 February	World Cancer Day
13 February	World Radio Day; National Women's Day (Sarojini Naidu's birthday)
21 February	International Mother Language Day
24 February	Central Excise Day
28 February	National Science Day (Raman effect, 1928)
8 March	International Women's Day
15 March	World Consumer Rights Day
21 March	World Forestry Day; International Day of Forests
22 March	World Water Day
23 March	World Meteorological Day; Shaheed Diwas (Bhagat Singh, Rajguru, Sukhdev)
24 March	World TB Day
7 April	World Health Day (WHO founded 1948)
13 April	Jallianwala Bagh Day
14 April	Ambedkar Jayanti
18 April	World Heritage Day
21 April	Civil Services Day
22 April	Earth Day
23 April	World Book & Copyright Day
24 April	National Panchayati Raj Day
1 May	International Labour Day; Maharashtra & Gujarat Day
3 May	World Press Freedom Day
8 May	World Red Cross Day (Henry Dunant's birthday)
11 May	National Technology Day (Pokhran-II, 1998)
15 May	International Day of Families
21 May	Anti-Terrorism Day (Rajiv Gandhi's death anniversary)
22 May	International Day for Biological Diversity
31 May	World No Tobacco Day
1 June	World Milk Day

Date	Day
5 June	World Environment Day
8 June	World Oceans Day
12 June	World Day Against Child Labour
14 June	World Blood Donor Day
21 June	International Yoga Day (from 2015)
1 July	National Doctors' Day (B.C. Roy); GST Day; CA Day
11 July	World Population Day
26 July	Kargil Vijay Diwas (1999)
29 July	International Tiger Day
7 August	National Handloom Day (Swadeshi movement 1905)
9 August	Quit India Day; International Day of the World's Indigenous Peoples
12 August	International Youth Day; World Elephant Day
15 August	Independence Day
29 August	National Sports Day (Dhyan Chand)
5 September	Teachers' Day (S. Radhakrishnan)
8 September	International Literacy Day
14 September	Hindi Divas
15 September	Engineers' Day (M. Visvesvaraya)
16 September	World Ozone Day
27 September	World Tourism Day
1 October	International Day of Older Persons
2 October	Gandhi Jayanti; International Day of Non-Violence
8 October	Indian Air Force Day
10 October	World Mental Health Day; National Postal Day (9 Oct World Post Day)
16 October	World Food Day (FAO)
24 October	United Nations Day
31 October	Rashtriya Ekta Diwas (Sardar Patel)
11 November	National Education Day (Maulana Azad)
14 November	Children's Day (Nehru); World Diabetes Day
19 November	International Men's Day; World Toilet Day
26 November	Constitution Day (Samvidhan Divas); National Milk Day (Verghese Kurien)
1 December	World AIDS Day
2 December	National Pollution Control Day (Bhopal)
3 December	International Day of Persons with Disabilities
4 December	Indian Navy Day
7 December	Armed Forces Flag Day
10 December	Human Rights Day
14 December	National Energy Conservation Day

Date	Day
16 December	Vijay Diwas (1971 war)
22 December	National Mathematics Day (Ramanujan)
23 December	Kisan Diwas (Charan Singh)
24 December	National Consumer Day
25 December	Good Governance Day (Vajpayee)

28 Feb – Science | 5 Jun – Environment | 21 Jun – Yoga | 1 Jul – Doctors | 29 Aug – Sports | 5 Sep – Teachers | 15 Sep – Engineers | 31 Oct – Ekta | 26 Nov – Constitution | 22 Dec – Mathematics

★ **PYQ-TREND:** National Science Day reason, Engineers' Day (Visvesvaraya), Teachers' Day, National Sports Day, Constitution Day, Yoga Day, World Environment Day, Kargil Vijay Diwas, Army/Navy/Air Force days, National Youth Day, Good Governance Day, Rashtriya Ekta Diwas.

14.3 Famous Books and Authors

Book	Author
Arthashastra	Kautilya
Indica	Megasthenes
Abhijnanashakuntalam, Meghaduta, Raghuvamsha	Kalidasa
Mudrarakshasa	Vishakhadatta
Mrichchhakatika	Shudraka
Panchatantra	Vishnu Sharma
Kamasutra	Vatsyayana
Rajatarangini	Kalhana (history of Kashmir; first historical chronicle)
Gita Govinda	Jayadeva
Prithviraj Raso	Chand Bardai
Ain-i-Akbari, Akbarnama	Abul Fazl
Baburnama (Tuzuk-i-Baburi)	Babur
Humayunnama	Gulbadan Begum
Padmavat	Malik Muhammad Jayasi
Ramcharitmanas	Tulsidas
Gitanjali, Gora, Ghare Baire, Kabuliwala	Rabindranath Tagore
Anandamath, Durgeshnandini	Bankim Chandra Chatterjee
Devdas, Parineeta, Srikanta	Sarat Chandra Chattopadhyay
Godan, Gaban, Nirmala, Rangbhoomi, Kafan	Munshi Premchand
Madhushala	Harivansh Rai Bachchan
Kamayani	Jaishankar Prasad
Rashmirathi, Urvashi	Ramdhari Singh 'Dinkar'
Hind Swaraj, My Experiments with Truth	Mahatma Gandhi
Discovery of India, Glimpses of World History, An Autobiography	Jawaharlal Nehru
Annihilation of Caste, Who Were the Shudras	B.R. Ambedkar
Wings of Fire, Ignited Minds, India 2020	A.P.J. Abdul Kalam (Wings of Fire with Arun Tiwari)

Book	Author
Poverty and Un-British Rule in India	Dadabhai Naoroji
India Wins Freedom	Maulana Abul Kalam Azad
Unhappy India	Lala Lajpat Rai
The Indian Struggle	Subhas Chandra Bose
Gita Rahasya	Bal Gangadhar Tilak
Why I Am an Atheist	Bhagat Singh
Train to Pakistan, Delhi	Khushwant Singh
Malgudi Days, The Guide, Swami and Friends	R.K. Narayan
Untouchable, Coolie	Mulk Raj Anand
The God of Small Things	Arundhati Roy (Booker 1997)
Midnight's Children	Salman Rushdie (Booker 1981, Booker of Bookers)
The White Tiger	Aravind Adiga (Booker 2008)
The Inheritance of Loss	Kiran Desai (Booker 2006)
Tomb of Sand (Ret Samadhi)	Geetanjali Shree (International Booker 2022; translated by Daisy Rockwell)
A Suitable Boy	Vikram Seth
The Argumentative Indian, Development as Freedom, Idea of Justice	Amartya Sen
India Divided	Rajendra Prasad
The Last Mughal, White Mughals	William Dalrymple
A Passage to India	E.M. Forster
The Jungle Book, Kim	Rudyard Kipling (Nobel 1907; born in Bombay)
War and Peace, Anna Karenina	Leo Tolstoy
Das Kapital, Communist Manifesto (with Engels)	Karl Marx
The Republic	Plato
Politics	Aristotle
The Prince	Niccolò Machiavelli
Origin of Species	Charles Darwin
Hamlet, Macbeth, Othello, Romeo and Juliet	William Shakespeare
Mein Kampf	Adolf Hitler
A Brief History of Time	Stephen Hawking
Long Walk to Freedom	Nelson Mandela
The Old Man and the Sea	Ernest Hemingway
Harry Potter series	J.K. Rowling
Gulliver's Travels	Jonathan Swift
Don Quixote	Miguel de Cervantes
1984, Animal Farm	George Orwell (born Motihari, Bihar)
Dreams from My Father, A Promised Land	Barack Obama
Playing It My Way	Sachin Tendulkar
The Test of My Life	Yuvraj Singh
Unbreakable	Mary Kom
The Race of My Life	Milkha Singh

Book	Author
Ace Against Odds	Sania Mirza
Straight from the Heart	Kapil Dev
Sunny Days	Sunil Gavaskar
281 and Beyond	V.V.S. Laxman
The Insider	P.V. Narasimha Rao
Exam Warriors	Narendra Modi

★ **PYQ-TREND:** Book–author matching (very frequent): Rajatarangini, Gita Govinda, Anandamath, Godan, Discovery of India, Wings of Fire, Midnight's Children, Tomb of Sand, India Wins Freedom, Untouchable. Sports autobiographies (Playing It My Way, Unbreakable, Race of My Life). Booker-winning Indians.

14.4 Awards and Honours

- Bharat Ratna (instituted 2 January 1954): highest civilian award; maximum 3 per year (exceeded in 2024 with 5). First recipients (1954): C. Rajagopalachari, S. Radhakrishnan, C.V. Raman. First posthumous – Lal Bahadur Shastri (1966); first woman – Indira Gandhi (1971); first foreigner – Khan Abdul Ghaffar Khan (1987, 'Frontier Gandhi'); Nelson Mandela (1990); first sportsperson – Sachin Tendulkar (2014); 2024 awardees – Karpoori Thakur, L.K. Advani, P.V. Narasimha Rao, Chaudhary Charan Singh, M.S. Swaminathan. It is a medallion shaped like a peepal leaf.
- Padma awards (1954): Padma Vibhushan > Padma Bhushan > Padma Shri; announced on Republic Day eve.
- Gallantry (wartime): Param Vir Chakra (highest; first – Major Somnath Sharma, 1947, posthumous; designed by Savitri Khanolkar), Maha Vir Chakra, Vir Chakra. Peacetime: Ashoka Chakra, Kirti Chakra, Shaurya Chakra.
- Literature: Jnanpith (1961; first awarded 1965 – G. Sankara Kurup, Malayalam, 'Odakkuzhal'; first woman – Ashapurna Devi, 1976, Bengali); Sahitya Akademi Award (1954); Saraswati Samman (K.K. Birla Foundation, 1991); Vyas Samman (Hindi); Booker Prize (UK); Pulitzer Prize (USA, journalism & literature).
- Cinema: Dadasaheb Phalke Award (1969; highest in Indian cinema; first – Devika Rani, 'First Lady of Indian Cinema'); National Film Awards; Oscars (Academy Awards, USA – first Indian winner Bhanu Athaiya, costume design, Gandhi 1983; Satyajit Ray honorary Oscar 1992; A.R. Rahman – 2 Oscars (Slumdog Millionaire, 2009); 'Naatu Naatu' (RRR) – best original song 2023; 'The Elephant Whisperers' – documentary short 2023).
- Nobel Prize (1901; Alfred Nobel – inventor of dynamite; awarded in Stockholm, Peace Prize in Oslo; economics added 1968 (Sveriges Riksbank prize)). Indians/Indian-origin laureates: Rabindranath Tagore (Literature 1913 – first Indian & first Asian), C.V. Raman (Physics 1930), Har Gobind Khorana (Medicine 1968), Mother Teresa (Peace 1979), S. Chandrasekhar (Physics 1983), Amartya Sen (Economics 1998), V.S. Naipaul (Literature 2001), Venkatraman Ramakrishnan (Chemistry 2009), Kailash Satyarthi (Peace 2014, with Malala Yousafzai), Abhijit Banerjee (Economics 2019).
- Ramon Magsaysay Award (Philippines; 'Asia's Nobel'): Vinoba Bhave (first Indian, 1958), Mother Teresa, Kiran Bedi, Arvind Kejriwal, Ravish Kumar, etc.
- Other: Templeton Prize (religion), Pritzker (architecture; B.V. Doshi first Indian, 2018), Abel Prize & Fields Medal (mathematics; Fields every 4 years, under 40 – Manjul Bhargava, Indian-origin, 2014), Kalinga Prize (UNESCO, popularisation of science; instituted by Biju

Patnaik), Gandhi Peace Prize (1995; first – Julius Nyerere), Indira Gandhi Peace Prize, Shanti Swarup Bhatnagar Prize (science; now replaced by Rashtriya Vigyan Puraskar from 2023, highest – Vigyan Ratna), Borlaug Award (agriculture), Tansen Samman (music), Kalidas Samman (MP), Lal Bahadur Shastri award, Hoover Medal (engineering for humanity).

Bharat Ratna 1954 (first: Rajaji, Radhakrishnan, Raman) | Jnanpith first – G. Sankara Kurup 1965 | Phalke first – Devika Rani 1969 | First Indian Nobel – Tagore 1913 | First Oscar – Bhanu Athaiya | Param Vir Chakra first – Somnath Sharma

★ **PYQ-TREND:** First Bharat Ratna recipients, first woman/foreigner/sportsperson. Jnanpith first recipient and language. Dadasaheb Phalke first recipient. Indian Nobel laureates with fields and years. Magsaysay as Asia's Nobel. Fields medal field (mathematics). Pritzker (architecture). Gallantry awards order. Recent Bharat Ratna (2024).

✖ **COMMON MISTAKE:** Every year new Padma, Phalke, Jnanpith, Nobel and Booker winners are announced. Check the last year's list from current affairs.

14.5 Indian Firsts

First	Person
President	Dr. Rajendra Prasad
Vice-President	Dr. S. Radhakrishnan
Prime Minister	Jawaharlal Nehru
Governor-General of free India	Lord Mountbatten (Indian: C. Rajagopalachari)
Chief Justice of India	H.J. Kania
Woman President	Pratibha Patil
Woman PM	Indira Gandhi
Woman Governor	Sarojini Naidu (UP, 1947)
Woman CM	Sucheta Kripalani (UP, 1963)
Woman Union Minister	Rajkumari Amrit Kaur (Health)
Woman Supreme Court judge	Fathima Beevi (1989)
Woman Speaker of Lok Sabha	Meira Kumar
Woman IPS officer	Kiran Bedi (1972)
Woman IAS officer	Anna Rajam Malhotra
Woman astronaut (Indian-origin)	Kalpana Chawla (Columbia, 1997; died 2003)
Indian in space	Rakesh Sharma (Soyuz T-11, 1984; 'Saare Jahan Se Achha')
Indian on the ISS	Shubhanshu Shukla (Axiom-4 mission, 2025)
Woman to climb Everest	Bachendri Pal
Miss Universe	Sushmita Sen (1994)
Miss World	Reita Faria (1966)
Woman to win Nobel (in India)	Mother Teresa
Muslim President	Dr. Zakir Hussain
Chief Election Commissioner	Sukumar Sen
Field Marshal	Sam Manekshaw (1973); second – K.M. Cariappa
Commander-in-Chief (Indian)	K.M. Cariappa
Indian to swim English Channel	Mihir Sen

First	Person
Indian Grandmaster	Viswanathan Anand
Indian ICS officer	Satyendranath Tagore (1863/64)
Woman doctor (Western medicine)	Anandi Gopal Joshi / Kadambini Ganguly
Woman teacher	Savitribai Phule
Woman pilot	Sarla Thakral (1936)
Woman fighter pilots	Avani Chaturvedi, Bhawana Kanth, Mohana Singh (2016)
Indian woman in Antarctica	Mehar Moosa (1976)
Indian woman to win Olympic medal	Karnam Malleswari
Indian talkie film	Alam Ara (1931; Ardeshir Irani)
Indian feature film	Raja Harishchandra (1913; Dadasaheb Phalke)
Indian colour film	Kisan Kanya (1937)
Indian satellite	Aryabhata (19 April 1975, Soviet rocket)
Indian-built launch from India (satellite)	Rohini (SLV-3, 1980)
Nuclear test	Pokhran, 18 May 1974 ('Smiling Buddha'); Pokhran-II 11 & 13 May 1998 (Operation Shakti)
Indian bank	Bank of Hindostan (1770)
Newspaper	Hicky's Bengal Gazette (1780; James Augustus Hicky)
Indian university (modern)	Calcutta, Bombay, Madras (1857)
Railway	Bombay–Thane (1853)
Metro	Kolkata (1984)
Airline	Tata Airlines (1932; J.R.D. Tata first licensed pilot)
Telegraph line	Calcutta–Diamond Harbour (1851)
Television	Delhi (1959)
Indian woman to preside over the UN General Assembly	Vijaya Lakshmi Pandit (1953)

First Indian in space – Rakesh Sharma 1984 | First satellite – Aryabhata 1975 | First talkie – Alam Ara 1931 | First film – Raja Harishchandra 1913 | First woman CM – Sucheta Kripalani | First woman Governor – Sarojini Naidu | First Field Marshal – Manekshaw

★ **PYQ-TREND:** Women firsts (CM, Governor, SC judge, IPS, Union minister). First Indian in space. First talkie and feature film. First satellite. Field Marshals. First CJI. Pokhran tests. First newspaper.

14.6 International Organisations

Organisation	HQ	Key facts
United Nations	New York	Founded 24 Oct 1945; 193 members; Charter signed in San Francisco (June 1945); six principal organs; Security Council – 15 members (5 permanent with veto: USA, UK, France, Russia, China); India founding member
International Court of Justice	The Hague	15 judges, 9-year term

Organisation	HQ	Key facts
UNESCO	Paris	Education, science, culture; World Heritage
UNICEF	New York	Children (1946); Nobel 1965
WHO	Geneva	7 April 1948
ILO	Geneva	1919; Nobel 1969
WTO	Geneva	1 Jan 1995
FAO	Rome	Food & agriculture (16 Oct)
IFAD, WFP	Rome	Agriculture development; World Food Programme (Nobel 2020)
IMF, World Bank	Washington DC	Bretton Woods 1944
IAEA	Vienna	Atomic energy ('Atoms for Peace')
UNIDO, OPEC	Vienna	–
UNEP	Nairobi	Environment (1972)
UN-Habitat	Nairobi	–
UNHCR	Geneva	Refugees (Nobel twice)
Red Cross (ICRC)	Geneva	Henry Dunant, 1863; Nobel three times
Interpol	Lyon (France)	Police cooperation
Amnesty International	London	Human rights
WWF	Gland (Switzerland)	Panda logo
Greenpeace	Amsterdam	Environment
IUCN	Gland	Red List
NATO	Brussels	1949; military alliance; latest members Finland (2023), Sweden (2024)
European Union	Brussels	Euro currency
ASEAN	Jakarta	1967 (Bangkok Declaration); 10 long-standing members, Timor-Leste being admitted as the 11th
SAARC	Kathmandu	1985 (Dhaka); 8 members (Afghanistan joined 2007)
BIMSTEC	Dhaka	1997; 7 members
SCO	Beijing	2001; India joined 2017
BRICS	– (NDB in Shanghai)	Brazil, Russia, India, China, South Africa + new members (Egypt, Ethiopia, Iran, UAE, Indonesia)
G20	No permanent HQ	India presided 2023 (New Delhi summit; African Union admitted as member)
Commonwealth	London	56 members
African Union	Addis Ababa	–
APEC	Singapore	India not a member
OECD	Paris	–

Organisation	HQ	Key facts
OIC	Jeddah	–
Asian Development Bank	Manila	1966
AIIB	Beijing	2016
International Solar Alliance	Gurugram (India)	2015; India–France initiative; first international body with HQ in India
International Big Cat Alliance	India	2023
World Economic Forum	Cologny/Geneva	Davos meetings
FATF	Paris	Money laundering
International Olympic Committee	Lausanne	–
FIFA	Zurich	–
ICC (cricket)	Dubai	–
International Criminal Court	The Hague	–

- UN Secretaries-General: Trygve Lie (first), Dag Hammarskjöld, U Thant, Kurt Waldheim, Javier Pérez de Cuéllar, Boutros Boutros-Ghali, Kofi Annan, Ban Ki-moon, António Guterres (term to 2026).
- UN official languages (6): English, French, Spanish, Russian, Chinese, Arabic.

UN 24 Oct 1945 (193 members) | P5 – USA, UK, France, Russia, China | ICJ – The Hague | WHO, ILO, WTO – Geneva | UNESCO – Paris | FAO – Rome | IAEA – Vienna | UNEP – Nairobi | ISA – Gurugram

★ **PYQ-TREND:** HQ of organisations (WHO, UNESCO, UNEP, IAEA, FAO, Interpol, Amnesty, WWF, ASEAN, SAARC). UN Security Council P5. UN founding date. First international organisation with HQ in India (ISA). SAARC members. G20 presidency of India. UN official languages.

14.7 Indian Organisations, HQs and Defence

Organisation	HQ
ISRO	Bengaluru
DRDO	New Delhi
BARC	Mumbai (Trombay)
RBI, SEBI, NABARD, BSE, NSE	Mumbai
SIDBI	Lucknow
NITI Aayog, ICAR, ICMR, CSIR	New Delhi
Indian Institute of Science	Bengaluru
Botanical Survey of India	Kolkata
Zoological Survey of India	Kolkata
Geological Survey of India	Kolkata
Survey of India	Dehradun
Archaeological Survey of India	New Delhi
Forest Research Institute	Dehradun
Central Rice Research Institute	Cuttack
Central Drug Research Institute	Lucknow
National Dairy Research Institute	Karnal

Organisation	HQ
Central Potato Research Institute	Shimla
National Institute of Oceanography	Goa
Indian Institute of Astrophysics	Bengaluru
Physical Research Laboratory	Ahmedabad
National Remote Sensing Centre	Hyderabad
Vikram Sarabhai Space Centre	Thiruvananthapuram
Satish Dhawan Space Centre (launch site)	Sriharikota, Andhra Pradesh
Indian Meteorological Department	New Delhi (founded 1875, Calcutta)
Coal India Ltd.	Kolkata
ONGC	New Delhi (registered), Dehradun
BHEL, GAIL, NTPC, SAIL	New Delhi
HAL	Bengaluru
Bharat Electronics	Bengaluru
Indian Railways zones	– (see Geography)

- Defence: Army (HQ New Delhi; 7 commands), Navy (Indian Navy Day 4 Dec – Operation Trident 1971), Air Force (founded 8 October 1932). President is the Supreme Commander. Chief of Defence Staff post created 2019 (first – Gen. Bipin Rawat, 1 Jan 2020). Department of Military Affairs (2020).
- Training: National Defence Academy (Khadakwasla, Pune), Indian Military Academy (Dehradun), Air Force Academy (Dundigal, Hyderabad), Indian Naval Academy (Ezhimala, Kerala), Officers Training Academy (Chennai), Defence Services Staff College (Wellington), National Defence College (New Delhi), High Altitude Warfare School (Gulmarg), Counter Insurgency & Jungle Warfare School (Vairengte, Mizoram).
- Paramilitary/CAPFs: BSF (1965; borders with Pakistan, Bangladesh), ITBP (1962; China border), SSB (Nepal & Bhutan borders), Assam Rifles (oldest – 1835; Myanmar border; 'Sentinels of the North-East'), CRPF (largest CAPF; 1939), CISF (industrial security, airports, metros), NSG ('Black Cats', 1984), Coast Guard (1977; under Ministry of Defence).
- Missiles (IGMDP – Dr. Kalam): Prithvi (surface-to-surface), Agni (ballistic; Agni-V ICBM range > 5,000 km; MIRV test 'Mission Divyastra' 2024), Trishul, Akash (surface-to-air), Nag (anti-tank), BrahMos (supersonic cruise – India–Russia; Brahmaputra + Moskva), Astra (air-to-air), Nirbhay (subsonic cruise), Dhanush (ship-launched), K-series (submarine-launched). Rafale (France), Tejas (indigenous Light Combat Aircraft – HAL), Sukhoi-30 MKI (Russia). Aircraft carriers: INS Vikramaditya, INS Vikrant (first indigenous, commissioned 2022). Nuclear submarines: INS Arihant (first indigenous SSBN), INS Arighaat.
- Military exercises (examples): Malabar (India–USA–Japan–Australia), Yudh Abhyas (USA), Garuda & Shakti (France), Indra (Russia), Hand-in-Hand (China, earlier), Mitra Shakti (Sri Lanka), Surya Kiran (Nepal), Sampriti (Bangladesh), Nomadic Elephant (Mongolia), SIMBEX (Singapore), Varuna (France – navy), Dharma Guardian (Japan), AUSINDEX (Australia), Ekuverin (Maldives), Tarang Shakti (multinational air exercise hosted by India, 2024).

ISRO – Bengaluru | BARC – Mumbai | Sriharikota – launch site | IAF founded 8 Oct 1932 | First CDS – Bipin Rawat | Assam Rifles – oldest CAPF | BSF 1965 | INS Vikrant – first indigenous carrier | BrahMos – India + Russia

★ **PYQ-TREND:** HQs of Indian institutes (ISRO, BARC, CRRI Cuttack, NDRI Karnal, Survey of India Dehradun, BSI/ZSI Kolkata). Defence academies' locations. Missile types (Prithvi, Agni, Akash, Nag, BrahMos). Military exercise partners (Malabar, Yudh Abhyas, Surya Kiran, Mitra Shakti, Sampriti). CAPF border responsibilities. First CDS.

14.8 Inventions and Discoveries

Invention / discovery	Inventor / discoverer
Telephone	Alexander Graham Bell (1876)
Electric bulb, phonograph, motion-picture camera	Thomas Alva Edison
Radio	Guglielmo Marconi (also J.C. Bose – wireless demonstration)
Television	John Logie Baird
Aeroplane	Wright brothers (1903)
Steam engine (improved)	James Watt (Thomas Newcomen earlier)
Railway locomotive	George Stephenson
Printing press (movable type)	Johannes Gutenberg
Dynamite	Alfred Nobel
Computer (analytical engine)	Charles Babbage ('father of computer')
World Wide Web	Tim Berners-Lee (1989)
Penicillin	Alexander Fleming (1928)
Vaccination (smallpox)	Edward Jenner
Pasteurisation, rabies vaccine	Louis Pasteur
Polio vaccine	Jonas Salk (injectable), Albert Sabin (oral)
X-rays	Wilhelm Röntgen
Radioactivity	Henri Becquerel
Radium, polonium	Marie & Pierre Curie (Marie – Nobel in Physics & Chemistry)
Electron	J.J. Thomson
Neutron	James Chadwick
Proton / nucleus	Rutherford
Theory of relativity	Albert Einstein
Law of gravitation, laws of motion	Isaac Newton
Telescope (astronomical use)	Galileo Galilei
Microscope	Zacharias Janssen; Leeuwenhoek improved
Thermometer	Galileo; mercury thermometer – Fahrenheit
Barometer	Evangelista Torricelli
Dynamo / electromagnetic induction	Michael Faraday
AC motor, Tesla coil	Nikola Tesla
Diesel engine	Rudolf Diesel
Petrol car (motor car)	Karl Benz
Bicycle	Kirkpatrick Macmillan
Sewing machine	Elias Howe (Thimonnier earlier)
Safety lamp (miners)	Humphry Davy

Invention / discovery	Inventor / discoverer
Lift (safety elevator)	Elisha Otis
Ballpoint pen	László Bíró
Revolver	Samuel Colt
Machine gun	Richard Gatling (Gatling gun); Hiram Maxim
Refrigerator	Jacob Perkins / Carl von Linde
Air conditioner	Willis Carrier
Mobile phone	Martin Cooper (Motorola, 1973)
Electric battery	Alessandro Volta
Periodic table	Dmitri Mendeleev
Oxygen	Joseph Priestley / Carl Scheele
Hydrogen	Henry Cavendish
Insulin (discovery)	Frederick Banting & Charles Best
Blood circulation	William Harvey
Blood groups	Karl Landsteiner
Cholera & TB bacteria	Robert Koch
Malaria parasite in mosquito	Ronald Ross
DNA structure	Watson & Crick
Laws of heredity	Gregor Mendel
Zero, decimal system (concept)	Ancient Indian mathematicians (Aryabhata, Brahmagupta)
Raman effect	C.V. Raman
Crescograph (plant growth)	Jagadish Chandra Bose
Bose–Einstein statistics	Satyendra Nath Bose (boson named after him)
Chandrasekhar limit	S. Chandrasekhar
Shampoo (introduced in Britain)	Sake Dean Mahomed
Optical fibre pioneer	Narinder Singh Kapany
USB co-inventor	Ajay Bhatt
Pentium chip	Vinod Dham
Hotmail	Sabeer Bhatia (with Jack Smith)

★ **PYQ-TREND:** Inventor matching (telephone – Bell, radio – Marconi, television – Baird, WWW – Berners-Lee, dynamite – Nobel, printing press – Gutenberg, barometer – Torricelli, penicillin – Fleming, crescograph – J.C. Bose). Indian scientists and their contributions (Raman, J.C. Bose, S.N. Bose, Chandrasekhar).

CHAPTER 15

Science & Technology, Space and Computers

15.1 Indian Space Programme

- INCOSPAR set up in 1962 (Vikram Sarabhai – 'father of the Indian space programme'); ISRO formed 15 August 1969; Department of Space 1972. First rocket launched from Thumba (TERLS, Kerala) on 21 November 1963 (Nike-Apache). Satish Dhawan Space Centre, Sriharikota – main launch site (SHAR). A second spaceport is being built at Kulasekarapattinam (Tamil Nadu) for SSLV.
- Launch vehicles: SLV-3 (1980, Rohini – first Indian launch from Indian soil; project director A.P.J. Abdul Kalam), ASLV, PSLV ('workhorse'; polar & sun-synchronous orbits; record 104 satellites in one launch, Feb 2017; launched Chandrayaan-1 and Mangalyaan), GSLV (geosynchronous; cryogenic upper stage), GSLV Mk III / LVM3 ('Bahubali'; heaviest; launched Chandrayaan-2 & 3; OneWeb satellites), SSLV (small satellites, 2022–23). Agnibaan & Vikram-S – private rockets (Skyroot's Vikram-S – first private launch, Nov 2022).

Mission	Year	Key fact
Aryabhata	1975	First Indian satellite (Soviet Kosmos rocket)
Bhaskara-I	1979	Earth observation
Rohini	1980	First launched by an Indian rocket (SLV-3)
APPLE	1981	First experimental communication satellite
INSAT-1B	1983	Multipurpose (communication, weather)
IRS-1A	1988	First indigenous remote-sensing satellite
Chandrayaan-1	22 Oct 2008	First lunar mission; discovered water molecules on the Moon (Moon Impact Probe); PSLV-C11
Mangalyaan (Mars Orbiter Mission)	Launched 5 Nov 2013; reached Mars 24 Sept 2014	India first Asian nation to reach Mars orbit and first in the world on maiden attempt; cheapest
Astrosat	2015	First dedicated multi-wavelength space observatory
NavIC (IRNSS)	2013–18	Indian regional navigation system (7 satellites; now NVS series)
GSAT-11	2018	Heaviest Indian satellite (launched by Ariane-5)
Chandrayaan-2	22 July 2019	Orbiter works; Vikram lander crash-landed

Mission	Year	Key fact
Chandrayaan-3	Launched 14 July 2023; landed 23 August 2023	First soft landing near the lunar south pole; lander Vikram, rover Pragyan; landing site 'Shiv Shakti point'; 23 August = National Space Day
Aditya-L1	Launched 2 Sept 2023; reached L1 Jan 2024	First Indian solar observatory at Lagrange point L1 (≈ 1.5 million km)
XPoSat	1 Jan 2024	X-ray polarimetry of cosmic sources
SpaDeX	Dec 2024 – Jan 2025	Space docking experiment; India fourth country to dock satellites
Gaganyaan	Upcoming	First Indian human spaceflight; crew module tests; astronaut-designates trained (Shubhanshu Shukla flew to the ISS on Axiom-4 in 2025)
Bharatiya Antariksh Station	Planned (~2035)	Indian space station

- NISAR – NASA–ISRO SAR satellite (2025). Shukrayaan (Venus orbiter) planned. Chandrayaan-4 (sample return) planned.
- Satellites types: geostationary (≈ 36,000 km, 24 h – communication, weather: INSAT/GSAT), polar/sun-synchronous (≈ 500–900 km – remote sensing, mapping: IRS, Cartosat, Oceansat, RISAT).
- ISRO chairmen: Vikram Sarabhai (first), M.G.K. Menon, Satish Dhawan, U.R. Rao, K. Kasturirangan, G. Madhavan Nair, K. Radhakrishnan, A.S. Kiran Kumar, K. Sivan, S. Somanath, V. Narayanan (from January 2025).
- World space firsts: Sputnik-1 (USSR, 4 Oct 1957 – first artificial satellite); Laika (first animal in orbit, 1957); Yuri Gagarin (first human in space, 12 April 1961, Vostok-1); Valentina Tereshkova (first woman, 1963); Alexei Leonov (first spacewalk, 1965); Neil Armstrong & Buzz Aldrin (Moon, Apollo 11, 1969); Hubble Space Telescope (1990); James Webb Space Telescope (2021, at L2). International Space Station (1998 onwards). Sunita Williams – Indian-origin astronaut with long stays on ISS.

ISRO 15 Aug 1969 | Father – Vikram Sarabhai | Aryabhata 1975 | Chandrayaan-1 2008 – water on Moon | Mangalyaan 2014 | Chandrayaan-3 landing 23 Aug 2023 (south pole) | Aditya-L1 – Sun, L1 | NavIC – navigation

★ **PYQ-TREND:** First Indian satellite and rocket. Father of Indian space programme. Chandrayaan-1/2/3 achievements and dates; Vikram & Pragyan; National Space Day. Mangalyaan's distinction. Aditya-L1 destination. NavIC as regional navigation. PSLV 104 satellites record. Sputnik, Gagarin, Tereshkova firsts. Launch site Sriharikota and first launch from Thumba.

15.2 Defence & Nuclear Science

- Indian atomic programme: Homi J. Bhabha ('father of Indian nuclear programme'; TIFR 1945; Atomic Energy Commission 1948; BARC). Three-stage nuclear programme: PHWR (natural uranium) → fast breeder reactors (plutonium; Kalpakkam PFBR) → thorium-based reactors (India has large thorium reserves in monazite sands of Kerala).
- Pokhran-I (18 May 1974, 'Smiling Buddha'); Pokhran-II (11 & 13 May 1998, 'Operation Shakti'; Kalam & R. Chidambaram). India's nuclear doctrine: No First Use, credible minimum

deterrence. Nuclear triad completed with INS Arihant (2018).

- Integrated Guided Missile Development Programme (IGMDP, 1983) led by A.P.J. Abdul Kalam ('Missile Man of India').
- Anti-satellite missile test – Mission Shakti (27 March 2019); India fourth country with ASAT capability.

Bhabha – nuclear programme | Pokhran-I 1974, Pokhran-II 1998 | Mission Shakti ASAT 2019 | Three-stage programme ends with thorium

★ **PYQ-TREND:** Father of Indian nuclear programme. Pokhran test code names and years. Thorium-based third stage. Mission Shakti (ASAT). Kalam and IGMDP.

15.3 Computer Fundamentals

- Charles Babbage – 'father of the computer' (Difference Engine, Analytical Engine); Ada Lovelace – first programmer. Alan Turing – father of theoretical computer science/AI (Turing machine). John von Neumann – stored-program architecture.

Generation	Technology	Example
First (1940–56)	Vacuum tubes	ENIAC (first general-purpose electronic computer, 1946), UNIVAC
Second (1956–63)	Transistors	IBM 1401
Third (1964–71)	Integrated circuits	IBM 360
Fourth (1971–present)	Microprocessors (VLSI)	Intel 4004 (first microprocessor, 1971), PCs
Fifth (present & beyond)	AI, ULSI, parallel processing, quantum	–

- India's first computer: TIFRAC (1960, TIFR, Mumbai); first computer installed – HEC-2M at ISI Kolkata (1956). India's first supercomputer: PARAM 8000 (1991, C-DAC Pune; Vijay Bhatkar – father of Indian supercomputers). National Supercomputing Mission; AIRAWAT (AI supercomputer).
- Types: supercomputer (fastest; weather forecasting), mainframe (large organisations, banks), mini, micro (PC), embedded. Analog, digital, hybrid computers.
- Hardware: Input devices (keyboard, mouse – Douglas Engelbart, scanner, joystick, light pen, microphone, webcam, OMR, OCR, MICR – cheques, barcode reader, biometric sensor), output devices (monitor – VDU, printer – impact (dot matrix) & non-impact (laser, inkjet), plotter, speaker, projector), CPU (ALU – arithmetic & logic; CU – control unit; registers). Motherboard. SMPS – power. UPS – backup.
- Memory: primary – RAM (volatile; SRAM faster used in cache, DRAM main memory) and ROM (non-volatile; PROM, EPROM – erased by UV, EEPROM – electrically, flash); cache (fastest after registers). Secondary – HDD, SSD, CD (700 MB), DVD (4.7 GB), Blu-ray (25 GB per layer), pen drive. Memory hierarchy (fastest to slowest): registers > cache > RAM > SSD > HDD > optical.

Unit	Equals
Bit	Binary digit (0 or 1)
Nibble	4 bits
Byte	8 bits
Kilobyte (KB)	1,024 bytes

Unit	Equals
Megabyte (MB)	1,024 KB
Gigabyte (GB)	1,024 MB
Terabyte (TB)	1,024 GB
Petabyte (PB)	1,024 TB
Exabyte, Zettabyte, Yottabyte	Next in order

- Number systems: binary (base 2), octal (8), decimal (10), hexadecimal (16; digits 0–9, A–F). ASCII (7-bit, 128 characters), Unicode (all scripts). 1's & 2's complement.
- Software: system software (operating systems – Windows (Microsoft – Bill Gates & Paul Allen), Linux (Linus Torvalds; open source), macOS, Android (Google; Linux-based), iOS, Unix (Ken Thompson & Dennis Ritchie), MS-DOS (CLI)), utility software (antivirus, compression), application software (MS Office – Word, Excel (spreadsheet – cells, formulas), PowerPoint (slides), Access (database)). Firmware (BIOS). Compiler (whole program at once), interpreter (line by line), assembler (assembly → machine code). Booting: cold boot / warm boot.
- Languages: machine language (binary), assembly (low-level), high-level – FORTRAN (first high-level, 1957; scientific), COBOL (business), BASIC, C (Dennis Ritchie, Bell Labs), C++ (Bjarne Stroustrup), Java (James Gosling, Sun Microsystems; 'write once, run anywhere'), Python (Guido van Rossum), HTML (web pages, Tim Berners-Lee), SQL (databases), JavaScript.
- Shortcut keys (Windows/MS Office): Ctrl+C copy, Ctrl+X cut, Ctrl+V paste, Ctrl+Z undo, Ctrl+Y redo, Ctrl+S save, Ctrl+P print, Ctrl+A select all, Ctrl+F find, Ctrl+H replace, Ctrl+B bold, Ctrl+I italic, Ctrl+U underline, Ctrl+N new, Ctrl+O open, Ctrl+W/Ctrl+F4 close document, Alt+F4 close window, F1 help, F2 rename, F5 refresh/slide show, F7 spell check (Word), F12 save as, Ctrl+Alt+Del task manager/security, Windows+D desktop, Windows+L lock, Ctrl+Home – beginning of document, Shift+F3 change case.

Father of computer – Babbage | First programmer – Ada Lovelace | ENIAC 1946 | Intel 4004 – first microprocessor 1971 | PARAM 8000 – 1991 | 1 byte = 8 bits; 1 KB = 1,024 B | RAM volatile, ROM non-volatile | Cache – fastest memory after registers

★ **PYQ-TREND:** Father of computer. Computer generations and technologies. RAM vs ROM. Memory units conversion. Input vs output devices (MICR for cheques, OMR for answer sheets). Compiler vs interpreter. First supercomputer of India. Shortcut keys (Ctrl+Z, F7, F12, Alt+F4). Linux open-source. Inventors of C, Java, Python. Binary base. MS Excel/PowerPoint usage.

15.4 Internet, Networks and Cyber Security

- ARPANET (1969, USA) – precursor of the internet. TCP/IP – Vint Cerf & Bob Kahn ('fathers of the internet'). World Wide Web – Tim Berners-Lee (1989, CERN). First web browser – WorldWideWeb/Nexus; Mosaic – first popular graphical browser. Internet in India: VSNL launched public internet on 15 August 1995.
- Networks: LAN (building), MAN (city), WAN (country/world; internet is the largest WAN), PAN (personal, Bluetooth). Topologies: bus, star (most common; central hub/switch), ring, mesh, tree. Devices: hub, switch, router (connects networks), modem (modulator–demodulator; digital ↔ analog), gateway, repeater, bridge, NIC. Wi-Fi (IEEE 802.11), Bluetooth (short range, Ericsson), NFC, Li-Fi (light; Harald Haas).
- Protocols: HTTP/HTTPS (web; S = secure, SSL/TLS), FTP (file transfer), SMTP (sending email), POP3/IMAP (receiving email), DNS (domain name → IP address), DHCP (assigns IP), TCP (reliable) and UDP (fast, connectionless), IP address – IPv4 (32-bit), IPv6 (128-bit). URL –

Uniform Resource Locator. Domains: .com (commercial), .org, .edu, .gov, .in (India), .net.

- Email: first sent by Ray Tomlinson (1971; chose @). CC (carbon copy), BCC (blind carbon copy). Spam – unsolicited bulk email.
- Cyber threats: virus (attaches to programs; needs host), worm (self-replicating, spreads over network), Trojan horse (disguised as useful software), ransomware (encrypts data and demands ransom – WannaCry 2017), spyware, adware, phishing (fraudulent emails/sites to steal details), vishing (voice), smishing (SMS), keylogger, botnet, DDoS (denial of service), hacking (white hat – ethical; black hat), cookies (small data files stored by websites), firewall (blocks unauthorised access), antivirus, encryption, two-factor authentication, CAPTCHA (distinguishes humans from bots). First computer virus – Creeper (1971); first PC virus – Brain (1986, Pakistan).
- India: IT Act 2000 (cyber law; amended 2008); CERT-In (national cyber security agency, 2004); Digital Personal Data Protection Act 2023; Aadhaar (UIDAI, 2009; 12 digits; first issued to Ranjana Sonawane, 2010); DigiLocker, UPI (2016), BHIM, CoWIN, ONDC, Bharat Net (rural broadband), 5G launch in India (1 October 2022).
- Emerging tech: AI (term coined by John McCarthy, 1956), machine learning, blockchain (cryptocurrency – Bitcoin by 'Satoshi Nakamoto', 2009), Internet of Things, cloud computing (SaaS, PaaS, IaaS), quantum computing (qubits; National Quantum Mission 2023), 3D printing, augmented & virtual reality, metaverse, Large Language Models (ChatGPT, 2022), semiconductor fabs (India Semiconductor Mission).

ARPANET 1969 | WWW – Berners-Lee 1989 | Internet in India – 15 Aug 1995 (VSNL) | IPv4 32-bit, IPv6 128-bit | IT Act 2000 | 5G India – 1 Oct 2022 | Email – Ray Tomlinson 1971 | AI term – John McCarthy 1956

★ **PYQ-TREND:** Full forms (HTTP, URL, DNS, LAN, WAN, ISP, Wi-Fi, PDF, CAPTCHA, UPS). Inventor of WWW. Email protocols (SMTP send, POP3 receive). Virus vs worm vs Trojan. Phishing. Firewall role. IT Act year. Network topologies and devices (router, modem). IPv4/IPv6 size. Father of AI.

15.5 Important Full Forms

Abbreviation	Full form
ISRO	Indian Space Research Organisation
DRDO	Defence Research and Development Organisation
NASA	National Aeronautics and Space Administration
PSLV / GSLV	Polar / Geosynchronous Satellite Launch Vehicle
GPS	Global Positioning System
NavIC	Navigation with Indian Constellation
LASER	Light Amplification by Stimulated Emission of Radiation
RADAR	Radio Detection and Ranging
SONAR	Sound Navigation and Ranging
MRI	Magnetic Resonance Imaging
CT scan	Computed Tomography
ECG	Electrocardiogram
DNA / RNA	Deoxyribonucleic acid / Ribonucleic acid
AIDS / HIV	Acquired Immuno Deficiency Syndrome / Human Immunodeficiency Virus

Abbreviation	Full form
BCG	Bacillus Calmette–Guérin
ORS	Oral Rehydration Solution
CFC	Chlorofluorocarbon
LPG / CNG / PNG	Liquefied Petroleum Gas / Compressed Natural Gas / Piped Natural Gas
LED / LCD	Light Emitting Diode / Liquid Crystal Display
CPU / ALU	Central Processing Unit / Arithmetic Logic Unit
RAM / ROM	Random Access Memory / Read Only Memory
BIOS	Basic Input Output System
USB	Universal Serial Bus
PDF	Portable Document Format
HTML	HyperText Markup Language
HTTP / HTTPS	HyperText Transfer Protocol (Secure)
URL	Uniform Resource Locator
ISP	Internet Service Provider
Wi-Fi	Wireless Fidelity
OTP	One Time Password
ATM	Automated Teller Machine
PIN	Personal Identification Number
IFSC	Indian Financial System Code
NEFT / RTGS / IMPS	National Electronic Funds Transfer / Real Time Gross Settlement / Immediate Payment Service
UPI	Unified Payments Interface
KYC	Know Your Customer
GDP / GNP	Gross Domestic / National Product
GST	Goods and Services Tax
FDI / FII	Foreign Direct Investment / Foreign Institutional Investor
SEBI	Securities and Exchange Board of India
NABARD	National Bank for Agriculture and Rural Development
MGNREGA	Mahatma Gandhi National Rural Employment Guarantee Act
NITI	National Institution for Transforming India
CAG	Comptroller and Auditor General
UPSC / SSC	Union Public Service Commission / Staff Selection Commission
CBI / NIA / RAW	Central Bureau of Investigation / National Investigation Agency / Research and Analysis Wing
NSG / BSF / CRPF / CISF / ITBP / SSB	National Security Guard / Border Security Force / Central Reserve Police Force / Central Industrial Security Force / Indo-Tibetan Border Police / Sashastra Seema Bal
NDRF	National Disaster Response Force

Abbreviation	Full form
UNESCO	United Nations Educational, Scientific and Cultural Organization
UNICEF	United Nations International Children's Emergency Fund (now United Nations Children's Fund)
WHO / ILO / WTO	World Health / International Labour / World Trade Organization
SAARC	South Asian Association for Regional Cooperation
ASEAN	Association of Southeast Asian Nations
BRICS	Brazil, Russia, India, China, South Africa
BIMSTEC	Bay of Bengal Initiative for Multi-Sectoral Technical and Economic Cooperation
NATO	North Atlantic Treaty Organization
OPEC	Organization of the Petroleum Exporting Countries
FIFA	Fédération Internationale de Football Association
PIL	Public Interest Litigation
FIR	First Information Report
EVM / VVPAT	Electronic Voting Machine / Voter Verifiable Paper Audit Trail
NOTA	None of the Above
IPC / BNS	Indian Penal Code (replaced by Bharatiya Nyaya Sanhita, 1 July 2024)
CrPC / BNSS	Code of Criminal Procedure (replaced by Bharatiya Nagarik Suraksha Sanhita, 2024)
Evidence Act / BSA	Indian Evidence Act (replaced by Bharatiya Sakshya Adhiniyam, 2024)

BNS, BNSS, BSA replaced IPC, CrPC, Evidence Act on 1 July 2024

★ **PYQ-TREND:** Full forms are asked in nearly every shift (LASER, RADAR, SONAR, ISRO, NABARD, SEBI, MGNREGA, UPI, NEFT, RTGS, BIOS, HTML, URL). New criminal laws replacing IPC/CrPC/Evidence Act (2024).

CHAPTER 16

PYQ Concept Recall Bank

Each row is a concept SSC has examined repeatedly, written as a short question with a crisp answer. Cover the answer column and test yourself. This is active recall, not MCQ practice.

R1. History

Concept question	Answer
Largest Harappan site in India?	Rakhigarhi (Haryana)
Harappan site with a dockyard?	Lothal
Who discovered Harappa?	Daya Ram Sahni (1921)
Oldest Veda?	Rigveda
Source of 'Satyameva Jayate'?	Mundaka Upanishad
Veda linked to music?	Samaveda
Only Mahajanapada in South India?	Assaka (Ashmaka)
Where did Buddha give his first sermon?	Sarnath
Fourth Buddhist Council was held under?	Kanishka (Kashmir)
24th Tirthankara?	Mahavira
Who added Brahmacharya to Jain vows?	Mahavira
Author of Arthashastra?	Kautilya (Chanakya)
Who deciphered Ashokan edicts?	James Prinsep (1837)
Which edict describes the Kalinga war?	Major Rock Edict XIII
Who is called the 'Napoleon of India'?	Samudragupta
Founder of Nalanda University?	Kumaragupta I
Chinese traveller in Harsha's time?	Hiuen Tsang
Builder of Kailasa temple, Ellora?	Krishna I (Rashtrakuta)
Builder of Brihadeshwara temple?	Rajaraja Chola I
Year of the Second Battle of Tarain?	1192
Who began Qutub Minar?	Qutbuddin Aibak
Who shifted the capital to Daulatabad?	Muhammad bin Tughlaq
Who introduced market control?	Alauddin Khilji
Who founded Agra?	Sikandar Lodi
Founders of Vijayanagara?	Harihara and Bukka (1336)
Battle that destroyed Vijayanagara?	Talikota (1565)
Gol Gumbaz is in?	Bijapur
First Battle of Panipat fought between?	Babur and Ibrahim Lodi (1526)
Third Battle of Panipat winner?	Ahmad Shah Abdali (1761)
Battle of Haldighati year?	1576
Who introduced the Rupiya?	Sher Shah Suri
Akbar's new religion?	Din-i-Ilahi (1582)

Concept question	Answer
Taj Mahal built by?	Shah Jahan
Founder of Khalsa?	Guru Gobind Singh (1699)
Guru who compiled Adi Granth?	Guru Arjan Dev
First European to arrive by sea route?	Vasco da Gama (Calicut, 1498)
Battle of Plassey year?	1757
Battle that gave Diwani rights?	Buxar (1764), Treaty of Allahabad 1765
Subsidiary Alliance introduced by?	Lord Wellesley
Doctrine of Lapse introduced by?	Lord Dalhousie
Permanent Settlement introduced by?	Lord Cornwallis (1793)
Sati abolished by?	Lord William Bentinck (1829)
First railway line in India?	Bombay–Thane, 1853
Father of Local Self-Government?	Lord Ripon
Partition of Bengal by?	Lord Curzon (1905)
Where did the Revolt of 1857 begin?	Meerut
Leader of 1857 at Lucknow?	Begum Hazrat Mahal
Founder of Brahmo Samaj?	Raja Ram Mohan Roy
Founder of Arya Samaj?	Swami Dayanand Saraswati
First INC president?	W.C. Bonnerjee
INC session of Purna Swaraj?	Lahore 1929
First woman INC president?	Annie Besant (1917)
Gandhi's first satyagraha in India?	Champaran (1917)
Date of Jallianwala Bagh massacre?	13 April 1919
Incident that ended Non-Cooperation?	Chauri Chaura (1922)
Dandi March started on?	12 March 1930
Round Table Conference attended by Gandhi?	Second (1931)
'Do or Die' slogan movement?	Quit India (1942)
Who wrote Hind Swaraj?	Mahatma Gandhi
Who founded the Azad Hind government?	Subhas Chandra Bose (Singapore, 1943)
Last Viceroy of India?	Lord Mountbatten

R2. Polity

Concept question	Answer
President of the Constituent Assembly?	Dr. Rajendra Prasad
Chairman of the Drafting Committee?	Dr. B.R. Ambedkar
Constitution adopted on?	26 November 1949
Source of DPSP?	Ireland
Source of Fundamental Duties?	USSR
Source of the Concurrent List?	Australia
Words added to the Preamble in 1976?	Socialist, Secular, Integrity
Article for the right to constitutional remedies?	Article 32
Article abolishing untouchability?	Article 17

Concept question	Answer
Article for the right to education?	Article 21A (86th Amendment)
Writ against illegal detention?	Habeas Corpus
Writ to prevent unlawful holding of public office?	Quo Warranto
Article for Uniform Civil Code?	Article 44
Article for village panchayats (DPSP)?	Article 40
Committee for Fundamental Duties?	Swaran Singh Committee
Number of Fundamental Duties?	11
Schedule with anti-defection law?	Tenth Schedule
Minimum age for President?	35 years
Who administers oath to the President?	Chief Justice of India
Article for the President's ordinance power?	Article 123
Article for national emergency?	Article 352
Ex-officio chairman of Rajya Sabha?	Vice-President
Who certifies a money bill?	Speaker of Lok Sabha
Who presides over a joint sitting?	Speaker of Lok Sabha
Max nominated members in Rajya Sabha?	12
Minimum age for Rajya Sabha?	30 years
Retirement age of SC judges?	65 years
Article for SC's advisory jurisdiction?	Article 143
Case that gave the basic structure doctrine?	Kesavananda Bharati (1973)
First state to adopt Panchayati Raj?	Rajasthan (Nagaur, 1959)
Amendment for Panchayati Raj?	73rd (1992)
Article for the Election Commission?	Article 324
Article for the CAG?	Article 148
Article for the Finance Commission?	Article 280
'Mini Constitution' amendment?	42nd (1976)
Amendment that lowered voting age to 18?	61st (1989)
GST amendment?	101st (2016)
Women's reservation amendment?	106th (2023)
NITI Aayog established on?	1 January 2015
First Chief Election Commissioner?	Sukumar Sen

R3. Geography

Concept question	Answer
How many states does the Tropic of Cancer cross?	8
Standard meridian of India passes through?	Mirzapur (82°30'E)
Country sharing longest border with India?	Bangladesh
Strait between India and Sri Lanka?	Palk Strait
Channel separating Andaman and Nicobar?	Ten Degree Channel
Highest peak in India?	Kangchenjunga
Highest peak in South India?	Anamudi

Concept question	Answer
Oldest fold mountains in India?	Aravallis
India's only active volcano?	Barren Island
Pass connecting Srinagar and Leh?	Zoji La
Pass in Sikkim on the old silk route?	Nathu La
Origin of Narmada?	Amarkantak
Longest peninsular river?	Godavari
'Sorrow of Bihar'?	Kosi
Highest dam in India?	Tehri
Hirakud dam is on?	Mahanadi
Largest freshwater lake in India?	Wular
Largest saline lake in India?	Sambhar
Floating national park?	Keibul Lamjao (Loktak, Manipur)
Place with highest rainfall?	Mawsynram
Best soil for cotton?	Black (regur) soil
National park for one-horned rhino?	Kaziranga
Only home of Asiatic lions?	Gir
First national park of India?	Jim Corbett (Hailey NP, 1936)
Where were cheetahs reintroduced?	Kuno National Park
First biosphere reserve?	Nilgiri
Largest coffee producer state?	Karnataka
Oldest oil refinery?	Digboi
Largest coking coal field?	Jharia
Hottest planet?	Venus
Largest planet?	Jupiter
Layer containing the ozone layer?	Stratosphere
Deepest point of the oceans?	Mariana Trench (Challenger Deep)
Longest river in the world?	Nile
Strait between Spain and Morocco?	Strait of Gibraltar
Parliament of Japan?	Diet
Parliament of Israel?	Knesset
Currency of Bhutan?	Ngultrum
Capital of Myanmar?	Naypyidaw
Grassland of Argentina?	Pampas

R4. Economy

Concept question	Answer
Father of economics?	Adam Smith
GNP = ?	GDP + net factor income from abroad
Who first estimated India's national income?	Dadabhai Naoroji
Model of the Second Five-Year Plan?	P.C. Mahalanobis
When was RBI established?	1 April 1935

Concept question	Answer
Who issues the ₹1 note?	Ministry of Finance (Government of India)
Rate at which RBI lends to banks?	Repo rate
Inflation target of RBI?	CPI 4% ± 2%
When were 14 banks nationalised?	19 July 1969
Regulator of securities market?	SEBI
Number of stocks in Sensex?	30
GST launch date?	1 July 2017
Fiscal deficit equals?	Total borrowings of the government
Year of LPG reforms?	1991
WTO HQ?	Geneva
Days of guaranteed work under MGNREGA?	100
PM-KISAN yearly support?	₹6,000
Census 2011 sex ratio?	943
State with highest literacy (2011)?	Kerala

R5. Science

Concept question	Answer
SI unit of pressure?	Pascal
Unit of power of a lens?	Dioptre
Instrument measuring humidity?	Hygrometer
Instrument measuring blood pressure?	Sphygmomanometer
Escape velocity of Earth?	11.2 km/s
Mirror used as a rear-view mirror?	Convex
Lens for myopia?	Concave
Why is the sky blue?	Scattering of light
Why does a diamond sparkle?	Total internal reflection
Speed of sound is maximum in?	Solids
Frequency used by bats?	Ultrasonic
Water's maximum density temperature?	4 °C
Fuse wire property?	Low melting point
Filament metal in bulbs?	Tungsten
Household wiring connection?	Parallel
Discoverer of the neutron?	James Chadwick
Energy source of the Sun?	Nuclear fusion
Chemical name of baking soda?	Sodium bicarbonate
Formula of Plaster of Paris?	$CaSO_4 \cdot \frac{1}{2}H_2O$
Acid in ant sting?	Formic acid
Acid in tomato?	Oxalic acid
pH of human blood?	≈ 7.4
Main gas in LPG?	Butane (with propane)
Gas in Bhopal tragedy?	Methyl isocyanate

Concept question	Answer
Brass is an alloy of?	Copper and zinc
Liquid non-metal?	Bromine
Most electronegative element?	Fluorine
Powerhouse of the cell?	Mitochondria
Who discovered the cell?	Robert Hooke
Vitamin causing scurvy when deficient?	Vitamin C
Vitamin for blood clotting?	Vitamin K
Vitamin made by sunlight in the skin?	Vitamin D
Largest gland?	Liver
Master gland?	Pituitary
Universal blood donor?	O (negative)
Smallest bone?	Stapes
Malaria is caused by?	Plasmodium (via female Anopheles)
Test for typhoid?	Widal test
Father of genetics?	Gregor Mendel
Edible part of potato?	Stem (tuber)
Hormone for fruit ripening?	Ethylene

R6. Static GK & Technology

Concept question	Answer
Number of spokes in the Ashoka Chakra?	24
Playing time of the national anthem?	≈ 52 seconds
National aquatic animal?	Ganges river dolphin
National Science Day?	28 February
Engineers' Day?	15 September
First Indian individual Olympic gold?	Abhinav Bindra (2008)
Players in a kabaddi team on court?	7
Ranji Trophy sport?	Cricket
Thomas Cup sport?	Badminton
'Flying Sikh'?	Milkha Singh
First Indian woman to climb Everest?	Bachendri Pal
First Bharat Ratna recipients?	Rajagopalachari, Radhakrishnan, C.V. Raman (1954)
First Dadasaheb Phalke awardee?	Devika Rani
First Indian Nobel laureate?	Rabindranath Tagore (1913)
Author of Rajatarangini?	Kalhana
Author of Wings of Fire?	A.P.J. Abdul Kalam
UN founded on?	24 October 1945
UNESCO HQ?	Paris
HQ of International Solar Alliance?	Gurugram
First Indian satellite?	Aryabhata (1975)

Concept question	Answer
Chandrayaan-3 landed on?	23 August 2023
Father of Indian space programme?	Vikram Sarabhai
Father of the computer?	Charles Babbage
Inventor of the World Wide Web?	Tim Berners-Lee
1 byte = ?	8 bits
Volatile memory?	RAM
Protocol for sending email?	SMTP
India's cyber law?	IT Act 2000
Inventor of the telephone?	Alexander Graham Bell
Inventor of dynamite?	Alfred Nobel

CHAPTER 17

Last-Minute Revision: Tables & Traps

17.1 State-Wise Quick Map (Capital • Dance • Festival • Park)

State	Capital	Folk dance	Festival	Famous park / site
Andhra Pradesh	Amaravati	Kuchipudi (classical), Burrakatha	Ugadi	Papikonda NP; Tirupati
Arunachal Pradesh	Itanagar	Bardo Chham	Losar, Solung	Namdapha NP
Assam	Dispur	Bihu, Sattriya (classical)	Bihu	Kaziranga, Manas
Bihar	Patna	Jat-Jatin, Bidesia	Chhath	Valmiki NP; Bodh Gaya, Nalanda
Chhattisgarh	Raipur (Nava Raipur)	Panthi, Raut Nacha	Bastar Dussehra	Indravati NP; Chitrakote falls
Goa	Panaji	Dekhni, Fugdi	Carnival, Shigmo	Bhagwan Mahavir WLS; Dudhsagar
Gujarat	Gandhinagar	Garba, Dandiya	Navratri, Uttarayan	Gir; Rani ki Vav; Statue of Unity
Haryana	Chandigarh	Phag, Saang	Lohri, Teej	Sultanpur NP; Rakhigarhi
Himachal Pradesh	Shimla (winter – Dharamshala)	Nati	Kullu Dussehra	Great Himalayan NP
Jharkhand	Ranchi	Chhau, Karma	Sarhul	Betla NP; Hundru falls
Karnataka	Bengaluru	Yakshagana	Mysuru Dasara	Bandipur, Nagarhole; Hampi
Kerala	Thiruvananthapuram	Kathakali, Mohiniyattam, Theyyam	Onam	Periyar, Silent Valley, Eravikulam
Madhya Pradesh	Bhopal	Matki, Tertali	Khajuraho Dance Festival	Kanha, Bandhavgarh, Kuno; Sanchi
Maharashtra	Mumbai (winter – Nagpur)	Lavani, Tamasha	Ganesh Chaturthi, Gudi Padwa	Tadoba; Ajanta, Ellora
Manipur	Imphal	Manipuri (classical), Thang-Ta	Sangai, Lai Haraoba	Keibul Lamjao
Meghalaya	Shillong	Shad Suk Mynsiem, Wangala	Wangala	Nokrek; Mawsynram
Mizoram	Aizawl	Cheraw	Chapchar Kut	Dampa Tiger Reserve
Nagaland	Kohima	Chang Lo, Zeliang	Hornbill	Intanki NP

State	Capital	Folk dance	Festival	Famous park / site
Odisha	Bhubaneswar	Odissi (classical), Chhau	Rath Yatra, Nuakhai	Simlipal, Bhitarkanika; Konark
Punjab	Chandigarh	Bhangra, Giddha	Baisakhi, Lohri	Harike wetland; Golden Temple
Rajasthan	Jaipur	Ghoomar, Kalbelia	Pushkar Fair, Teej	Ranthambore, Keoladeo, Sariska
Sikkim	Gangtok	Singhi Chham	Losar, Saga Dawa	Khangchendzonga NP
Tamil Nadu	Chennai	Bharatanatyam (classical), Kummi	Pongal	Mudumalai; Mahabalipuram
Telangana	Hyderabad	Perini	Bathukamma, Bonalu	Ramappa temple
Tripura	Agartala	Hojagiri	Kharchi Puja	Clouded Leopard NP
Uttar Pradesh	Lucknow	Kathak (classical), Nautanki	Kumbh (Prayagraj)	Dudhwa; Taj Mahal
Uttarakhand	Dehradun (summer – Gairsain)	Chholiya	Kumbh (Haridwar)	Jim Corbett, Valley of Flowers
West Bengal	Kolkata	Gambhira, Baul (music)	Durga Puja	Sundarbans; Santiniketan

UT	Capital / note
Andaman & Nicobar	Sri Vijaya Puram (Port Blair)
Chandigarh	Shared capital of Punjab & Haryana
Dadra & Nagar Haveli and Daman & Diu	Daman
Delhi (NCT)	New Delhi
Jammu & Kashmir	Srinagar (summer), Jammu (winter)
Ladakh	Leh (also Kargil)
Lakshadweep	Kavaratti
Puducherry	Puducherry

★ **PYQ-TREND:** State-linked dance/festival/park questions combine these columns. Capitals with two seats (HP, Maharashtra, Uttarakhand, J&K). Andhra Pradesh capital Amaravati.

17.2 Commonly Confused Pairs

A	B
Kalinga war – Ashoka (Rock Edict XIII)	Kalinga's Kharavela – Hathigumpha inscription
Fa-Hien – Chandragupta II	Hiuen Tsang – Harsha
Al-Biruni – Mahmud of Ghazni	Ibn Battuta – Muhammad bin Tughlaq
Panipat I 1526	Panipat II 1556
Plassey 1757 – Siraj	Buxar 1764 – Mir Qasim + Shuja + Shah Alam
Ryotwari – Munro	Mahalwari – Holt Mackenzie
First GG of Bengal – Warren Hastings	First GG of India – William Bentinck

A	B
Brahmo Samaj – Ram Mohan Roy	Prarthana Samaj – Atmaram Pandurang
Surat split – 1907	Lucknow Pact – 1916
Habeas Corpus – illegal detention	Quo Warranto – illegal office holding
Article 352 – national emergency	Article 356 – President's Rule
73rd – Panchayats (29 subjects)	74th – Municipalities (18 subjects)
Speaker – certifies money bill	Chairman RS – Vice-President
Repo – RBI lends	Reverse repo – RBI borrows
CRR – kept with RBI	SLR – kept by banks themselves
Direct tax – income tax	Indirect tax – GST
Kharif – rice, cotton	Rabi – wheat, mustard
Wular – largest freshwater lake	Sambhar – largest saline lake
Kosi – Sorrow of Bihar	Damodar – Sorrow of Bengal
Convex mirror – vehicles	Concave mirror – shaving/dentist
Myopia – concave lens	Hypermetropia – convex lens
Fission – reactors, atom bomb	Fusion – Sun, hydrogen bomb
Baking soda – $NaHCO_3$	Washing soda – $Na_2CO_3 \cdot 10H_2O$
POP – $CaSO_4 \cdot \frac{1}{2}H_2O$	Gypsum – $CaSO_4 \cdot 2H_2O$
Brass – Cu + Zn	Bronze – Cu + Sn
Malaria – Anopheles	Dengue – Aedes; Filaria – Culex
Vitamin A – night blindness	Vitamin D – rickets
Insulin – lowers sugar (β cells)	Glucagon – raises sugar (α cells)
RAM – volatile	ROM – non-volatile
Virus – needs host program	Worm – self-replicating over network
SMTP – sending mail	POP3 – receiving mail
Thomas Cup – men's badminton	Uber Cup – women's badminton
Davis Cup – men's tennis	Billie Jean King Cup – women's tennis
Ramon Magsaysay – Asia's Nobel	Jnanpith – Indian literature
WHO – Geneva	UNESCO – Paris; UNEP – Nairobi

17.3 Numbers to Remember

Fact	Number
States / UTs	28 / 8
Scheduled languages	22
Classical languages	11 (after Oct 2024)
Classical dances (SNA)	8
Fundamental Duties	11
Fundamental Rights (categories)	6
Schedules of the Constitution	12
Lok Sabha elected seats	543
Rajya Sabha max / nominated	250 / 12
Supreme Court judges (max)	34

Fact	Number
High Courts	25
UN members	193
UNESCO sites in India	43 (2024)
Bones in adult human	206
Human chromosomes	46
Teeth (adult / milk)	32 / 20
Ashoka Chakra spokes	24
Flag ratio	3:2
Anthem duration	52 seconds
Olympic hockey golds	8
Players: kabaddi / kho-kho / polo / volleyball	7 / 9 / 4 / 6
Speed of light	3×10^8 m/s
Speed of sound in air	≈ 343 m/s
Escape velocity	11.2 km/s
Normal BP / heartbeat	120/80 mm Hg / 72 per min
RBC lifespan	120 days
Census 2011 sex ratio / literacy	943 / 74.04%

17.4 Exam-Day Strategy for GA

- Do GA first; spend at most 10–12 minutes. Answer what you know immediately; mark doubtful questions and come back only if time remains.
- Eliminate options. If two options remain and you know the concept area, a guess is reasonable; if you have no idea, skip (negative marking).
- Watch for word traps: 'first' vs 'last', 'largest' vs 'longest', 'founded' vs 'built', 'Governor-General of Bengal' vs 'of India', 'oldest' vs 'first'.
- Read current affairs of the last 6–8 months separately: awards, sports winners, appointments, summits, schemes, space missions, indices and rankings.

✔ **TOPPER TRICK:** Revise this book with the "three-pass" method: pass 1 – full chapters; pass 2 – only yellow strips and blue boxes (Appendices A & B); pass 3 – the recall bank and this chapter on the last two days.

✖ **COMMON MISTAKE:** Do not learn shaky "facts" from unverified social-media PDFs in the last week. If two sources disagree, prefer NCERT, official government websites and the Constitution's text.

APPENDIX

Appendix A – One-Place Key-Fact Sheet

Every yellow key-fact strip from the chapters, collected chapter by chapter for the final revision.

Ancient Indian History

Bhimbetka – rock paintings (MP, UNESCO) | Mehrgarh – earliest agriculture | Burzahom – pit houses, dog burials | Koldihwa – early rice

Rigveda – oldest, 10 mandalas, 1028 hymns | Gayatri Mantra – Rigveda 3rd mandala | Satyameva Jayate – Mundaka Upanishad | Samaveda – music

Buddha: Lumbini birth – Bodh Gaya enlightenment – Sarnath first sermon – Kushinagar death | Mahavira: Kundagrama – Jrimbhikagrama – Pavapuri

Arthashastra – Kautilya | Indica – Megasthenes | Kalinga war 261 BCE | Edicts deciphered by James Prinsep 1837 | Rock Edict XIII – Kalinga

Shaka Samvat 78 CE = national calendar | Vikram Samvat 57 BCE | Junagadh inscription – Rudradaman (first Sanskrit) | Tirukkural – Thiruvalluvar

Prayag Prashasti – Harisena (Samudragupta) | Nalanda – Kumaragupta I | Fa-Hien – Chandragupta II | Hiuen Tsang – Harsha | Harshacharita – Banabhatta

Kailasa temple Ellora – Krishna I | Brihadeshwara – Rajaraja I | Aihole inscription – Ravikirti (Pulakeshin II) | Vikramashila – Dharmapala

Medieval Indian History

712 – Sindh (Qasim) | 1025 – Somnath raid (Mahmud) | 1191/1192 – Tarain I/II | 1194 – Chandawar

Qutub Minar – Aibak began, Iltutmish completed | Market reforms – Alauddin | Capital to Daulatabad, token currency – Muhammad bin Tughlaq | Agra founded – Sikandar Lodi (1504)

Vijayanagara 1336 (Harihara & Bukka) | Talikota 1565 | Bahmani 1347 (Hasan Gangu) | Gol Gumbaz – Bijapur | Charminar – Golconda/Hyderabad

Panipat I 1526 (Babur v Ibrahim Lodi) | Panipat II 1556 (Akbar v Hemu) | Panipat III 1761 (Abdali v Marathas) | Haldighati 1576 | Khanwa 1527

Shivaji coronation 1674 Raigad | Khalsa 1699 (Guru Gobind Singh) | Adi Granth 1604 (Guru Arjan Dev) | Amritsar founded – Guru Ram Das

Modern Indian History

Vasco da Gama – Calicut 1498 | EIC – 1600 | Surat factory – 1613 | Wandiwash – 1760 | Goa liberated – 1961

Plassey 1757 (Clive v Siraj) | Buxar 1764 | Diwani 1765 (Allahabad) | Subsidiary alliance – Wellesley | Doctrine of Lapse – Dalhousie

First GG of Bengal – Warren Hastings | First GG of India – William Bentinck | First Viceroy – Canning | Last Viceroy – Mountbatten | Only Indian GG – C. Rajagopalachari

1857 began at Meerut on 10 May | Mangal Pandey – Barrackpore | GoI Act 1858 ended Company rule

Brahmo Samaj 1828 – Ram Mohan Roy | Arya Samaj 1875 – Dayanand | Ramakrishna Mission 1897 – Vivekananda | Satyashodhak 1873 – Phule

INC 1885 Bombay (Hume; WC Bonnerjee) | Surat split 1907 | Lucknow Pact 1916 | Purna Swaraj – Lahore 1929 | Jana Gana Mana first sung – Calcutta 1911

Champaran 1917 | Jallianwala 13 Apr 1919 | Chauri Chaura 1922 | Dandi 12 Mar–6 Apr 1930 | Quit India 8 Aug 1942 | Independence 15 Aug 1947

Art & Culture

Bharatanatyam – TN | Kathakali & Mohiniyattam – Kerala | Kuchipudi – AP | Odissi – Odisha | Kathak – UP | Manipuri – Manipur | Sattriya – Assam

Konark – Narasimhadeva I | Kailasa (Ellora) – Krishna I | Humayun's Tomb – first garden tomb | Gateway of India – 1924 | Rani ki Vav – ₹100 note | Statue of Unity – 182 m

22 scheduled languages | Classical: Tamil 2004 first; 11 in total after 2024 additions | Sahitya Akademi 1954

Indian Polity I – Constitution and Rights

First meeting 9 Dec 1946 | Objectives Resolution 13 Dec 1946 | Adopted 26 Nov 1949 | In force 26 Jan 1950 | Original: 395 Articles, 22 Parts, 8 Schedules

42nd Amendment 1976 – Socialist, Secular, Integrity | Kesavananda 1973 – Preamble is part, basic structure

Schedules: 7th – lists | 8th – languages | 9th – 1st Amdt | 10th – anti-defection (52nd) | 11th – Panchayats (29) | 12th – Municipalities (18)

28 states + 8 UTs | Telangana – 29th state, 2014 | J&K and Ladakh UTs – 31 Oct 2019 | Sikkim – 36th Amendment 1975

Art 14 equality | 17 untouchability | 19 six freedoms | 21 life & liberty | 21A education (86th, 2002) | 32 remedies ('heart & soul') | Property → 300A (44th, 1978)

Art 40 – village panchayats | 44 – Uniform Civil Code | 45 – early childhood care | 48A – environment | 50 – separation of judiciary | 51 – international peace

Fundamental Duties – 42nd Amdt 1976, Swaran Singh Committee, 10 + 1 (86th, 2002) = 11

Indian Polity II – Institutions

Art 52 President | Art 61 impeachment | Art 72 pardon | Art 123 ordinance | Art 352/356/360 emergencies | Min age 35

Art 63 VP | Art 64 VP = RS Chairman | Art 74 CoM aids President | Art 75 PM appointment, 15% cap (91st) | Art 76 Attorney General

LS max 552, RS max 250 | Ages 25/30 | Art 108 joint sitting | Art 110 money bill | Art 112 budget | Quorum 1/10 | RS 12 nominated (Art 80)

SC judges retire 65, HC judges 62 | Art 32 & 226 writs | Art 131 original | Art 143 advisory | Kesavananda 1973 – basic structure | Lok Adalat Act 1987

Panchayati Raj first – Nagaur, 2 Oct 1959 | 73rd (Panchayat, 29 subjects) & 74th (Municipal, 18 subjects) Amendments 1992 | Legislative Councils in 6 states

Art 148 CAG | Art 280 Finance Commission | Art 315 UPSC | Art 324 Election Commission | Art 279A GST Council | NITI Aayog 1 Jan 2015

42nd – mini constitution | 44th – property removed | 52nd – anti-defection | 61st – voting age 18 | 73rd/74th – local bodies | 101st – GST | 103rd – EWS | 106th – women's reservation

First general election 1951–52 | EVM – Paravur 1982 | NOTA 2013 | National Voters' Day 25 Jan

Indian Geography

Tropic of Cancer – 8 states | Standard meridian 82°30'E (Mirzapur) | Longest border – Bangladesh | Largest state – Rajasthan | Smallest – Goa | Longest coastline – Gujarat

Highest peak in India – Kangchenjunga | South India – Anamudi | Aravalli – oldest fold mountains (Guru Shikhar) | Barren Island – only active volcano | Chilika – largest brackish lagoon

Longest river – Ganga | Longest peninsular – Godavari | Highest dam – Tehri | Largest freshwater lake – Wular | Largest saline lake – Sambhar | Largest river island – Majuli

Monsoon onset Kerala ~1 June | Highest rainfall – Mawsynram | Kalbaisakhi – Bengal | Mango showers – Kerala/Karnataka | NE monsoon – Tamil Nadu winter rain

Black soil – cotton | Laterite – leaching, tea/coffee/cashew | Alluvial – most widespread | Largest forest area – MP | Highest % forest – Mizoram

First NP – Jim Corbett | Rhino – Kaziranga | Asiatic lion – Gir | Hangul – Dachigam | Sangai – Keibul Lamjao | Cheetah – Kuno | First biosphere reserve – Nilgiri | First Ramsar – Chilika & Keoladeo

Kharif – monsoon (rice, cotton) | Rabi – winter (wheat, mustard) | White revolution – Verghese Kurien | Green – Swaminathan | Coffee – Karnataka | Rubber – Kerala | Jute – West Bengal

Oldest refinery – Digboi | Largest coking coal – Jharia | First nuclear plant – Tarapur | Jaduguda – uranium | Panna – diamonds | TISCO 1907

First train 16 Apr 1853 | NH-44 longest | NW-1 Ganga | JNPT – largest container port | Kandla – tidal | Visakhapatnam – deepest landlocked harbour

World & Physical Geography

Largest planet – Jupiter | Smallest – Mercury | Hottest – Venus | Red – Mars | Largest moon – Ganymede | Dwarf planet – Pluto (2006) | Sunlight to Earth ≈ 8 min 20 s

1° longitude = 4 min | 21 June – longest day (N) | 22 Dec – shortest day (N) | Equinoxes 21 Mar & 23 Sep | Perihelion ~3 Jan

Moho – crust/mantle | Gutenberg – mantle/core | Crust – oxygen most abundant | Limestone → marble | P waves fastest; S waves not through liquid | Richter scale

N_2 78% | O_2 21% | Ar 0.93% | CO_2 0.04% | Ozone – stratosphere | Weather – troposphere | Meteors burn – mesosphere | Radio reflection – ionosphere

Largest & deepest ocean – Pacific | Deepest point – Mariana Trench (Challenger Deep) | Smallest ocean – Arctic | Gulf Stream – warm | Humboldt – cold | Dead Sea – lowest land point

Longest river – Nile | Largest by volume – Amazon | Largest lake – Caspian | Deepest lake – Baikal | Highest waterfall – Angel | Driest – Atacama | Largest hot desert – Sahara | Largest island – Greenland

Indian Economy

Adam Smith – Wealth of Nations 1776 | Keynes – General Theory 1936 | Giffen & Veblen – exceptions to law of demand | Monopsony – one buyer

GNP = GDP + NFIA | NNP = GNP − depreciation | Base year 2011–12 | Inflation target CPI 4% ± 2% | First NI estimate – Dadabhai Naoroji

Planning Commission 15 Mar 1950 | NITI Aayog 1 Jan 2015 | 2nd Plan – Mahalanobis | Plan holiday 1966–69 | 12th Plan – last

RBI 1 Apr 1935, nationalised 1949 | ₹1 note issued by GoI | MPC – 6 members | SBI 1 Jul 1955 | 14 banks nationalised 19 Jul 1969 | NABARD 1982 | Sensex 30, Nifty 50

Revenue deficit = revenue expenditure − revenue receipts

Fiscal deficit = total expenditure − (revenue receipts + non-debt capital receipts) = total borrowings

Primary deficit = fiscal deficit − interest payments

Effective revenue deficit = revenue deficit − grants for creation of capital assets

GST 1 July 2017 (101st Amdt) | Budget date 1 Feb since 2017 | First budget of free India – R.K. Shanmukham Chetty | FRBM 2003

LPG 1991 | WTO 1 Jan 1995 (Geneva) | IMF & World Bank – Bretton Woods 1944 | FEMA 1999 | IBC 2016 | Demonetisation 8 Nov 2016

MGNREGA – 100 days | PM-KISAN ₹6,000/yr | Ayushman Bharat ₹5 lakh | Jan Dhan 2014 | Ujjwala 2016 | Swachh Bharat 2 Oct 2014

Census 2011: 121 crore | sex ratio 943 | literacy 74.04% | density 382 | Kerala highest sex ratio & literacy | Bihar highest density

General Science – Physics

7 base SI units | 1 kWh = 3.6 MJ | 1 hp = 746 W | 1 nautical mile = 1,852 m | −40 °C = −40 °F | Body temp 98.6 °F

F = ma | g = 9.8 m/s² (max at poles) | Escape velocity 11.2 km/s | Geostationary ≈ 36,000 km, 24 h | T = 2π√(l/g) | KE ∝ v²

Pascal – hydraulic lift/brakes | Archimedes – buoyancy, ships | Lactometer – milk | Bernoulli – aeroplane lift | Water boils below 100 °C at hills

Max density of water at 4 °C | Latent heat of ice 80 cal/g; steam 540 cal/g | Sea breeze by day, land breeze at night | Thermos – vacuum + silvered walls

Sound needs medium | v: solid > liquid > gas | Air ≈ 343 m/s | Audible 20 Hz–20 kHz | Echo min distance ≈ 17 m | Bats – ultrasonic

c = 3 × 10^8 m/s | Rear-view mirror – convex | Shaving/dentist mirror – concave | Myopia – concave lens | Hypermetropia – convex | Blue sky – scattering | Diamond sparkle – TIR | Raman effect – 28 Feb

V = IR | P = VI | 1 unit = 1 kWh | India 230 V, 50 Hz AC | Fuse – low melting point, in series | Filament – tungsten | Heater – nichrome | Transformer – AC only

Electron – Thomson | Neutron – Chadwick | Nucleus – Rutherford | Radioactivity – Becquerel | Fission – reactors/atom bomb | Fusion – Sun/H-bomb | X-rays – Röntgen | E = mc²

Hygrometer – humidity | Anemometer – wind | Sphygmomanometer – BP | Lactometer – milk | Seismograph – earthquakes | Fathometer – sea depth | Pyrometer – high temperature

General Science – Chemistry

Plasma – 4th state | BEC – 5th | Sublimation – camphor, naphthalene, iodine, dry ice | Isotopes – same Z | Avogadro 6.022 × 10^{23} | 2n^2 rule

Modern periodic law – Moseley (atomic number) | 18 groups, 7 periods | Most electronegative – F | Liquid metal – Hg | Liquid non-metal – Br | Densest – Os | Highest melting – W

Baking soda $NaHCO_3$ | Washing soda $Na_2CO_3 \cdot 10H_2O$ | Bleaching powder $CaOCl_2$ | POP $CaSO_4 \cdot \frac{1}{2}H_2O$ | Gypsum $CaSO_4 \cdot 2H_2O$ | Laughing gas N_2O | Indelible ink – $AgNO_3$ | Aqua regia 3 HCl : 1 HNO_3

Most malleable/ductile – gold | Best conductor – silver | Na & K stored in kerosene | White phosphorus in water | Pencil 'lead' = graphite | Diamond hardest | Brass Cu+Zn | Bronze Cu+Sn | Solder Pb+Sn | 24 carat pure gold

LPG – butane + propane | CNG – methane | Urea 46% N | Bhopal 1984 – methyl isocyanate | Minamata – Hg | Itai-itai – Cd | Teflon – non-stick | Nylon – first synthetic fibre | Pyrex – borosilicate

General Science – Biology

Cell – Robert Hooke 1665 | Nucleus – Robert Brown | Powerhouse – mitochondria | Suicide bag – lysosome | Protein factory – ribosome | DNA double helix – Watson & Crick 1953 | 46 chromosomes

Father of taxonomy – Linnaeus | Five kingdoms – Whittaker 1969 | TMV – first virus | Largest phylum – Arthropoda | Bryophytes – amphibians of plant kingdom | Platypus – egg-laying mammal

A – night blindness | B1 – beriberi | B3 – pellagra | B12 – pernicious anaemia (cobalt) | C – scurvy | D – rickets | K – clotting | Iodine – goitre | Fat-soluble: A, D, E, K

Largest gland – liver | Master gland – pituitary | Largest organ – skin | 206 bones; femur longest; stapes smallest | RBC life 120 days | Universal donor O, recipient AB | BP 120/80 | Insulin – β cells | Graveyard of RBCs – spleen

Malaria – Plasmodium via female Anopheles | Dengue – Aedes | Filaria – Culex | TB – Koch | Smallpox vaccine – Jenner | Rabies vaccine – Pasteur | Penicillin – Fleming 1928 | Widal – typhoid | ELISA – HIV

O_2 in photosynthesis from water | Ethylene – ripening (gaseous) | ABA – stress | Gibberellin – stem elongation | Potato – stem | Ginger – rhizome | Saffron – stigma | Clove – flower bud | Rhizobium – legumes

Father of genetics – Mendel (pea) | 3:1 and 9:3:3:1 | Dolly – 1996 | Origin of Species – Darwin 1859 | 10% law – Lindeman | Ecosystem – Tansley | Chipko – 1973

Static GK I – Sports

Modern Olympics – Athens 1896 (Coubertin) | IOC HQ – Lausanne | First individual gold – Abhinav Bindra 2008 | First athletics gold – Neeraj Chopra 2021 | Hockey golds – 8 | National Sports Day – 29 Aug

Cricket pitch 22 yards | Kho-kho 9, Kabaddi 7, Polo 4, Volleyball 6, Basketball 5 | Chess & badminton – India origin | Basketball – Naismith

Ranji – cricket | Durand – football (1888) | Beighton – hockey | Davis – tennis | Thomas & Uber – badminton | Wimbledon – grass, oldest | Largest cricket stadium – Narendra Modi Stadium

Flying Sikh – Milkha Singh | Payyoli Express – PT Usha | Wizard – Dhyan Chand | First Indian GM – Anand | First Indian woman on Everest – Bachendri Pal 1984 | Khel Ratna first – Anand

Static GK II – Books, Awards, Days, Firsts, Organisations

Flag 3:2, 24 spokes, 22 July 1947 | Anthem 24 Jan 1950 (52 s) | Emblem – Sarnath, 26 Jan 1950 | National calendar – Saka (22 Mar 1957) | Tiger 1973 | Dolphin 2009 | ₹ symbol 2010

28 Feb – Science | 5 Jun – Environment | 21 Jun – Yoga | 1 Jul – Doctors | 29 Aug – Sports | 5 Sep – Teachers | 15 Sep – Engineers | 31 Oct – Ekta | 26 Nov – Constitution | 22 Dec – Mathematics

Bharat Ratna 1954 (first: Rajaji, Radhakrishnan, Raman) | Jnanpith first – G. Sankara Kurup 1965 | Phalke first – Devika Rani 1969 | First Indian Nobel – Tagore 1913 | First Oscar – Bhanu Athaiya | Param Vir Chakra first – Somnath Sharma

First Indian in space – Rakesh Sharma 1984 | First satellite – Aryabhata 1975 | First talkie – Alam Ara 1931 | First film – Raja Harishchandra 1913 | First woman CM – Sucheta Kripalani | First woman Governor – Sarojini Naidu | First Field Marshal – Manekshaw

UN 24 Oct 1945 (193 members) | P5 – USA, UK, France, Russia, China | ICJ – The Hague | WHO, ILO, WTO – Geneva | UNESCO – Paris | FAO – Rome | IAEA – Vienna | UNEP – Nairobi | ISA – Gurugram

ISRO – Bengaluru | BARC – Mumbai | Sriharikota – launch site | IAF founded 8 Oct 1932 | First CDS – Bipin Rawat | Assam Rifles – oldest CAPF | BSF 1965 | INS Vikrant – first indigenous carrier | BrahMos – India + Russia

Science & Technology, Space and Computers

ISRO 15 Aug 1969 | Father – Vikram Sarabhai | Aryabhata 1975 | Chandrayaan-1 2008 – water on Moon | Mangalyaan 2014 | Chandrayaan-3 landing 23 Aug 2023 (south pole) | Aditya-L1 – Sun, L1 | NavIC – navigation

Bhabha – nuclear programme | Pokhran-I 1974, Pokhran-II 1998 | Mission Shakti ASAT 2019 | Three-stage programme ends with thorium

Father of computer – Babbage | First programmer – Ada Lovelace | ENIAC 1946 | Intel 4004 – first microprocessor 1971 | PARAM 8000 – 1991 | 1 byte = 8 bits; 1 KB = 1,024 B | RAM volatile, ROM non-volatile | Cache – fastest memory after registers

ARPANET 1969 | WWW – Berners-Lee 1989 | Internet in India – 15 Aug 1995 (VSNL) | IPv4 32-bit, IPv6 128-bit | IT Act 2000 | 5G India – 1 Oct 2022 | Email – Ray Tomlinson 1971 | AI term – John McCarthy 1956

BNS, BNSS, BSA replaced IPC, CrPC, Evidence Act on 1 July 2024

APPENDIX

Appendix B – PYQ-Trend Concept Map

All PYQ-Trend boxes in one place, chapter-wise. These are the concepts SSC JE has tested repeatedly; revise this list in the last week.

Ancient Indian History

★ Bhimbetka location and its rock paintings. Mehrgarh as earliest farming site. Burzahom pit dwellings. Microliths belong to the Mesolithic age.

★ Great Bath at Mohenjo-daro, dockyard at Lothal, ploughed field at Kalibangan, Dholavira's water management and UNESCO status, Rakhigarhi largest Indian site, discoverers of Harappa and Mohenjo-daro, Pashupati seal animals, cotton first grown here.

★ Satyameva Jayate source (Mundaka Upanishad). Samaveda linked to music. Gayatri Mantra in Rigveda. Ancient names of Punjab rivers. Founders of the six schools of philosophy (Kanada – Vaisheshika, Kapila – Samkhya).

★ Capitals of Mahajanapadas (especially Avanti–Ujjain, Kosala–Shravasti, Vatsa–Kaushambi, Anga–Champa). Only southern Mahajanapada = Assaka. Udayin founded Pataliputra. Battle of Hydaspes.

★ Place of first sermon (Sarnath). Fourth council under Kanishka and the Hinayana–Mahayana split. Third council under Ashoka. Mahavira added Brahmacharya. 23rd and 24th Tirthankaras. Tripitaka components. Shravanabelagola and Gomateshwara.

★ Author of Arthashastra and Indica. Kalinga war and Rock Edict XIII. Decipherment by James Prinsep. Script of most edicts (Brahmi). Sarnath lion capital as national emblem. Ashoka's children sent to Sri Lanka.

★ Kanishka and Shaka era; Fourth Buddhist Council. Gandhara vs Mathura art. Rudradaman's Junagadh inscription. Indo-Greeks first gold coins. Silappatikaram and Manimekalai authors. Karikala Chola built Kallanai.

★ Samudragupta 'Napoleon of India'. Kalidasa's works. Aryabhata's Aryabhatiya. Nalanda founder. Chinese travellers and their kings (Fa-Hien–Chandragupta II, Hiuen Tsang–Harsha). Harsha defeated by Pulakeshin II.

★ Builder of Kailasa temple at Ellora and Brihadeshwara temple. Tripartite struggle participants. Uttaramerur inscription on local self-government. Mahabalipuram under Pallavas. Vikramashila founder.

Medieval Indian History

★ Battles of Tarain and their winners. Al-Biruni's Kitab-ul-Hind. Muhammad bin Qasim and Sindh. Prithviraj Raso by Chand Bardai.

★ Muhammad bin Tughlaq's experiments (capital shift, token currency). Alauddin's market control and Dagh system. Iltutmish's tanka and jital. Razia only woman ruler. Ibn Battuta. Amir Khusrau. Founder of Agra. Balban's theory of kingship (Sijda, Paibos).

★ Founders of Vijayanagara and Bahmani. Krishnadeva Raya's Amuktamalyada and Ashtadiggajas. Battle of Talikota 1565. Gol Gumbaz location. Foreign travellers to Vijayanagara.

★ Philosophies of Ramanuja (Vishishtadvaita), Madhva (Dvaita), Shankara (Advaita). Shankara's four mathas. Disciples of Ramananda. Founder of Chishti order in India

(Moinuddin Chishti, Ajmer). Shankardev and Sattriya. Basavanna and Lingayats.

★ Three battles of Panipat with year and combatants – asked very often. Haldighati 1576. Akbar's Din-i-Ilahi, Mansabdari, Ibadat Khana. Sher Shah's Rupiya and GT Road. Shah Jahan's buildings. Bibi ka Maqbara. Thomas Roe at Jahangir's court. Baburnama language (Turkish). Humayunnama by Gulbadan Begum.

★ Shivaji's Ashtapradhan and Chauth–Sardeshmukhi. Founder of Khalsa and year. Guru who compiled Adi Granth; Guru who founded Amritsar; Guru who introduced Gurmukhi. Treaty of Bassein. Maharaja Ranjit Singh's capital.

Modern Indian History

★ First Europeans to come and last to leave (Portuguese). Vasco da Gama's arrival at Calicut. Battle of Wandiwash. Founder of Calcutta. Bombay as dowry.

★ Plassey vs Buxar participants and years. Treaty of Allahabad. Subsidiary Alliance (Wellesley) and first state to accept it (Hyderabad). Doctrine of Lapse states. Fourth Anglo-Mysore War and Tipu's death.

★ Partition of Bengal (Curzon, 1905). Ripon and local self-government. Sati abolition (Bentinck, 1829). First railway 1853 (Dalhousie). Vernacular Press Act (Lytton). Capital shift to Delhi 1911. Permanent Settlement (Cornwallis). Wood's Despatch 1854.

★ Permanent Settlement by Cornwallis; Ryotwari by Munro; Mahalwari by Holt Mackenzie.

★ Leaders at each centre of 1857 (especially Lucknow – Begum Hazrat Mahal, Bihar – Kunwar Singh, Jhansi – Lakshmibai, Kanpur – Nana Saheb). Starting place Meerut. Savarkar called it the first war of independence.

★ Founders of Brahmo Samaj, Arya Samaj, Prarthana Samaj, Satyashodhak Samaj. Satyarth Prakash author. Vivekananda at Chicago 1893. Vidyasagar and widow remarriage. Sir Syed Ahmed Khan and Aligarh.

★ First president and venue of INC. Sessions: Surat split, Lucknow Pact, Lahore Purna Swaraj, Belgaum (Gandhi), first woman president (Annie Besant), first Indian woman president (Sarojini Naidu), Haripura (Bose). Drain Theory by Naoroji. Founding of Muslim League in Dhaka. Home Rule Leagues.

★ Champaran as Gandhi's first satyagraha in India. Jallianwala Bagh date and General Dyer. Chauri Chaura and withdrawal of NCM. Dandi March dates. 'Do or Die' and Quit India. Second Round Table – only one Gandhi attended. Poona Pact. First individual satyagrahi. Kakori and Chittagong. Bhagat Singh's execution date. Simon Commission and Lala Lajpat Rai.

★ Newspapers and their founders (Kesari, Al-Hilal, Young India). Authors of Hind Swaraj, Discovery of India, Anandamath. Slogans and their authors (Inquilab Zindabad, Jai Hind).

Art & Culture

★ Classical dance–state matching is one of the most frequent culture questions. Sattriya's founder (Shankardev). Natyashastra author. Kathak as the only northern classical form. Famous dancers and their forms (Birju Maharaj – Kathak, Kelucharan Mohapatra – Odissi, Yamini Krishnamurthy – Bharatanatyam/Kuchipudi, Sonal Mansingh – Odissi).

★ Lavani (Maharashtra), Garba (Gujarat), Bihu (Assam), Cheraw (Mizoram), Yakshagana (Karnataka), Theyyam (Kerala), Ghoomar & Kalbelia (Rajasthan), Hojagiri (Tripura), Nati (Himachal), Raut Nacha & Panthi (Chhattisgarh), Jhumar (Punjab/Jharkhand), Rouf (J&K).

★ Musician–instrument matching (very frequent): Bismillah Khan – shehnai, Shivkumar Sharma – santoor, Hariprasad Chaurasia – flute, Amjad Ali Khan – sarod, Zakir Hussain – tabla, Ravi Shankar – sitar. Carnatic trinity. First musician with Bharat Ratna (M.S.

Subbulakshmi).

★ Bihu–Assam, Onam–Kerala, Pongal–TN, Hornbill–Nagaland, Hemis–Ladakh, Chapchar Kut–Mizoram, Wangala–Meghalaya, Sangai–Manipur, Nuakhai–Odisha. UNESCO ICH (Kumbh Mela, Yoga, Durga Puja, Garba).

★ Monument–builder–city matching. Nagara/Dravida/Vesara examples. Konark 'Black Pagoda'. Rani ki Vav on ₹100 note. GoI Gumbaz whispering gallery. Ahmedabad first Indian World Heritage City. Recently added UNESCO sites (Dholavira, Ramappa, Santiniketan, Hoysala temples, Moidams). Jantar Mantar builder.

★ Madhubani–Bihar, Warli–Maharashtra, Kalamkari–AP, Pattachitra–Odisha, Phad–Rajasthan, Gond–MP, Tanjore–TN. Kathputli–Rajasthan. First GI tag – Darjeeling tea.

★ Which amendment added which language (71st – Konkani, Manipuri, Nepali; 92nd – Bodo, Dogri, Maithili, Santhali). First classical language (Tamil). Ol Chiki script for Santhali. Number of scheduled languages (22).

Indian Polity I – Constitution and Rights

★ Chairman of Drafting Committee (Ambedkar), President of Constituent Assembly (Rajendra Prasad), temporary president (Sachchidananda Sinha). Adoption and enforcement dates. Objectives Resolution by Nehru. Constitutional adviser B.N. Rau. Flag adoption date.

★ Source of DPSP (Ireland), Fundamental Duties (USSR), Concurrent List (Australia), Fundamental Rights and judicial review (USA), amendment procedure (South Africa), residuary powers (Canada), Emergency suspension (Germany). GoI Act 1935 as the main source.

★ Words added by the 42nd Amendment. Preamble part of the Constitution per Kesavananda Bharati. Source of liberty-equality-fraternity (France). Order of words in Preamble (Sovereign Socialist Secular Democratic Republic).

★ Schedule contents (especially 7th, 8th, 9th, 10th, 11th). Subjects moved to Concurrent List by the 42nd Amendment. Number of subjects in the 11th and 12th schedules. Part III and IV article ranges.

★ Article 1 wording; Article 3 for new states. First linguistic state (Andhra). Fazl Ali Commission. Newest state (Telangana). Number of states and UTs. Sikkim's merger amendment.

★ Article 32 'heart and soul'. Writs and their meanings (Habeas Corpus, Mandamus, Quo Warranto). Article 17 untouchability, Article 21A education, Article 24 child labour. Right to property removal (44th Amendment) and new Art 300A. Articles 20 and 21 cannot be suspended in emergency. Privacy as FR (2017).

★ Article 44 (UCC), Article 40 (panchayats), Article 48A (environment), Article 50 (separation of judiciary from executive), Article 39A (free legal aid). DPSP source (Ireland). Non-justiciable nature.

★ Committee recommending Fundamental Duties (Swaran Singh). Number of duties (11). 11th duty added by 86th Amendment. Article 51A.

Indian Polity II – Institutions

★ Electoral college composition (who votes, who doesn't). Oath by CJI; resignation to VP. Impeachment ground. Art 123 ordinance, Art 72 pardon. Pocket veto (Zail Singh). Emergency articles. First woman / Dalit / tribal President. Rajendra Prasad only President elected twice.

★ VP is ex-officio Chairman of Rajya Sabha. 15% limit on ministry size (91st Amendment). Attorney General can speak but not vote. Art 74 and 75. First non-Congress PM (Morarji Desai).

★ Who certifies money bill (Speaker). Joint sitting presided by Speaker; held thrice. RS can hold money bill for 14 days. Zero Hour is an Indian innovation. PAC composition and chairman. RS powers under Art 249 and 312. First Speaker G.V. Mavalankar. Minimum ages. Members nominated to RS (12).

★ Retirement ages of SC and HC judges. Art 143 advisory jurisdiction. Basic structure doctrine case. Oldest High Court. Collegium and NJAC. First Lok Adalat at Junagadh. PIL originators. Removal procedure of judges.

★ Balwant Rai Mehta Committee and three tiers. First state with Panchayati Raj (Rajasthan, Nagaur). 73rd/74th Amendments details. Women's reservation 1/3 in local bodies. States having Legislative Councils. Governor's pardon power excludes death sentence. Art 239AA Delhi.

★ Articles for CAG, Finance Commission, Election Commission, UPSC. First CEC Sukumar Sen. NITI Aayog date and nature (non-constitutional). NHRC year. GST Council article and chairman. CAG called guardian of public purse.

★ 42nd as 'mini constitution'. 61st lowered voting age to 18. 52nd anti-defection. 86th education. 101st GST. 103rd EWS 10%. 106th women's reservation. 24th assent compulsory. 36th Sikkim. Article 368.

★ First use of EVM. NOTA year. National Voters' Day date. Articles 324–329 on elections. Voting age 18 (61st Amendment).

Indian Geography

★ States crossed by Tropic of Cancer. Standard meridian city. Neighbour with longest/shortest border. Ten Degree Channel. McMahon/Durand/Radcliffe lines. Southernmost point Indira Point. Palk Strait.

★ Pass–state matching (Zoji La, Nathu La, Shipki La, Rohtang, Bomdi La, Lipulekh). Highest peaks (Kangchenjunga, Anamudi, Guru Shikhar, Dhupgarh). Bhabar/Terai/Bhangar/Khadar. Barren Island volcano. Chilika and Wular lakes. Nilgiris junction of Ghats.

★ River origins (Narmada & Son – Amarkantak; Godavari – Trimbakeshwar; Kaveri – Talakaveri; Krishna – Mahabaleshwar). West-flowing rivers through rift valleys (Narmada, Tapi). 'Sorrow of Bihar' (Kosi) and 'Sorrow of Bengal' (Damodar). Dam–river matching (Hirakud, Tehri, Bhakra, Sardar Sarovar). Majuli. Waterfalls (Jog, Chitrakote, Dudhsagar). Loktak floating park. Indus Waters Treaty year.

★ Local winds (Loo, Mango showers, Kalbaisakhi). Mawsynram highest rainfall. Tamil Nadu's winter rainfall from NE monsoon. Western disturbances and rabi crops. El Niño effect.

★ Black soil and cotton (regur). Laterite soil formed by leaching. Red colour due to iron oxide. Gypsum for alkaline soil. Sundarbans mangroves and sundari tree. State with largest forest cover (MP).

★ National park–state–animal matching (very frequent): Kaziranga, Gir, Dachigam, Keibul Lamjao, Silent Valley, Eravikulam, Hemis, Kuno. Project Tiger year and first reserve. Ramsar first sites. First biosphere reserve Nilgiri. Largest tiger reserve.

★ Kharif vs rabi crop identification. Leading producer states (coffee, rubber, jute, tea, sugarcane). Revolutions and their products/fathers. Operation Flood. International Year of Millets 2023.

★ Mineral–location matching (Jharia, Kolar, Khetri, Jaduguda, Panna, Mumbai High, Digboi). Steel plants and foreign collaboration (Bhilai–USSR, Rourkela–Germany, Durgapur–UK). Nuclear plants. Industrial city nicknames.

★ First train date and route. Railway zone headquarters. Longest NH. Port characteristics (Kandla tidal, Visakhapatnam deepest, Kolkata riverine, Mormugao iron ore). National waterways. Atal tunnel.

World & Physical Geography

★ Hottest planet Venus (not Mercury). Planets rotating clockwise (Venus, Uranus). Largest moon Ganymede. Red planet. Pluto's reclassification. Light year as distance. Brightest star Sirius. Solar vs lunar eclipse positions.

★ Longest day 21 June. Equinox dates. Time-longitude relation. Countries on the Equator and tropics. Seasons due to axial tilt. International Date Line.

★ Most abundant element in crust (oxygen) and metal (aluminium). SIAL/SIMA/NIFE. Metamorphic rock examples (marble, slate, quartzite). P and S waves. Ring of Fire. Wegener's continental drift. Richter scale.

★ Layer with ozone (stratosphere) and weather (troposphere). Ionosphere reflects radio waves. Composition percentages. Local winds and their regions (Chinook, Foehn, Sirocco, Harmattan, Mistral). Cyclone names by region. Roaring Forties. Ozone Day.

★ Deepest trench. Warm vs cold currents. Gulf Stream effect on Europe. Grand Banks fishing ground. Ocean named after a country (Indian). Saltiest water bodies. Great Barrier Reef.

★ Strait–countries matching (Gibraltar, Hormuz, Malacca, Bering, Dover). Boundary lines (38th parallel, 49th parallel, Durand). Grasslands names. Highest waterfall and longest river. Suez/Panama. Indian Antarctic stations. Lake Baikal deepest. Everest revised height.

★ Parliament names (Diet – Japan, Knesset – Israel, Majlis – Iran/Maldives, Jatiya Sangsad – Bangladesh, Storting – Norway, Riksdag – Sweden, Folketing – Denmark, Bundestag – Germany, Duma – Russia). Currencies of neighbours (Ngultrum, Taka, Kyat, Rufiyaa). Capitals of Myanmar (Naypyidaw) and Sri Lanka. Three capitals of South Africa.

Indian Economy

★ Father of economics. Market forms (monopsony, oligopoly, duopoly). Giffen goods. Law of demand. Sector classification of activities.

★ GDP vs GNP definitions. First person to estimate national income (Naoroji). HDI components and founder (Mahbub ul Haq). CPI and WPI agencies. Inflation target. Stagflation meaning.

★ Second Plan's Mahalanobis model. First Plan – Harrod-Domar, agriculture. Plan holiday years. Rolling plan concept. NITI Aayog date and chairman. Bombay Plan.

★ RBI establishment and nationalisation years. Who issues ₹1 note (Finance Ministry). Repo/CRR/SLR definitions and their effect on inflation. Bank nationalisation dates. NABARD functions. SEBI as market regulator. Sensex/Nifty composition. Narasimham committee. UPI and NPCI. First Indian bank (PNB). M3 as broad money.

★ Fiscal deficit definition (= borrowings). Primary deficit. Direct vs indirect tax examples. GST launch date, amendment, first state to ratify. Rail budget merger 2017. First budget presenter. Laffer curve. FRBM Act.

★ 1991 reforms (LPG) and leaders. WTO formation and HQ. Bretton Woods institutions. FEMA replaced FERA. IDA as soft-loan window. HQs of ADB, AIIB, NDB. SDR basket.

★ Scheme–purpose matching (MGNREGA days, Ujjwala LPG, PM-KISAN amount, Ayushman Bharat cover, Jal Jeevan). Tendulkar committee on poverty. Disguised unemployment in

agriculture. Launch places (Beti Bachao – Panipat).

★ Census 2011 sex ratio, literacy, density, highest/lowest states. First synchronous census 1881. Most and least populous states.

General Science – Physics

★ SI units of physical quantities (pressure – pascal, power – watt, magnetic field – tesla, radioactivity – becquerel, luminous intensity – candela). Light year/parsec as distance. Nautical mile and knot usage. Temperature conversion −40. Scalar vs vector identification. Unit of power of lens.

★ Examples of Newton's laws (rocket – 3rd law; inertia examples). g at poles vs equator. Weightlessness in satellites. Escape velocity value and its independence from mass. Pendulum period independent of mass. Work done zero when force □ displacement. Doubling velocity quadruples KE.

★ Pascal's law applications. Archimedes' principle and floatation. Why it's easier to swim in sea water. Surface tension examples (soap reduces it). Bernoulli examples (aeroplane, roof blown off). Pressure cooker cooks faster (raises boiling point). Barometer reading and weather.

★ Anomalous expansion of water (4 °C). Steam burns more than boiling water (latent heat). Pressure cooker principle. Salt raises boiling point and lowers melting point. Sea/land breeze timing. Earthen pot cooling by evaporation. Thermos flask design. Why mercury in thermometers. White clothes in summer. Woollen clothes keep warm (trapped air).

★ Sound cannot travel in vacuum. Speed of sound highest in solids. Audible range. Bats and dolphins use ultrasound. SONAR principle. Doppler effect example. Pitch depends on frequency. Decibel unit. Soldiers break step on bridges (resonance).

★ Mirror used in vehicles (convex), shaving (concave). Correction lenses for eye defects. Blue sky and red sunset (scattering). Twinkling of stars (refraction). Diamond's brilliance and optical fibres (total internal reflection). Rainbow formation. Mirage. Raman effect and National Science Day. Lateral inversion. Danger lights red. Power of lens in dioptre.

★ Household wiring is parallel. Fuse wire properties. Tungsten in bulbs; nichrome in heaters. Commercial unit of energy (kWh). Transformer works on AC only and principle of mutual induction. Generator vs motor energy conversion. Ammeter in series, voltmeter in parallel. Frequency of AC in India (50 Hz). Silver best conductor.

★ Discoverers of electron, proton, neutron, X-rays, radioactivity. Fission vs fusion. Energy source of the Sun (fusion). Carbon dating. Order of electromagnetic spectrum. TV remote uses infrared. Microwave oven principle. LASER full form. Moderator and control rods. Universal gates. Photoelectric effect and Einstein's Nobel.

★ Instrument–quantity matching is asked in almost every SSC paper (hygrometer, anemometer, altimeter, pyrometer, tachometer, sphygmomanometer, lactometer, fathometer).

General Science – Chemistry

★ Examples of sublimation. Colloids and Tyndall effect (milk, fog). Separation techniques (cream from milk – centrifugation; petroleum – fractional distillation; dyes – chromatography). Isotopes vs isobars. Avogadro number. Plasma state.

★ Father of the periodic table (Mendeleev) and modern periodic law (Moseley). Group names (halogens, alkali, noble gases). Extremes (most electronegative, liquid non-metal, lightest metal, best conductor). Noble gas uses (helium for diving & balloons, neon for signs). Periodic trends.

★ Common name–formula matching (very frequent). Natural acids (ant sting – formic, tomato – oxalic, curd – lactic, tamarind – tartaric, vinegar – acetic). pH of blood, gastric juice, pure water. Indicator colours (phenolphthalein pink in base). Aqua regia composition. Rusting conditions and galvanisation. Catalysts in Haber/Contact processes. Hydrogenation with nickel. Antacids.

★ Alloy compositions (brass, bronze, solder, German silver, duralumin, stainless steel). Ores of metals (bauxite – Al, galena – Pb, cinnabar – Hg, haematite – Fe). Allotropes of carbon and their properties. Metals stored in kerosene. Reactivity series. Froth flotation. Pure gold carat. Vulcanisation with sulphur. Thermite welding.

★ Main constituents of LPG, CNG, biogas. Bhopal gas (MIC). Diseases from pollutants (Minamata – mercury, Itai-itai – cadmium, blue baby – nitrate). Polymers and uses (Teflon, Bakelite, PVC, nylon, rayon). Hard water and its removal. Saponification. CO poisoning. Ethene ripens fruits; calcium carbide banned. Acid rain & Taj Mahal. Penicillin discovery. Fire extinguisher types. Glass colouring oxides.

General Science – Biology

★ Discoverer of cell and nucleus. Powerhouse (mitochondria), suicide bags (lysosomes), protein synthesis (ribosomes). Organelles with own DNA. Cell wall material. Mitosis vs meiosis. Watson & Crick. Number of human chromosomes. Largest/longest human cell.

★ Binomial nomenclature and Linnaeus. Five-kingdom classification. Scientific names of common organisms (Homo sapiens, Panthera leo/tigris, Mangifera indica, Oryza sativa, Triticum aestivum, Azadirachta indica – neem, Ficus benghalensis – banyan). Viruses as a link. Animals wrongly called fish. Heart chambers by class. Bryophytes/pteridophytes characteristics.

★ Vitamin–deficiency disease matching is one of the most frequent GA questions. Chemical names (retinol, calciferol, ascorbic acid, tocopherol). Fat- vs water-soluble vitamins. Vitamin made in skin by sunlight (D). Vitamin by gut bacteria (K). Mineral in B12 (cobalt). Iodine and goitre. Kwashiorkor vs marasmus. Richest source of vitamin C (amla).

★ Largest gland, master gland, largest organ. Blood groups – universal donor/recipient and Landsteiner. Hormone–gland–disorder matching (insulin – diabetes, thyroxine – goitre, ADH – diabetes insipidus, adrenaline – emergency). Function of cerebellum and medulla. Number of bones; smallest and largest bones. RBC lifespan, site of formation and destruction. Bile has no enzymes. HCl in stomach. Heart chambers and pacemaker. Nephron as kidney unit. Haemophilia.

★ Disease–pathogen–vector matching (malaria, dengue, kala-azar, sleeping sickness, filaria). Bacterial vs viral diseases. Tests (Widal, ELISA, Mantoux). Vaccine pioneers (Jenner, Pasteur, Salk). Polio-free India 2014. HIV attacks T-helper cells. Occupational diseases (silicosis, byssinosis). Genetic disorders (Down syndrome trisomy 21). Ronald Ross and malaria.

★ Edible parts of plants (potato stem, carrot root, ginger rhizome, saffron stigma, clove bud, cauliflower inflorescence, apple thalamus). Plant hormones and functions (ethylene ripening, auxin phototropism, ABA stress). Source of oxygen in photosynthesis. Xylem vs phloem. Rhizobium and legumes. Mimosa movement. Bt cotton. Annual rings for age. Insectivorous plants (Nepenthes – pitcher plant, Drosera, Venus flytrap) grow in nitrogen-poor soil.

★ Mendel's ratios and pea plant. Darwin vs Lamarck. Homologous vs analogous organs. Vestigial organs. 10% law of energy transfer. Biomagnification. Biodiversity hotspots of India. Chipko and Bishnoi movements. Dolly the sheep. DNA fingerprinting in India (Lalji Singh). Sex determination by father.

Static GK I – Sports

★ First Indian individual gold (Bindra), first woman medallist (Malleswari), first individual medal (K.D. Jadhav). Neeraj Chopra's javelin gold. Olympic ring colours. Coubertin. First Asian Games host (Delhi 1951). National Sports Day and Dhyan Chand. Number of hockey golds.

★ Number of players per team (kabaddi 7, kho-kho 9, polo 4, volleyball 6, netball 7). Terms belonging to sports (googly – cricket; bully – hockey; chukker – polo; birdie – golf; deuce – tennis; castling – chess). Origin of games. National games of countries. India has no official national game.

★ Trophy–sport matching (Ranji, Durand, Santosh, Beighton, Thomas, Uber, Swaythling, Davis, Ryder). Grand Slam surfaces. India's World Cup wins and captains. Stadium–city matching (Eden Gardens, Wankhede, Kalinga, Chepauk). Oldest football tournament in Asia (Durand).

★ Nicknames (Flying Sikh, Payyoli Express, Dhing Express, Master Blaster, Little Master, The Wall). Sport of famous players (Pankaj Advani – billiards, Dipa Karmakar – gymnastics, Mary Kom – boxing, Mirabai – weightlifting). First Indian woman on Everest. First Grandmaster. Arjuna/Dronacharya award categories. D. Gukesh world champion.

Static GK II – Books, Awards, Days, Firsts, Organisations

★ Flag ratio and spokes, designer Pingali Venkayya. Anthem duration (52 s) and first singing. Source of 'Satyameva Jayate'. National aquatic animal (dolphin), heritage animal (elephant). ₹ symbol designer. Vande Mataram source (Anandamath).

★ National Science Day reason, Engineers' Day (Visvesvaraya), Teachers' Day, National Sports Day, Constitution Day, Yoga Day, World Environment Day, Kargil Vijay Diwas, Army/Navy/Air Force days, National Youth Day, Good Governance Day, Rashtriya Ekta Diwas.

★ Book–author matching (very frequent): Rajatarangini, Gita Govinda, Anandamath, Godan, Discovery of India, Wings of Fire, Midnight's Children, Tomb of Sand, India Wins Freedom, Untouchable. Sports autobiographies (Playing It My Way, Unbreakable, Race of My Life). Booker-winning Indians.

★ First Bharat Ratna recipients, first woman/foreigner/sportsperson. Jnanpith first recipient and language. Dadasaheb Phalke first recipient. Indian Nobel laureates with fields and years. Magsaysay as Asia's Nobel. Fields medal field (mathematics). Pritzker (architecture). Gallantry awards order. Recent Bharat Ratna (2024).

★ Women firsts (CM, Governor, SC judge, IPS, Union minister). First Indian in space. First talkie and feature film. First satellite. Field Marshals. First CJI. Pokhran tests. First newspaper.

★ HQ of organisations (WHO, UNESCO, UNEP, IAEA, FAO, Interpol, Amnesty, WWF, ASEAN, SAARC). UN Security Council P5. UN founding date. First international organisation with HQ in India (ISA). SAARC members. G20 presidency of India. UN official languages.

★ HQs of Indian institutes (ISRO, BARC, CRRI Cuttack, NDRI Karnal, Survey of India Dehradun, BSI/ZSI Kolkata). Defence academies' locations. Missile types (Prithvi, Agni, Akash, Nag, BrahMos). Military exercise partners (Malabar, Yudh Abhyas, Surya Kiran, Mitra Shakti, Sampriti). CAPF border responsibilities. First CDS.

★ Inventor matching (telephone – Bell, radio – Marconi, television – Baird, WWW – Berners-Lee, dynamite – Nobel, printing press – Gutenberg, barometer – Torricelli, penicillin – Fleming, crescograph – J.C. Bose). Indian scientists and their contributions (Raman, J.C. Bose, S.N. Bose, Chandrasekhar).

Science & Technology, Space and Computers

★ First Indian satellite and rocket. Father of Indian space programme. Chandrayaan-1/2/3 achievements and dates; Vikram & Pragyan; National Space Day. Mangalyaan's distinction. Aditya-L1 destination. NavIC as regional navigation. PSLV 104 satellites record. Sputnik, Gagarin, Tereshkova firsts. Launch site Sriharikota and first launch from Thumba.

★ Father of Indian nuclear programme. Pokhran test code names and years. Thorium-based third stage. Mission Shakti (ASAT). Kalam and IGMDP.

★ Father of computer. Computer generations and technologies. RAM vs ROM. Memory units conversion. Input vs output devices (MICR for cheques, OMR for answer sheets). Compiler vs interpreter. First supercomputer of India. Shortcut keys (Ctrl+Z, F7, F12, Alt+F4). Linux open-source. Inventors of C, Java, Python. Binary base. MS Excel/PowerPoint usage.

★ Full forms (HTTP, URL, DNS, LAN, WAN, ISP, Wi-Fi, PDF, CAPTCHA, UPS). Inventor of WWW. Email protocols (SMTP send, POP3 receive). Virus vs worm vs Trojan. Phishing. Firewall role. IT Act year. Network topologies and devices (router, modem). IPv4/IPv6 size. Father of AI.

★ Full forms are asked in nearly every shift (LASER, RADAR, SONAR, ISRO, NABARD, SEBI, MGNREGA, UPI, NEFT, RTGS, BIOS, HTML, URL). New criminal laws replacing IPC/CrPC/Evidence Act (2024).